MIL-T-60857A(MU)
15 January 1973
SUPERSEDING
MIL-T-60857(MU)
10 November 1969

MILITARY SPECIFICATION

TEST SET, RADAR: AN/TPM-22

1. SCOPE

1.1 This specification covers Radar Test Set AN/TPM-22, support maintenance shop set, hereinafter referred to as SMSS, a console and rack mounted test set used by direct support and general support maintenance (3rd and 4th echelons) for testing all items of Radar Set AN/VPS-2 except Antenna AS-2049/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction: M18 Manufacturing Processes and Inspection of
MIL-T-50332	Test Set, Subassembly MX-8047/TPM-22: 10554526
MIL-T-50333	Test Set, Subassembly MX-8048/TPM-22: 10554527
MIL-T-50334	Test Set, Subassembly MX-8049/TPM-22: 10554528
MIL-T-50335	Test Set, Subassembly MX-8051/TPM-22: 10554530
MIL-T-50339	Test Set, Subassembly MX-8050/TPM-22: 11726306
MIL-P-50346	Power supply subassembly: 10554531

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MIL-T-50752	Test Set, Subassembly MX-8046A/TPM-22: 11739273

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

B10554524	Test Set, Radar AN/TPM-22
F10554526	Test Set, Subassembly MX-8047/TPM-22
F10554527	Test Set, Subassembly MX-8048/TPM-22
F10554528	Test Set, Subassembly MX-8049/TPM-22
F10554530	Test Set, Subassembly MX-8051/TPM-22
F10554531	Power Supply Subassembly PP6324/TPM-22
F10554646	Bench Console Assembly
F10554772	Vertical Console Assembly
F10554809	Overhead Console Assembly
F11726306	Test Set, Subassembly MX-8050/TPM-22
F11739273	Test Set, Subassembly MX8046A/TPM-22

PACKAGING DATA SHEET

10554524	Packaging of Test Set, Radar AN/TPM-22

(Copies of specification, standards, drawings and packaging data sheets required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Fabrication.- The SMSS shall be manufactured in accordance with Drawing B10554524 and drawings pertaining thereto. The SMSS consists of vertical console assembly 1A1 (Dwg. B10554772), overhead console assembly 1A2 (Dwg. F10554809), and bench console assembly 1A3 (Dwg. F10554646). The subassemblies of the SMSS shall have been inspected and accepted in accordance with their respective drawings or specifications prior to their installation in the SMSS. The following subassemblies are covered by the indicated specifications:

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UNIT NO.	NAME	DRAWING	SPECIFICATION
1A1A2	Test Set, Subassembly MX-8047/TPM-22	10554526	MIL-T-50332
1A1A3	Test Set, Subassembly MX-8051/TPM-22	10554530	MIL-T-50335
1A1A4	Power Supply Subassembly PP6324/TPM-22	10554531	MIL-P-50346
1A2A1	Test Set, Subassembly MX-8048/TPM-22	10554527	MIL-T-50333
1A2A3	Test Set, Subassembly MX-8049/TPM-22	10554528	MIL-T-50334
1A3A2	Test Set, Subassembly MX-8050/TPM-22	11726306	MIL-T-50339
1A1A1	Test Set, Subassembly MX-8046A/TPM-22	11739273	MIL-T-50752

3.2 General specification.- Where applicable, the SMSS shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishings
d. Part identification and marking
e. Workmanship
f. Soldering electrical connections

3.2.2 MIL-C-46357.

a. Etched circuit assemblies
b. Wiring
c. Threaded fasteners
d. Marking
e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration.- The SMSS shall be mounted on the vibration table in normal service position. The SMSS shall meet the requirements of 3.4 and 3.5 after being subject to continuous vibration for two hours along each of three mutually perpendicular axes within the frequency range and amplitude as follows:

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Frequency Range	Double Amplitude
5 to 35	0.06 inch
15 to 25	0.04 inch
25 to 55	0.02 inch

The rate of change of frequency at each frequency range shall not exceed one cycle-per-second per second (1 cps/sec). Vibration tests shall be conducted under ambient room conditions.

3.3.2 Shock.- The SMSS shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G's) and a pulse duration of 11 ± 1 milliseconds.

3.3.3 Altitude.

3.3.3.1 Operating.- The SMSS shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg.).

3.3.3.2 Nonoperating.- The SMSS shall meet the requirements of 3.4 and 3.5 after exposure for not less than 1 hour to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg.).

3.3.4 Temperature.

3.3.4.1 Operating.- The SMSS shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between +32°F and +131°F.

3.3.4.2 Nonoperating (storage).- The SMSS shall not suffer permanent damage when subjected to the temperature extremes of -80°F and +185°F, and shall meet the requirements of 3.4 and 3.5 when turned to room ambient.

3.3.5 Humidity-nonoperating.- The SMSS shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical.- Each conductor shall be electrically continuous as specified on the schematic diagram.

3.4.1 Test Set, Subassembly MX-8049/TPM-22.- The Digital Voltmeter and Oscilloscope thumb wheel switches of Test Set, Subassembly MX-8049/TPM-22 shall establish electrical continuity between the points indicated below when set to the positions shown below.

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SIGNAL MONITORING SWITCH CONTINUITY

DIGITAL VOLTMETER SWITCH

SWITCH NUMBER	POSITION	TERMINAL POINTS FROM	TO
	0	J4-E (Conductor)	
	1	J1-A	
	2	J1-H	
	3	J1-F	
	4	J1-C	
	5	J2-H	
	6	J2-K	
S7	7	J2-G	
High	8	J2-A	J10
	9	J2-J	
	10	J3-A	
	11	J3-C	
	12	J3-B	
	13	J3-J	
	14	J3-H	
	15	No Connection	
	0	No Connection	
	1	J1-B	
	2	J1-G	
	3	J1-G	
	4	J1-K	
	5	J2-F	
	6	J2-F	
S8	7	J2-F	
Return	8	J2-B	J11
	9	J2-B	
	10	J3-L	
	11	J3-L	
	12	J3-L	
	13	J3-K	
	14	J3-K	
	15	No Connection	

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SIGNAL MONITORING SWITCH CONTINUITY (Continued)

OSCILLOSCOPE SWITCH

SWITCH NUMBER	POSITION	TERMINAL POINTS FROM	TO
	0	J4-B*	J6*
	0	J4-B**	J6**
	1	J4-C*	J6*
	1	J4-C**	J6**
	2	J4-D*	J6*
	2	J4-D**	J6**
	3	J4-E*	J6*
	3	J4-E**	J6**
S5	4	J4-F*	J6*
Chan. A	4	J4-F**	J6**
	5	J4-G*	J6*
	5	J4-G**	J6**
	6	J3-A*	J6*
	6	J3-L**	J6**
	7	J3-J*	J6*
	7	J3-K**	J6**
	8	J5-E*	J6*
	8	J5-F**	J6**
	9	J25-16	J6*
	9	J25-14	J6**
	0	J4-B*	J7*
	0	J4-B**	J7**
	1	J4-C*	J7*
	1	J4-C**	J7**
	2	J4-D*	J7*
	2	J4-D**	J7**
	3	J4-E*	J7*
	3	J4-E**	J7**
S6	4	J4-F*	J7*
Chan. B	4	J4-F**	J7**
	5	J4-G*	J7*
	5	J4-G**	J7**
	6	J3-A*	J7*
	6	J3-L**	J7**
	7	J3-J*	J7*
	7	J3-K**	J7**
	8	J5-E*	J7*
	8	J5-F**	J7**
	9	J25-16	J7*
	9	J25-14	J7**

*Conductor
**Shield

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3.4.2 Test Set, Subassembly MX-8051/TPM-22.

3.4.2.1 Overload test circuitry.- Test Set, Subassembly MX-8051/TPM-22 shall have the resistance values shown below between the indicated connection points with controls set as indicated.

OVERLOAD TEST CIRCUITY RESISTANCE

TERMINAL POINTS		CONTROL SETTINGS LOAD SWITCH		VARIABLE LOADS		RESISTANCE BETWEEN TERMINAL POINTS
FROM	TO	S10	S11	R27	R29	OHMS ± 10%
E17	E18	Min.	N.A.*	Extreme CW	N.A.	30
		Max.		Extreme CW		Less than 1
		Max.		Extreme CCW		3
E17	E18	NL	N.A.	N.A.	N.A.	
E19	E20	N.A.	Min.	N.A.	Extreme CW	500
			Max.		Extreme CW	Less than 1
			Max.		Extreme CCW	50
E19	E20		NL	N.A.	N.A.	

*N.A. - Not Applicable

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3.4.2.2 Converter loading circuitry.- The resistance of Test Set, Subassembly MX-8051/TPM-22 between TP27 and TP28 shall be as shown below with S13 in CONVERTER position and with controls set as shown below.

CONVERTER LOADING CIRCUITRY RESISTANCE

CONVERTER SWITCH S14 POSITION	CONVERTER LOAD SWITCH NUMBER	CONVERTER LOAD SWITCH POSITION	RESISTANCE BETWEEN TP27 AND TP28 OHMS ± 5%
OFF	--	--	0
+24V	--	--	18*
+22V	S15	NL	∞
+22V	S15	FL	22
-22V	S15	NL	∞
-22V	S15	FL	22
+30V	S16	NL	∞
+30V	S16	FL	30
-30V	S16	NL	∞
-30V	S16	FL	30
+40V	S17	NL	∞
+40V	S17	FL	40
-40V	S17	NL	∞
-40V	S17	FL	40
+40V	S18	NL	∞
+40V	S18	FL	40
+24V	S18	NL	∞
+24V	S18	FL	∞

*Tolerance is ± 12 ohms.

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3.4.2.3 Servo loading circuitry.- Resistance values of Test Set, Subassembly MX-8051/TPM-22 shall be as shown below between the indicated points.

SERVO LOADING CIRCUITRY RESISTANCE

TERMINAL POINTS		RESISTANCE BETWEEN POINTS
J1-m	TP17	1000 ohms ± 1%
J1-r	TP18	1000 ohms ± 1%
J7-a	TP19	1000 ohms ± 1%
J7-m	TP20	1000 ohms ± 1%
J7-g	TP21	1000 ohms ± 1%
J7-g	TP22	1000 ohms ± 1%
J1-m	J1-r	5 ohms ± 5%
J7-a	J7-m	5 ohms ± 5%
J7-g	J7-g	5 ohms ± 5%

3.5 Performance.- The SMSS shall perform the functions described hereinafter when power, as required, is applied as described below, and when all circuit breakers and interlock switches are closed.

3.5.1 Power Requirements.- The SMSS shall operate on electrical power meeting the following requirements.

3.5.1.1 General use primary power.

	Characteristics	Terminals
(a)	24 Vdc ± 2V at 50 amps (negative side grounded at source only).	P20-0&R P20-P(Ret)
(b)	115V ± 10%, 400 Hz ± 5%, 1 phase at 10 amps.	P23-X & Y
(c)	280V ± 10%, 400 Hz ± 5%, 3 phase, 4 wire wye at 5 amps.	P22-W(ØC), X(ØB), Y(ØA), Z(Ret)

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3.5.1.2 Special purpose power.

	Characteristics	Terminals
(a)	115V ± 5% rms 400 Hz ± 1%, 1 phase at 300 mA, Servo Power	P20-L P20-K(Ret)
(b)	208V ± 5% peak-to-peak, 400 Hz ± 1%, 1 phase, center tap, at 700 mA, quasi-squarewave servo power.	P20-Z & a, P20-d(CT)
(c)	± 15 Vdc ± 0.5V at 300 mA servo power.	P20-I, (+) P20-H(RLT) P20-J(-)

3.5.1.3 Output power.- The SMSS shall provide power and signals meeting the following requirements for use in conjunction with GE DS/CS equipment:

	Characteristics	Terminals
(a)	+5 Vdc ± 5% at 2 amps maximum	P20-b, P20-c(Ret)
(b)	-23 Vdc ± 5% at 0.5 amps maximum	P20-e,P20-f(Ret)
(c)	-10 Vdc ± 5% at 3 mA maximum	P20-Y,P20-X(Ret)
(d)	+10 Vdc ±5% at 5 mA maximum	P20-V,P20-X(Ret)
(e)	+20 Vdc ± 5% at 3 mA maximum	P20-W,P20-X(Ret)

3.5.1.4 Interlock switch 1A1S1.- When subassembly 1A1A3 is not in its fully closed position (secured in the vertical console), interlock switch 1A1S1 shall serve to disconnect a 24 volt power control signal from subassembly 1A1A3. When the switch shaft is pulled out to the service position, the power control signal shall be reapplied to the subassembly. The switch shall be connected to a pilot light indicating when the switch is in the service position.

3.5.1.5 Circuit breaker CB1.- Three-phase circuit breaker CB1 shall be closed when reset button is depressed so white band is concealed. All three legs of the breaker shall be open when reset button is pulled out manually to expose white band.

3.5.2 Power supply subassembly 1A1A4.- This subassembly supplies power for all subassemblies of the SMSS except the blower subassembly 1A1A5.

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3.5.2.1 Circuit breakers.- Each circuit breaker (CB1 through CB5) listed below shall be closed when reset push-button is depressed so white band is concealed. Each circuit breaker shall be open when reset pushbutton is pulled out (manually or by overload) to expose white band.

Panel Marking	Reference Designation
115 V - 400 Hz (2A)	CB1
PWR SUP 1 (35A)	CB2
TESTER CKT (10A)	CB3
CONTROL CKT (2A)	CB4
PWR SUP 2 (25A)	CB5

3.5.2.2 115V, 400 Hz ON-OFF Switch S1.- With, TEST MODE switch S2 in any position except FC CONSOLE, 115V, 400 Hz switch S1 shall serve to change indicator DS1 from OFF to ON and the reverse, and to cause 115 ± 10 volts, 400 Hz power to be applied to J2-A and L in accordance with indicator DS1.

3.5.2.3 +24 Vdc ON-OFF switch S3.- With, TEST MODE switch S2 in any position except FC CONSOLE, +24 Vdc switch S3 shall serve to change indicator DS3 from OFF to ON and the reverse, and to cause +24 volts dc power to be applied to points specified on the schematic drawing, in accordance with indicator DS3, and shall cause blower 1A1A5 to operate.

3.5.2.4 Converter No. 1 voltages.- With S3 ON, TEST MODE switch S2 in any position except FC CONSOLE, and S5 thrown to position 1, VOLTAGE selector switch S4 shall serve to register voltages of converter No. 1 on VOLTS meter M1, in accordance with switch positions and indicated voltages of Table I.

3.5.2.5 Converter No. 2 voltages.- With S5 thrown to position 2, and either S3 ON or TEST MODE switch S2 in the FC CONSOLE position, VOLTAGE selector switch S4 shall serve to register the voltages of converter No. 2 on VOLTS meter M1, in accordance with the switch positions and indicated voltages of Table I, with the exception that in the +6 position the reading shall be +5 ± 1.

TABLE I - CONVERTER VOLTAGES

POSITION OF VOLTAGE SWITCH (S4)	READING ON VOLTS METER (M1)
OFF	0 ± 1
+24V	+24 ± 2
+6	+6 ± 1
+15	+15 ± 1

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-15	-15 ± 1
+23	+23 ± 1
-23	-23 ± 1
+29	+29 ± 1
+33	+33 ± 1
-33	-33 ± 1

3.5.2.6 DC voltage outputs.- With S3 ON, and TEST MODE switch S2 in any position except FC CONSOLE, the voltages of Table II shall appear at the terminals indicated in Table II.

TABLE II - OUTPUT VOLTAGES

CONNECTOR TERMINALS	NO LOAD DC VOLTS
J3-P(+) to J3-R(RET)	+24 ± 2
J3-S(+) to J3-T(RET)	+24 ± 2
J3-A(+) to J3-B(RET)	+15 ± 1
J3-E(-) to J3-F(RET)	-15 ± 1
J3-C(+) to J3-H(RET)	+23 ± 1
J3-c(-) to J3-W(RET)	-23 ± 1
J3-h(+) to J3-d(RET)	+33 ± 1
J3-j(-) to J3-e(RET)	-33 ± 1
J3-U(+) to J3-0(RET)	+29 ± 1
J3-K(+) to J3-f(RET)	+6 ± 0.5
J4-M(+) to J4-W(RET)	+33 ± 1
J4-N(-) to J4-A(RET)	-33 ± 1
J4-B(+) to J4-C(RET)	+29 ± 1
J4-E(+) to J4-G(RET)	+5 ± 0.5

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3.5.2.7 Power control voltage.- TEST MODE switch S2 shall serve, when set in turn to switch positions shown in Table III, to apply a 24 volt power control signal between indicated terminal of J2 and +24 volts return at J1-E.

TABLE III - POWER CONTROL VOLTAGE

POSITION OF TEST MODE SWITCH (S2)	J2 TERMINAL CARRYING 24 VOLT POWER CONTROL VOLTAGE
PWR/SUP SERVO	N
MICROWAVE/KLYSTRON	E
PWR SUP	
RCVR/XMTR	D
RCVR	F
RANGL COMPUTER	R
PC CARDS	P

3.5.2.8 FC console voltages.- With TEST MODE switch S2 in FC CONSOLL position, the voltages of Table IV shall appear at the terminals indicated in Table IV.

TABLE IV - FC CONSOLE VOLTAGES

TEST POINTS	J4 TERMINALS	DC VOLTS NO LOAD
+10 (TP3) to COM (TP6)	J4-X(+) to J4-S (COM)	+10 ± 0.1
-10 (TP4) to COM (TP6)	J4-U(-) to J4-S (COM)	-10 ± 0.1
+20 (TP5) to COM (TP6)	J4-T(+) to J4-S (COM)	+20 ± 0.2
	J4-F(+) to J4-G(COM)	+5 ± 0.1

3.5.3 Test Set, Subassembly MX-8046A/TPM-22.- With Test Mode switch of Power Supply 1A4 set to RCVR/XMTR position, the Test Set, Subassembly MX8046A/TPM-22 shall operate as follows:

3.5.3.1 Power Switch S1.

3.5.3.1.1 OFF.- When S1 is in normally closed position the power and off legend shall be illuminated. All other indicators shall be extinguished.

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3.5.3.1.2 ON.- When S1 is in normally open position power and on legend of S1 shall be illuminated and the following conditions exist.

a. Stalo Pulse generator and noise source legends are extinguished when their corresponding switches are in the normally closed position.

b. Off legend on Radiate switch shall be illuminated in its normally closed position.

c. Internal legend on internal external switch shall be illuminated in its normally closed position.

d. If amplifier on/off switch indicator shall be extinguished in its normally closed position.

e. See Table V for the following voltage requirements:

TABLE V

From (+)	TO (-)	Voltage
J5-1	J5-5	+28 V ± .1%
J7-2	J7-4	+28 V ± .1%
J7-7	J7-6	+33 V ± .5%
J7-1	J7-3	+15 V ± .5%
J7-3	J7-5	+15 V ± .5%
C2	FL 8	+24 V ± 4 Vdc
J4-U	J4-S	+33 V ± .5%
J4-N	J4-K	+15 V ± .5%
J4-K	J4-M	+15 V ± .5%
J4-X	J4-W	+28 V ± .1%

f. When there is no voltage on pin R of J-13 and all other voltages are applied in accordance with Table V, all legends except S-1 shall be extinguished.

FROM (+)	TO (-)	VOLTAGE
J4-V	J4-W	+28V ± Vdc
J4-G	J4-T	+24V ± Vdc
J4-G	J4-C	+24V ± 4 Vdc
J4-G	J4-H	+24F ± 4 Vdc

3.5.3.2 Radiate switch S2.

3.5.3.2.1 With conditions stated in 3.5.3.1.2 and S2 in normally open position, ON legend of S2 shall be illuminated and OFF be extinguished. The ON legend of S2 shall be illuminated for 2 minutes ± 12 sec after which RADIATE and ON legend shall be illuminated and 24 Vdc ± 4 Vdc shall be present between J4-R (+) and J4-F(Ret).

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3.5.3.2.2 The RADIATE light should extinguish when the NOISE FIGURE switch is in the normally open position and the NOISE FIGURE legend is illuminated.

3.5.3.2.3 The time delay cycle shall be reactivated when the NOISE FIGURE switch is placed in the normally closed position and the NOISE FIGURE legend extinguished.

3.5.3.3 Noise Figure switch S3.

3.5.3.3.1 With conditions stated in 3.5.3.1.2 the legend of S3 shall not be illuminated when S3 is in normally closed position. When S3 is in normally open position the entire NOISE FIGURE legend shall be illuminated.

3.5.3.4 STALO switch S4.

3.5.3.4.1 With conditions stated in 3.5.3.1.2 the STALO legend of S4 will be illuminated when S4 is in the normally open position.

3.5.3.5 PULSE GENERATOR/PULSE/SIG switch S5.

3.5.3.5.1 With conditions stated in 3.5.3.1.2 and the DRIVER switch S10 in the normally open position, the PULSE GENERATOR and SIG legend of S5 will be illuminated when S5 is in the normally open position.

3.5.3.5.2 When the DRIVER switch S10 is in the normally closed position the entire legend of S5 will be illuminated.

3.5.3.6 INTERNAL/EXTERNAL switch S6.

3.5.3.6.1 With conditions stated in 3.5.3.1.2 the INTERNAL legend of S6 will be illuminated when S6 is in the normally closed position.

3.5.3.6.2 The EXTERNAL legend of S6 will be illuminated and the INTERNAL legend will be extinguished when S6 is in the normally open position.

3.5.3.7 IF AMPLIFIER PWR Switch S11.

3.5.3.7.1 With conditions stated in 3.5.3.1.2 the S11 indicator will be extinguished in the normally open position.

3.5.3.7.2 The S11 indicator will be illuminated in the normally closed position.

3.5.3.8 IF AMPLIFIER A3.

3.5.3.8.1 With conditions stated in 3.5.3.1.2 and S11 in the normally closed position the amplifier shall meet the following requirements.

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a. The gain of the amplifier shall be not less than 60 db min and not greater than 85 db.

3.5.3.9 Attenuator AT1.

3.5.3.9.1 Attenuator AT1 shall provide a continuously variable attenuation of the signal present at output jack J3. The amount of attenuation shall be controlled by attenuation control and indicated on control dial in db and meet the following requirements:

a. Insertion loss - at 0db dial setting shall not be greater than .7db.

b. Attenuation - 0 to 50 db ± 2%.

3.5.3.10 AGC ADJ R1.

3.5.3.10.1 Voltage between J19 (center contact) and ground shall be 0 ± .1 Vdc (minimum), variable by operation of AGC ADJ control R1 with conditions stated in 3.5.1.3.

3.5.3.11 PULSE GENERATOR A5.

3.5.3.11.1 With conditions stated in 3.5.3.5.1 the pulse generator shall provide an adjustable pulse present at output jack J6 and meet the following requirements with S9 in Internal position (NC).

a. Pulse width 0.3 ± 0.1 usec.

b. Output: Output shall be 6.5V ± 1.5V at max setting of level Adj control R3. (See Fig. 1). Min output shall be .5 ± .05V at min setting of R3. (See Fig. 2).

c. Frequency: HIGH frequency cutoff should be a repetition rate of 35 ± 1 KHz; low frequency cutoff should be a max. PRF of 4 KHz. When an external pulse is applied at EXTERNAL SYNC jack high frequency cutoff should be a min. rate of 90 KHz low frequency cutoff should be a max. rate of 1 KHz.

3.5.3.12 Insertion loss between attenuator in jack (J1) and pulse out jack (J3).

3.5.3.12.1 With condition stated in 3.5.3.6.1 and Mod Mode switch S8 in the ON position and a frequency between 9.14 GHz and 9.245 GHz inserted at J1 the maximum insertion loss shall be 3.5 db max. measured at J3.

3.5.3.13 MICROWAVE POWER SOURCE.

3.5.3.13.1 With conditions stated in 3.5.3.4.1 the power source shall meet the following requirements:

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a. Power Output no less than 50 mW.

b. Channel output frequency MHz ± 1 MHz

1	9145.000
2	9153.000
3	9161.000
4	9169.000
5	9177.000
6	9185.000
7	9204.936
8	9212.936
9	9220.936
10	9228.936
11	9236.936
12	9244.936

c. From 1KHz to 110 KHz there shall be no spurious noise greater than -50 db below the carrier.

d. With a square wave frequency between 6000 HZ and 20 KHz 4V peak to peak applied to MOD in jack (J23) spurious fm noise shall be greater than -50db below the carrier.

3.5.3.14 Microwave Chassis (A2) Insertion Loss and Attenuation.

3.5.3.14.1 With the power measurement in 3.5.3.13.1 the Microwave chassis (A2), shall meet the following requirements:

a. With conditions stated in 3.5.3.4.1 the power the power output at monitor jack J9 shall be 20 db ± 1 down from the specification in 3.5.3.1.2a.

b. With conditions stated in 3.5.3.4.1 and MODE switch S8 in on position the power output at spectrum analyzer jack (J24) shall be 32 db ± 1 db down from the power measurement in 3.5.3.13.1a.

c. With conditions stated in 3.5.3.4.1 and MODE switch S8 is in ON position the power output at pulse out Jack J3 shall be no greater than 3 db down from the power measurement in 3.5.3.13.1a.

3.5.3.15 SPST Modulator (A2K2).

3.5.3.15.1 With conditions stated in 3.5.3.4.1 and MODE switch S8 is off position the power output at pulse out Jack J3 shall be no greater than -40 db down from the measurement in 3.5.3.13.1a.

3.5.3.15.2 With conditions stated in 3.5.3.5.2 and 3.5.3.4.1 and PRF of 25 KHz, 20 KHz, and 10 KHz (R2 Adjust), the following frequency spectrum will be obtained for all 12 channels. (See figure 3a).

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3.5.3.16 NOISE SOURCE output (A2A1).- With conditions stated in 3.5.3.3.1 the noise figure at output jack J3 shall be 25.0 ± .5 db after minature microwave attenuator (A2AT1) on microwave chassis has been adjusted and attenuator set to 0db on front panel.

3.5.3.17 Mod Trigger J11, J15, J5-A1.- With a load of 95 ohms ± 5% at J15 and a square wave input signal 5 volt p-p max. with a PRF of 20 KHz min. there will be no phase distortion between the input pulse at J11 and the output at J15 and J5-A1.

3.5.3.18 60 MHz CW J12 and J16.- With a load of 50 ohms ± 5% at J16 and input signal of 60 MHz CW there will be no amplitude or phase distortion between J12 (input) and J16 (output).

3.5.3.19 TO Trigger J17, J14, J18.- With a load of 50 ohms at J17 and J18 apply 10 volts peak to peak at J14 with PRF at 20 KHz min. and a pulse width of .5 usec there will be no phase or amplitude distortion between the input pulse at J14 and the output pulse at J17 and J18.

3.5.3.20 +24 Vdc Control Out. With the voltages in Table V applied there will be +24 Vdc ± 4 Vdc at J13-W. S1 is normally open (paragraph 3.5.3.1.2).

3.5.4 Test Set, Subassembly MX-8047/TPM-22.- With TEST MODE switch S2 of power supply 1A4 set to MICROWAVE/KLYSTRON PWR SUP position, Test Set, Subassembly MX-8047/TPM-22 shall operate as follows:

3.5.4.1 MICROWAVE CHASSIS TEST section 2A1.

3.5.4.1.1 Interlock indicator lamp (DS7) and interlock circuit.- The interlock indicator lamp shall be illuminated when switch S9 is energized in the manual service position. Interlock circuit shall be closed and continuity shall exist between J8-S and TB2-4 under the following conditions:

a. 2A1/2A2 switch in 2A1 position

b. J2/J5 door interlock switch S7 in normally open position.

c. Interlock switch S9 energized in manual (service) or operate position

d. Interlock plate 10548925 properly installed on subassembly A1.

3.5.4.1.2 Blower motor (B1).- Blower motor B1 shall operate under conditions stated in paragraph 3.5.4.1.1.

3.5.4.1.3 POWER switch (S4).

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3.5.4.1.3.1 OFF legend.- OFF legend of S4 shall be illuminated when S4 is in normally closed position and conditions stated in 3.5.4.1.1 are met.

3.5.4.1.3.2 STANDBY legend.- STANDBY legend of power indicator lamp S4 shall be illuminated when S4 is in normally open position and conditions stated in 3.5.4.1.1 are met.

3.5.4.1.4 Output voltages (J1).- Under conditions listed in 3.5.4.1.3.2, output voltages at J1 shall be as listed in a through e below:

a. 28 Vdc between J1-1(+) and J1-2(Ret).

b. 28 Vdc between J1-3(+) and J1-2(Ret).

c. 28 Vdc between J1-5(+) and J1-6(Ret).

d. 33 Vdc between J1-7(-) and J1-8(Ret).

e. 0 to 6 V (variable by operation of R1) between J1-A2(+) and J1-4(Ret).

3.5.4.1.5 RADIATE/STANDBY switch (S5).

3.5.4.1.5.1 STANDBY legend.- STANDBY legend of S5 is illuminated under the following conditions:

a. Controls and conditions of 3.5.4.1.3.2 are met.

b. Switch S5 is in normally closed position.

3.5.4.1.5.2 RADIATE legend and radiate mode.- Radiate legend of S5 shall be illuminated and an output pulse as described e and f below shall be available between J5-E(-) and chassis ground (E4) under the following conditions:

a. Controls and cond. are met in accordance with 3.5.4.1.4 for more than 2 minutes ± 12 seconds. (Delay period verifies proper operation of time delay relay K1.)

b. S5 in normally open position.

c. P2 plugged into J5.

d. Shorted wire between K5 normally open and normally closed contacts.

e. Not greater than -14 kVdc at most negative point.

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f. 4.3 +0.5 kV amplitude from leading edge
-0.2 kV
to most positive point excluding positive overshoot.

3.5.4.1.6 KLYSTRON FILAMENT CURRENT meter (M1).- Meter M1 shall indicate 1.5A ± 5% for 20 ± 2 seconds after start of operation as stated in 3.5.4.1.5.2 and 2.0A ± 5% after 20 ± 2 seconds of operation.

3.5.4.1.7 2A1 OVERLOAD indicator (S6).

3.5.4.1.7.1 High voltage power supply overload.- HIGH VOLTAGE POWER SUPPLY legend of S6 shall be illuminated when 28 Vdc is present between P1-1 and chassis ground.

3.5.4.1.7.2 Modulator overload.- MODULATOR legend of S6 shall be illuminated when 28 Vdc is present between P1-3 and chassis ground.

3.5.4.1.8 2A1 OVERLOAD reset switch (S6).- Between P1-2(+) and chassis ground (Ret) 28 Vdc shall be present when conditions are as stated in 3.5.4.1.3.2 and S6 is actuated (depressed).

3.5.4.2 KLYSTRON POWER SUPPLY TEST section 2A2.

3.5.4.2.1 Interlock indicator lamp (DS7) and interlock circuit.- The interlock indicator lamp DS7 shall be illuminated, switch S9 is energized in the manual (service) position. Interlock circuit shall be closed and continuity shall exist between J8-S and TB2-4 under the following conditions.

a. 2A1/2A2 switch in 2A2 position.

b. J2/J5 door interlock switch S7 in normally open position.

c. Interlock switch S9 energized in manual (service) or operate position.

d. Interlock plate 10548925 properly installed on subassembly A1.

e. Shorting wire connected between S11 normally open and normally closed contacts.

f. Shorting wire connected between J3-9 and J3-10.

3.5.4.2.2 POWER switch (S1).

3.5.4.2.2.1 OFF legend.- OFF legend of POWER switch S1 shall be illuminated when S1 is in the normally closed position and conditions are as stated in 3.5.4.2.1.

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3.5.4.2.2.2 STANDBY legend.- STANDBY legend of POWER switch S1 shall be illuminated and 28 Vdc ± 3% shall be between J3-A3(+) and J3-A2(Ret) when S1 is in normally open position and conditions are as stated in 3.5.4.2.1.

3.5.4.2.3 OPERATE/STANDBY switch (S2).

3.5.4.2.3.1 STANDBY legend and standby mode.- KLYSTRON POWER SUPPLY TEST section of the high voltage tester is in the standby mode and STANDBY legend of S2 is illuminated when POWER switch STANDBY legend is illuminated as stated in 3.5.4.2.2.2 and S2 is in normally closed position.

3.5.4.2.3.2 OPERATE legend and operate mode.- KLYSTRON POWER SUPPLY TEST section of the high voltage tester is in the operate mode, OPERATE legend of S2 shall be illuminated, 28 Vdc shall be available between J3-A4(+) and J3-A5(Ret). 24 Vdc shall be available between J3-(+) and J3-5(Ret). -9.0 + 0.5/-0 KVdc shall be available between J2 and chassis ground under the following conditions.

a. controls and conditions of 3.5.4.2.2.2 are met.

b. S2 in normally open position.

c. 185 volt peak-to-peak symetrical square wave applied between J3-13 and J3-14.

d. switch S10 in Test B position.

3.5.4.2.4 KLYSTRON POWER SUPPLY CURRENT meter (M2).

3.5.4.2.4.1 Standby current.- M2 shall indicate 2 amperes when conditions are as stated in 3.5.4.2.2.2 and 14-ohm load is between J3-A3 and J3-A2.

3.5.4.2.4.2 Operate current.- M2 shall indicate 4 amperes when conditions are as stated in 3.5.4.2.3.2 and 14-ohm load is between:

a. J3-A3 and J3-A2

b. J3-A4 and J3-A5

3.5.4.2.5 2A2 OVERLOAD indicator (S3).

3.5.4.2.5.1 High voltage power supply overload.- When 28 Vdc is present between J3-1 and J3-16, HIGH VOLTAGE POWER SUPPLY legend of S3 shall be illuminated.

3.5.4.2.5.2 Modulator overload.- When 28 Vdc is present between J3-3 and J3-16, MODULATOR legend of S3 shall be illuminated.

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3.5.4.2.6 2A2 OVERLOAD reset switch (S3).- Between J3-2(+) and J3-16 (Ret) 28 Vdc shall be present when S3 is actuated and any of the following conditions are set.

a. 3.5.4.2.2.2

b. 3.5.4.2.3.1

c. 3.5.4.2.3.2

3.5.4.2.7 TEST A/B switch (S10).- Continuity exists between P2-C and chassis ground when conditions are as stated in 3.5.4.2.3.2 and S10 is in TEST A position.

3.5.5 Test Set, Subassembly MX-8048/TPM-22.- With TEST MODE switch S2 of power supply 1A4 set to RCVR position, Test Set, Subassembly MX-8048/TPM-22 shall operate as follows.

3.5.5.1 Power switch S1.- With IF DELAY ADJ switch (COARSE) in MIN position, and IF DELAY ADJ switch (FINE) turned fully counter-clockwise, POWER switch S1 shall serve to change indicator DS1 from OFF to ON and the reverse, and to cause the outputs specified in Tables VI and VII to appear at their respective terminals in accordance with POWER indicator DS1.

TABLE VI - OUTPUT CHARACTERISTICS

OUTPUT	CHARACTERISTIC	TERMINAL	TERMINATION
PRE RANGE TRIGGER	Pulse train 19,979 ±1 Hz, approx 0.46 us pulse width, not less than 2.4 V pulse amplitude, delayed approx 36.93 us from MODULATOR TRIGGER	J2-B	150 ohms
STC/DC UNIT 3	1) With AGC SELECT switch S6 in the DC position, not less than 2.4 Vdc.	J2-C	150 ohms
	2) With AGC SELECT switch S6 in the STC position, square wave approximately 20 Hz not less than 2.4 Vdc amplitude		

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TABLE VI - OUTPUT CHARACTERISTICS

OUTPUT	CHARACTERISTIC	TERMINAL	TERMINATION
E/L TRIGGER	Pulse train 19,979 ±1 Hz, approx 0.35 us pulse width, not less than 2.4 V pulse amplitude, delayed approx 38.64 us from MODULATOR TRIGGER	J2-D	150 ohms
MODULATOR TRIGGER UNIT 3	Pulse train 19,979 ±1 Hz, 0.25 us pulse width, not less than 12 V pulse amplitude, coincident with MODULATOR TRIGGER	J2-E	150 ohms
+15V 5% REG	+15 Vdc regulated to 5%	J3-g	N/A
-15V 5% REG	-15 Vdc regulated to 5%	J3-h	N/A
STALO OSC.	59.936 ± 0.001 MHz, not less than 2.24 volts rms amplitude, into 50 ohm load	J4-A	50 ohms
VIDEO PULSE 3A2	Pulse train 19,979 ±1 Hz, 0.2 us pulse width, delayed not greater than 2.0 us from MODULATOR TRIGGER. This signal shall vary in amplitude with changes in the position of VIDEO ADJUST control R1, from 0 volts when control is turned fully counterclockwise to not less than 2.4 V when control is turned fully clockwise.	J8	150 ohms
STC/DC 3A2	Same as STC/DC Unit 3 (see above).	J9	150 ohms

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TABLE VI - OUTPUT CHARACTERISTICS

OUTPUT	CHARACTERISTIC	TERMINAL	TERMINATION
3A2/3A5 TRIGGER	Same as PRE RANGE TRIGGER (see above) except amplitude shall vary with changes in position of 3A2/3A5 TRIGGER ADJUST control R2, from 0 V when control is turned fully counterclockwise, to not less than 2.4 V when control is turned fully clockwise.	J10	150 ohms
LO OUT (1)	59.936 ± 0.001 MHz, approx. 250 mV rms amplitude, into 50 ohm load.	J13	50 ohms
LO OUT (2)	Same as LO OUT (1) (see above)	J14	50 ohms
LO 3A5	59.936 ± 0.001 MHz, approx. 1.1 V rms amplitude, into 50 ohm load, with J23 open.	J20	50 ohms
3A9 TRIGGER	Same as MODULATOR TRIGGER UNIT 3 (see above).	J21	150 ohms
MODULATOR TRIGGER	Pulse train 19,979 ± 1 Hz, not less than 0.5 us pulse width, not less than +15 V pulse amplitude.	J22	150 ohms
LOCAL OSCILLATOR	59.936 ±0.001 MHz approx. 1.1 V rms amplitude into 50 ohm load, with J20 open.	J23	50 ohms
+25 V REG	25 Vdc ±0.5%	J3-M(+), J3-N(RET)	N/A
-15V REG	-15 Vdc ±0.5%	J3-a(-), J3-m(RET)	N/A
+15V REG	+15 Vdc ±0.5%	J3-b(+), J3-m(RET)	N/A

3.5.5.2 CLUTTER LOCK ON switch S2.- With POWER switch S1 turned ON, CLUTTER LOCK ON switch S2 shall serve to illuminate and extinguish CLUTTER LOCK ON indicator DS2 and the reverse, and to apply 24 Vdc +6, -2 signal to terminal J3-W in accordance with indicator DS2. It shall also perform functions discussed in paragraph 3.5.5.6.

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3.5.5.3 MID RANGE CALIBRATION switch S3.- With IF DELAY ADJ switch (COARSE) in MIN position, IF DELAY ADJ control (FINE) turned fully counterclockwise, and POWER switch S1 turned ON, MID RANGE CALIBRATION switch S3 shall serve to illuminate and extinguish MID RANGE CALIBRATION indicator DS3 and the reverse, to cause the outputs of Table VII to appear at their respective terminals in accordance with indicator DS3, and to perform functions discussed in paragraph 3.5.5.6.

TABLE VII - OUTPUT CHARACTERISTICS PECULIAR TO MID RANGE CALIBRATION

OUTPUT	CHARACTERISTICS	TERMINALS	TERMINATION
MID RANGE CALIBRATION CONTROL	With MID RANGE CALIBRATION indicator DS3 illuminated, +24 Vdc + 6, -2	J3-A(+) J3-T(RET)	N/A
RANGE CALIBRATION RF INPUT	(1) With MID RANGE CALIBRATION indicator DS3 illuminated, 59.936 ±0.001 MHz burst, 0.4 usec wide, approx 0.13 volts rms amplitude, into 50 ohm load, delayed not greater than 2.0 usec from MODULATOR TRIGGER.	J4-G	50 ohms
	(2) With MID RANGE CALIBRATION indicator DS3 extinguished, ground.		

3.5.5.4 RECEIVER NOISE switch S4.- With POWER switch S1 turned ON, RECEIVER NOISE switch S4 shall serve to illuminate and extinguish RECEIVER NOISE indicator DS4 and the reverse. It shall also perform functions discussed in paragraph 3.5.5.6.

3.5.5.5 3A6 TEST switch S5.- With POWER switch S1 turned ON, 3A6 TEST switch S5 shall serve to illuminate and extinguish 3A6 TEST indicator DS5 and the reverse. It shall also perform functions discussed in paragraph 3.5.5.6.

3.5.5.6 Combinations of switches S2, S3, S4 and S5.- With IF DELAY ADJ switch (COARSE) in MIN position, IF DELAY ADJ control (FINE) turned fully counterclockwise, IF AMPLITUDE ADJ control AT4 turned to the zero (00) and POWER switch S1 turned ON, switches S2, S3, S4, and S5 shall serve to cause the outputs of Table VIII to appear at their respective terminals in accordance with the combinations of indicators DS2, DS3, DS4, and DS5 shown in Table VIII.

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TABLE VIII - COMBINATIONS OF INDICATORS DS2, DS3, DS4, and DS5

NOTE: DS2 legend is CLUTTER LOCK ON.
DS3 legend is MID RANGE CALIBRATION.
DS4 legend is RECEIVER NOISE.
DS5 legend is 3A6 TEST.

In table below, "on" means legend is illuminated, or bright, and "off" means legend is extinguished or dark.

INDICATOR COMBINATIONS	COMBINATION LETTER CODE
DS2 on, DS3 on, DS4 on, DS5 on	A
DS2 on, DS3 on, DS4 on, DS5 off	B
DS2 on, DS3 on, DS4 off, DS5 on	C
DS2 on, DS3 off, DS4 on, DS5 on	D
DS2 off, DS3 on, DS4 on, DS5 on	E
DS2 on, DS3 on, DS4 off, DS5 off	F
DS2 on, DS3 off, DS4 on, DS5 off	G
DS2 on, DS3 off, DS4 off, DS5 on	H
DS2 off, DS3 on, DS4 on, DS5 off	I
DS2 off, DS3 on, DS4 off, DS5 on	J
DS2 off, DS3 off, DS4 on, DS5 on	K
DS2 on, DS3 off, DS4 off, DS5 on	L
DS2 off, DS3 on, DS4 off, DS5 off	M
DS2 off, DS3 off, DS4 on, DS5 off	N
DS2 off, DS3 off, DS4 off, DS5 on	O
DS2 off, DS3 off, DS4 off, DS5 off	P

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TABLE IX - OUTPUTS FOR COMBINATIONS OF INDICATORS DS2, DS3, DS4, and DS5

OUTPUT	CHARACTERISTICS	INDICATOR COMBINATION (TABLE VII)	TERMINAL	TERMINATION
IF INPUT UNIT 3	59.936 ±0.001 MHz burst 0.4 usec wide, approx. 0.05 volts rms amplitude into 50 ohm load, delayed not greater than 2.0 usec from MODULATOR TRIGGER.	G,L,N	J1-D	50 ohms
	59.941 ±0.001 MHz burst 0.4 usec wide, approx. 0.05 volts rms amplitude into 50 ohm load, delayed not greater than 2.0 usec from MODULATOR TRIGGER	P		
	Ground (not more than 2 ohms)	A,C,D,E,H,J,K,O		
	Ground (not less than 40 ohms)	B,F,I,M		

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TABLE IX - OUTPUTS FOR COMBINATIONS OF INDICATORS DS2, DS3, DS4, and DS5

OUTPUT	CHARACTERISTICS	INDICATOR COMBINATION (TABLE VII)	TERMINAL	TERMINATION
IF OUT	59.936 ±0.001 MHz burst 0.4 usec wide, approx. 0.04 volts rms amplitude, into 50 ohm load delayed not greater than 2.0 usec from MODULATOR TRIGGER.	D,H,K	J11	50 ohms
	59.941 ±0.001 MHz burst 0.4 usec wide, approx. 0.035 volts rms amplitude into 50 ohms load, delayed not greater than 2.0 usec from MODULATOR TRIGGER.	0		
	Ground (not more than 2 ohms)	B,F,G,I,L,M,N,P,		
	Ground (not less than 40 ohms)	A,C,E,J		
IF OUT 3A6	Same as IF OUT	Same as IF OUT	J12	50 ohms

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3.5.5.7 IF DELAY ADJ controls.- The IF DELAY ADJ controls (COARSE and FINE) shall serve to change the time delay of the following signals relative to the MODULATOR TRIGGER signal in accordance with Table X.

Signal	Table in which Described
RANGE CAL RF INPUT	VII
IF INPUT UNIT 3	IX
IF OUT	IX
IF OUT 3A6	IX

TABLE X - IF DELAY ADJUSTMENT

POSITION OF COARSE CONTROL	POSITION OF FINE CONTROL	DELAY RELATIVE TO MODULATOR TRIGGER
MIN	Fully CCW	Delay not greater than 2.0 us
	Intermediate	Delay more than 0.5 us and less than 9.0 us
	Fully CW	Delay not less than 9.0 us
PHASE	Fully CCW	Delay not greater than 35.0 us
	Intermediate	Delay more than 35.0 us and less than 44.0 us
	Fully CW	Delay not less than 44.0 us
MAX	Fully CCW	Delay not greater than 45.0 us
	Intermediate	Delay more than 45.0 us and less than 54.0 us
	Fully CW	Delay not less than 54.0 us

The IF DELAY ADJ controls (COARSE and FINE) shall serve to change the time delay of the VIDEO PULSE 3A2 signal of TABLE V relative to the MODULATOR TRIGGER signal in accordance with Table X, except that every delay of Table X (other than the first two figures given) shall be reduced by approximately 1.4 us.

3.5.5.8 IF AMPLITUDE ADJ control.- The IF AMPLITUDE ADJ control shall serve to change the amplitude of the signals of Table IX in accordance with Table XI.

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TABLE XI - IF AMPLITUDE ADJUSTMENT

SIGNAL	SETTING OF IF AMPLITUDE ADJ	AMPLITUDE OF SIGNAL
IF INPUT UNIT 3	00	As stated in Table IX.
	01 to 99	Same number of decibels down from the 00 setting as control is set for. For the G, L, N Indicator Combination, the 99 db setting shall yield an amplitude of approx. 1.3 uV rms. For the P Indicator Combination, the 99 db setting shall yield an amplitude of approx. 1.1 uV rms.
IF OUT	00	As stated in Table VIII.
	01 to 99	Same number of decibels down from the 00 setting as control is set for. For the D, H, K Indicator Combination, the 99 db setting shall yield an amplitude of approx. 0.9 uV rms. For the 0 Indicator Combination, the 99 db setting shall yield an amplitude of approx. 0.8 uV rms.
IF OUT 3A6	00 to 99	Same as IF OUT above.

3.5.6 Test Set, Subassembly MX-8049/TPM-22.- With TEST MODE switch S2 of power supply 1A4 in RANGE COMPUTER position, Test Set, Subassembly MX-8049/TPM-22 shall operate as follows.

3.5.6.1 POWER switch S1.

3.5.6.1.1 OFF.- OFF legend of DS4 shall be illuminated when S1 is in normally closed position.

3.5.6.1.2 ON.- ON legend of DS4 shall be illuminated and 24 Vdc ±5% shall be available between J5-D(+), J5-B(+), J5-T(+), and J8-P (Ret) when S1 is in normally open position.

3.5.6.2 24V circuit breaker CB1.- The 24V circuit breaker CB1 shall be closed when reset pushbutton is depressed so white band is not exposed. The circuit breaker shall be open when reset push-button is pulled out to expose white band.

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3.5.6.3 POWER indicator DS3 legends.

3.5.6.3.1 +33V legend.- The +33V legend of DS3 shall be illuminated when conditions of 3.5.6.1.2 are met.

3.5.6.3.2 -33V legend.- The -33V legend of DS3 shall be illuminated when conditions of 3.5.6.1.2 are met.

3.5.6.3.3 +23V legend.- The +23V legend of DS3 shall be illuminated when conditions of 3.5.6.1.2 are met.

3.5.6.3.4 -23V legend.- The -23V legend of DS3 shall be illuminated when conditions of 3.5.6.1.2 are met.

3.5.6.4 RADIATE switch S2.

3.5.6.4.1 DS5 OFF legend.- The OFF legend of DS5 shall be illuminated under conditions stated in 3.5.6.1.2 and S2 in normally closed position.

3.5.6.4.2 DS5 ON legend.- The ON legend of DS5 shall be illuminated and 24 Vdc ±5% shall be available between J5-J(+) and J5-K(Ret) under conditions stated in 3.5.6.1.2 and S2 in normally open position.

3.5.6.5 TARGET/NO TARGET switch S3.

3.5.6.5.1 DS6 TARGET legend.- The TARGET legend of DS6 shall be illuminated under conditions as stated in 3.5.6.1.2 and S3 in normally closed position.

3.5.6.5.2 DS6 NO TARGET legend.- The NO TARGET legend of DS6 shall be illuminated under conditions as stated in 3.5.6.1.2 and S3 in normally open position.

3.5.6.6 SEARCH/TRACK indicator DS2.

3.5.6.6.1 DS2 TRACK legend.- The TRACK legend of DS2 shall be illuminated when 24 Vdc ±5% is applied to J5-W(+) and J8-P(Ret).

3.5.6.6.2 DS2 SEARCH legend.- The SEARCH legend of DS2 shall be illuminated when 24Vdc ±5% is applied to J5-C(+) and J8-P(Ret).

3.5.6.7 DOPPLER indicator DS1.

3.5.6.7.1 DS1 NO ECM legend.- The NO ECM legend of DS1 shall be illuminated when 24 Vdc ± is applied to J5-A(+) and J8-P(Ret).

3.5.6.7.2 DS1 ECM legend.- The ECM legend of DS1 shall be illuminated when 24 Vdc ±5% is applied to J5-V(+) and J8-P(Ret).

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3.5.6.8 TARGET REJECT switch S4.

3.5.6.8.1 The TARGET REJECT legend of DS7 shall be illuminated white when conditions are as stated in 3.5.6.1.2 and S4 is in normally closed position.

3.5.6.8.2 The TARGET REJECT legend of DS7 shall be illuminated red and 24 Vdc ±5% shall be available between J5-G(+) and J5-H(Ret), when conditions are as stated in 3.5.6.1.2 and S4 is actuated.

3.5.6.9 60 MHz oscillator.- Under conditions stated in 3.5.6.1.2 the signal at J4-A (center contact to shield) shall be 59.936 MHz ± 1000 Hz at 2.0 Vac ± 1.0 Vac peak-to-peak into a 100 ohm load. The reading at J4-A shall be zero with S1 in the OFF position and DS 4 OFF legend illuminated.

3.5.6.10 Output signals.

3.5.6.10.1 Simulated target.- The simulated target pulse shall be present at J6 and shall have the characteristics described in Table XII, under the following conditions:

a. Controls and conditions met in accordance with 3.5.6.3.2.

b. Input signal 1 applied to J4-E as specified in Table XII.

c. OSCILLOSCOPE CHANNEL A switch S5 set to position 9.

3.5.6.10.1.1 COARSE RANGE switch S9.- The position of the COARSE RANGE switch and FINE RANGE control shall determine the amount of delay between input signal 1 and simulated target in three overlapping ranges as follows:

a. MIN = 1 ± 0.5 to 11 ± 1 usec

b. MID = 7 ± 1 to 18 ± 1 usec

c. MAX = 13 ± 1 to 24 ± 1 usec

3.5.6.10.1.2 FINE RANGE control R1.- The FINE RANGE control shall provide for a continuous adjustment of target simulator pulse delay within each range of the COARSE RANGE switch. Minimum delay shall occur with FINE RANGE control fully counterclockwise.

3.5.6.10.2 Sum and difference doppler.- The sum and difference doppler signals shall be present at J4-F and J4-G, respectively, and have the characteristics described in Table XII under the following conditions:

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a. Controls and conditions met in accordance with 3.5.6.3.2 and 3.5.6.5.1.

b. Input signal 1 applied to J4-E as specified in Table XII.

c. Input signal 2 applied to J4-D as specified in Table XII.

d. SUM level control R2 and DIFFERENCE level control R3 fully clockwise.

e. 150 ohms resistive load at J4-F and at J4-G.

3.5.6.10.2.1 SUM level control R2.- Under conditions stated in 3.5.6.10.2, the amplitude of the sum doppler signal shall be continuously variable from a minimum of .05V (p-p) to at least 1.5V (p-p) by operation of SUM level output R2. Maximum amplitude shall occur when R2 is fully clockwise.

3.5.6.10.2.2 DIFFERENCE level control R3.- Under conditions stated in 3.5.6.10.2, the amplitude of the difference doppler signal shall be continuously variable from a minimum of 0.5V (p-p) to at least 1.5 (p-p) by operation of DIFFERENCE level control R3. Maximum amplitude shall occur when R3 is fully clockwise.

TABLE XII - INPUT AND OUTPUT CHARACTERISTICS

REQUIREMENT	DESCRIPTION	TERMINALS
Input signals		
Signal 1	Positive pulse from zero volts to not less than 6 volts, 0.3 ± 0.1 usec wide, 40 nsec (max) rise time, 20 KHz ± 1% repetition rate.	J4-E
Signal 2	Positive pulse from zero volts to not less than 3 volts, 0.3 ± 0.1 usec wide, triggered by Signal 1.	J4-D
Output signals		
Simulated target	Positive pulse from zero volts to not less than 3 volts, 0.3 ± 0.1 usec wide, delayed 1 to 24 usec from Signal 1.	J6 or J7

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TABLE XII - INPUT AND OUTPUT CHARACTERISTICS (Cont'd)

REQUIREMENT	DESCRIPTION	TERMINALS
Output signals		
Sum doppler	2.5 KHz ± 1% square wave, from 0.5V(p-p) volts to equal to or greater than 1.5 V p-p). Amplitude controlled by SUM level control R2.	J4-F
Difference doppler	2.5 KHz ± 1% square wave, from .05V(p-p) volts to equal to or greater than 1.5V(p-p), in phase with sum doppler when simulated target appears before signal 2. Amplitude controlled by DIFFERENCE level control R3.	J4-G

3.5.7 Test Set, Subassembly MX-8050/TPM-22.- With TEST MODE switch S2 of power supply 1A1A4 set to PC CARDS position, Test Set, Subassembly MX-8050/TPM-22 shall operate as follows.

3.5.7.1 POWER/ON/OFF switch S1.

3.5.7.1.1 POWER/OFF legend.- The POWER/OFF legend of S1 shall be illuminated when S1 is in the normally closed position.

3.5.7.1.2 POWER/ON legend.- The POWER/ON legend of S1 shall be illuminated when S1 is in the normally open position.

3.5.7.2 Circuit Breakers.- Circuit breakers CB1 through CB8 shall be closed when their respective reset buttons are depressed so white bands are not exposed. Each circuit breaker shall be open when its reset button is pulled out manually to expose white band, and shall open automatically within 20 seconds when the indicated current overload is passed through it:

Circuit Breaker	Overload Current
CB1 -25V	2 Adc
CB2 +25V	2 Adc
CB3 -15V	2 Adc
CB4 +15V	2 Adc
CB5 +6.6V	6 Adc
CB6 +24V EXT	2 Adc
CB7 +24V INT	6 Adc
CB8 LINE	2 Adc

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3.5.7.3 LEVEL/AC/DC Switch S2.

3.5.7.3.1 LEVEL/DC Legend.- The LEVEL/DC legend of S2 shall be illuminated when all of the following conditions are met:

a. Conditions of 3.5.7.1.2 are met.

b. S2 in normally closed position.

3.5.7.3.2 LEVEL/AC legend.- The LEVEL/AC legend of S2 shall be illuminated when all of the following conditions are met.

a. Concitions of 3.5.7.3.1a are met.

b. S2 in normally open position.

3.5.7.4 OVERLOAD/OFF/ON Switch S7.

3.5.7.4.1 OVERLOAD/OFF Legend.- The OVERLOAD/OFF legend of S7 shall be illuminated when all of the following conditions are met:

a. Conditions of 3.5.7.3.1a are met.

b. S7 in normally closed position.

3.5.7.4.2 OVERLOAD/ON legend.- The OVERLOAD/ON legend of S7 shall be illuminated when all of the following conditions are met:

a. Conditions of 3.5.7.3.1a are met.

b. S7 in normally open position.

3.5.7.5 PROFILE/OFF/ON switch S3.

3.5.7.5.1 PROFILE/OFF legend.- The PROFILE/OFF legend of S3 shall be illuminated when all of the following conditions are met:

a. Conditions of 3.5.7.3.1a are met.

b. S3 in normally closed position.

3.5.7.5.2 PROFILE/ON legend.- The PROFILE/ON legend os S3 shall be illuminated when all of the following conditions are met:

a. Conditions of 3.5.7.3.1a are met.

b. S3 in normally open position.

3.5.7.6 NORMAL/HI/LOW switch S4.

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3.5.7.6.1 Normal legend.- The NORMAL legend of S4 shall be illuminated when all of the conditions stated in 3.5.7.5.1 are met.

3.5.7.6.2 HI Legend.- The HI legend of S4 shall be illuminated when all of the following conditions are met:

a. PROFILE/ON legend of S3 illuminated in accordance with 3.5.7.5.2.

b. S4 in normally closed position.

3.5.7.6.3 LOW Legend.- The LOW legend of S4 shall be illuminated when all of the following conditions are met:

a. PROFILE/ON legend of S3 illuminated in accordance with 3.5.7.5.2.

b. S4 in normally open position.

3.5.7.7 TEST/1/2 Switch S5.

3.5.7.7.1 TEST/1 Legend.- The TEST/1 legend of S5 shall be illuminated when all of the following conditions are met:

a. Conditions of 3.5.7.3.1a are met.

b. S5 in normally closed position.

3.5.7.7.2 TEST/2 Legend.- The TEST/2 legend of S5 shall be illuminated when all of the following conditions are met:

a. Conditions of 3.5.7.3.1a are met.

b. S5 in normally open position.

3.5.7.8 RESET/OFF/ON Switch S6.

3.5.7.8.1 RESET/OFF Legend.- The RESET/OFF legend of S6 shall be illuminated when all of the following conditions are met:

a. Conditions of 3.5.7.3.1a are met.

b. A blank test card or any test card among those listed in Table XIII properly inserted in card reader assembly and handle of card reader assembly thrown to the right-hand position.

c. S6 in normally closed position.

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TABLE XIII - TEST CARDS USED WITH PC CARD TESTER

Title/Description	Part Number
Test cards for Test Set, Subassembly MX-8050/TPM-22 (35/set)	11726471
Test cards for Radar Set AN/VPS-2 (37/set)	11726472
Test cards for Test Set, Radar AN/TPM-22 (19/set)	11726473
Test cards for Test Set AN/TSM-100 (173/set)	11727705

3.5.7.8.2 RESET/ON Legend.- The RESET/ON legend of S6 shall be illuminated under the following conditions:

a. Conditions of 3.5.7.3.1a and 3.5.7.8.1b are met.

b. S6 in normally open position.

3.5.7.9 Power indicators.- The power indicator lamps, +6.6V, -15V, +15V, +24V EXT, +24V INT, -25V, +25V and LINE shall be illuminated when conditions of 3.5.7.3.1a and 3.5.7.8.1b are met.

3.5.7.10 Digiswitch Lamps.- Two TEST POINT A and two TEST POINT B thumbwheel selector lamps shall be illuminated under conditions stated in 3.5.7.8.1a through 3.5.7.8.1c. Reading from left to right, the first switch in each group shall be illuminated and will indicate which other switch in the group is active (illuminated) as follows:

No. illuminated in first switch	Other switch illuminated
0	second
1	third
2	fourth
3	fifth
4	sixth
5	seventh

3.5.7.11 Test Points A and B, and TEST IDENT Lamp Outputs.- The outputs listed in Table XIV shall appear at test points A and B, and the TEST IDENT lamps shall be illuminated as given in Table XIV, under the following conditions:

a. Conditions of 3.5.7.3.1a, and 3.5.7.9 are met (with exceptions noted in Table XIV).

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b. Front panel switches are set as follows (with exceptions noted in Table XIV).

(1) LEVEL/DC
(2) OVERLOAD/OFF
(3) PROFILE/OFF
(4) NORMAL
(5) TEST/1
(6) RESET/OFF

c. Test cards listed in Table XIV are properly inserted in the card reader and the handle is thrown to the right-hand position.

d. Adapter CP6 inserted in test sockets A and B for test cards ST1 through ST33, and adapter CP4 inserted in test sockets B and C for test cards ST34 and 35.

The voltage levels of pulses shall be +3.5 -1.5V for a logic 1 and 0 +2.0V -0.8V for logic 0. Pulse outputs shown in Figures 4, 5, 6 and 7 are +0.4V signals at TEST POINT A in relation to signal at TEST POINT B.

Where more than one pin number is given in Table XIV under Test Point Selector A or B, each pin so numbered shall give the same result. For example, with the first test given under Test Card ST34, any combination of 24, 25, or 26 on Test Point Selector A, and 05, 06 or 07 on Test Point Selector B shall yield the same output.

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TABLE XIV - SELF TEST CARD OUTPUT

TEST CARD	TEST POINTS SELECTORS A	B	EXCEPTIONS TO SWITCHES SELECTED	APPLI-CABLE NOTES	TEST IDENT LAMPS	OUTPUT
ST 1	01-56	-		1, 8	ST, 1, A	4.5 ± 0.7 Vdc
	01	-	TEST/2	1, 8	ST, 1, A	0 ± 0.4 Vdc
ST 2	01-56	-	TEST/2 PROFILE/OFF	1, 8	ST, 2, C	4.5 ± 0.7 Vdc
ST 3	01-08	01	-	8, 10 Fig. 4	ST, 2, 1 128	T Pulse
ST 4	01-08	01		8, 10 Fig. 4	ST, 4, 64	$\overline{T}$ Pulse
ST 5	01-08	01		8, 10 Fig. 4	ST, 4, 1	T Pulse
ST 6	01-08	01		8, 10 Fig. 4	ST, 4, 2	$\overline{T}$ Pulse
ST 7	2,4,6,8	2		8, 10 Fig. 4	ST, 4, 2 1	T Pulse
	12	12		8, 10 Fig. 6	ST, 4, 2 1	$\overline{\text{STROBE}}$ PULSE
	19	19		4,8,10 Fig. 4	ST, 4, 2 1	$\overline{T}$ Pulse
	41	41		8, 10 Fig. 6	ST, 4 2, 1	STROBE PULSE
	43	-		8, 10 Fig. 4	ST, 4 2, 1	7.5 MHz CW. 2.0-5.5 vp-p. (use internal sync.)

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TABLE XIV - SELF TEST CARD OUTPUT (Cont'd)

TEST CARD	TEST POINTS SELECTORS A	B	EXCEPTIONS TO SWITCHES SELECTED	APPLICABLE NOTES	TEST IDENT LAMPS	OUTPUT
ST 8	2,4,6 8	2		8, 10 Fig. 4	ST, 8	$\overline{T}$ Pulse
	19	19		8, 10 Fig. 4	ST, 8	T Pulse
	37	47		8, 10 Fig. 7	ST, 8	F1 Pulse
	39	47		8, 10 Fig. 7	ST, 8	$\overline{F1}$ Pulse
	41	47		8, 10 Fig. 7	ST, 8	F2 Pulse
	43	47		8, 10 Fig. 7	ST, 8	$\overline{F2}$ Pulse
	47	47		8, 10 Fig. 7	ST, 8	$\overline{F4}$ Pulse
ST 9	2,4,6 8	2		8, 10 Fig. 4	ST, 8 1	$\overline{T}$ Pulse
ST 10	2,4,6 8	2		8,10 Fig. 4	ST, 8 2	$\overline{T}$ Pulse
ST 11	1-4	1		8, 10 Fig. 4	ST, 8 2, 1	$\overline{T}$ Pulse
	5	5	LEVEL/AC Freq Hz = X1K, X1	11, 12	ST, 8 2, 1	0to 3.0 volts p-p at 1KHz (amplitude variable with level control)

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TABLE XIV - SELF TEST CARD OUTPUT (cont'd)

TEST CARD	TEST POINTS SELECTORS A	B	EXCEPTIONS TO SWITCHES SELECTED	APPLI-CABLE NOTES	TEST IDENT LAMPS	OUTPUT
ST 11	5-8	24	LEVEL/DC	2, 9	ST, 8, 2, 1	-1.4 to +1.4 Vdc (positive when LEVEL control is fully clockwise)
	30	24	PROFILE/OFF	2, 9	ST, 8, 2 1	+5.0 Vdc ± 10%
	31	25	PROFILE/ON NORMAL/HIGH	2, 9	ST, 8, 2 1	5.5 ± .5 VDC
	32	26	PROFILE/ON NORMAL/LOW	2, 9	ST, 8, 2 2, 1	4.5 ± 0.5 Vdc
ST 12	2	1		2, 9	ST, 8, 4	-10 ± 1.5 Vdc
	3	1	PROFILE/OFF	2, 9	ST, 8, 4	3 ± 0.5 Vdc
	4	1		2, 9	ST, 8, 4	+14.85 to +15.15 Vdc
	4	5		2,9	ST, 8, 4	+14.85 to +15.15 Vdc
	4	6		2,9	ST, 8, 4	+14.85 to +15.15 Vdc
	4	7		2, 9	ST, 8, 4	+14.85 to +15.15 Vdc
	4	8		2,9	ST, 8, 4	+14.85 to +15.15 Vdc
	4	28		2, 9	ST, 8, 4	+14.85 to +15.15 Vdc

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TABLE XIV - SELF TEST CARD OUTPUT (Cont'd)

TEST CARD	TEST POINTS SELECTORS A	B	EXCEPTIONS TO SWITCHES SELECTED	APPLI-CABLE NOTES	TEST IDENT LAMPS	OUTPUT
ST 12	25	28		2, 9	ST, 8, 4	+14.85 to +15.15 Vdc
	14	28		2, 9	ST, 8, 4	-14.85 to -15.15 Vdc
	24	28		2, 9	ST, 8, 4	+24 ± 3 Vdc
	36	34		2, 9	ST, 8, 4	+14.85 to +15.15 Vdc
	37	35		2, 9	ST, 8, 4	+14.85 to +15.15 Vdc
	20	22		2, 9	ST, 8, 4	-14.85 to -15.15 Vdc
	21	23		2, 9	ST, 8, 4	-14.85 to -15.15 Vdc
	49	51		2, 9	ST, 8, 4	-24.75 to -25.25 Vdc
	50	52		2, 9	ST, 8, 4	-24.75 to -25.25 Vdc
	55	53		2, 9	ST, 8, 4	+24.75 to +25.25 Vdc
	56	54		2, 9	ST, 8, 4	+24.75 to +25.25 Vdc
ST 13	57	57	Freq = 1 Freq Hz = X1K, X1	11, 12	ST, 8, 4 1	1.7 ± 0.5V p-p sine wave at 1KHz

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TABLE XIV - SELF TEST CARD OUTPUT (Cont'd)

TEST CARD	TEST POINTS SELECTORS A	B	EXCEPTIONS TO SWITCHES SELECTED	APPLI-CABLE NOTES	TEST IDENT LAMPS	OUTPUT
ST 13	58	57	Freq = 1 Freq Hz = X1K, X1	11, 12	ST, 8, 4 1	1.7 ± 0.5V p-p sine wave at 1KHz
ST 14	55	-	MATRIX Selector 1 = position 12	1, 11	ST, 8, 4 2	+14.85 to +15.15 Vdc
	56	-	MATRIX Selector 2 position 12	1, 11	ST, 8, 4 2	+14.85 to +15.15 Vdc
	57	57	FREQ = 1 Freq Hz = X1K, X1	11, 12	ST, 8, 4 2	1.7 ± 0.5V p-p triangle wave at 1kHz
	58	58	FREQ = 1 Freq Hz X1K, X1	11, 12	ST, 8, 4 2	1.7 ± 0.5V p-p triangle wave at 1kHz
ST 15	55	-	MATRIX Selector 1 position 12	1, 11	ST, 8, 4 2, 1	-14.85 to -15.15 Vdc
	56	-	MATRIX Selector 2 position 12	1, 11	ST, 8, 4 2, 1	-14.85 to -15.15 Vdc
	57	57	Freq = 1 Freq Hz = X1K, X1	11, 12	ST, 8, 4 2, 1	1.7 ± 0.5V p-p square wave at 1kHz
	58	57	FREQ = 1 Freq Hz = X1K, X1	11, 12	ST, 8, 4, 2, 1	1.7 ± 0.5V p-p square wave at 1kHz

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TABLE XIV - SELF TEST CARD OUTPUT (Cont'd)

TEST CARD	TEST POINTS SELECTORS A	B	EXCEPTIONS TO SWITCHES SELECTED	APPLI-CABLE NOTES	TEST IDENT LAMPS	OUTPUT
ST 16	55	-	MATRIX Selector 1 position 12	1, 11	ST, 16	+24 ± 3 Vdc
	56	-	MATRIX Selector 2 position 12	1, 11	ST, 16	+24 ± 3 Vdc
	57	57	FREQ = 1 Freq Hz = X1K, X1	11, 12	ST, 16	.85 ± 0.3V p-p negative ramp waveform at 1kHz
	58	57	FREQ = 1 Freq Hz = X1K, X1	11, 12	ST, 16	.85 ± 0.3V p-p negative ramp waveform at 1kHz
ST 17	55	-	MATRIX Selector 1 position 12 0 to 40 Vdc applied between J2-A(+) and J2-B(Ret)	1, 11	ST, 16 1	Equal to voltage applied up to 36 ± 3 Vdc
	56	-	MATRIX Selector 2 position 12 0 to 40 Vdc applied between J2-A(+) and J2-B(Ret)	1, 11	ST, 16 1	Equal to voltage applied up to 36 ± 3 Vdc

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TABLE XIV - SELF TEST CARD OUTPUT (Cont'd)

TEST CARD	TEST POINTS SELECTORS A	B	EXCEPTIONS TO SWITCHES SELECTED	APPLI-CABLE NOTES	TEST IDENT LAMPS	OUTPUT
ST 17	57	57	FREQ = 1 Freq Hz = X1k, X1 function switch set to sine wave	11, 12	ST, 16 1	Sine wave, 0-8V p-p at 1kHz (amplitude varies with ATTN control. Maximum amplitude occurs with ATTN control fully CW.
	57	57	FREQ = 1 Freq Hz = X1K, X1 function switch set to square wave	11, 12	ST, 16, 1	Square wave 8V p-p at 1kHz with ATTN control fully clockwise
	57	57	FREQ = 1 Freq Hz = function switch set to triangle wave	11, 12	ST, 16, 1	Triangle wave 8V p-p at 1kHz with ATTN control fully clockwise
	57	57	FREQ = 1 Freq Hz = X1K, X1 function switch set to ramp waveform	11, 12	ST, 16, 1	Positive ramp waveform 4V p-p at 1kHz with ATTN control fully clockwise
	58	57	FREQ = 1 Freq Hz = X1K, X1 function switch set to ramp waveform	11, 12	ST, 16, 1	Positive ramp waveform 4V p-p at 1kHz with ATTN control fully clockwise

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TABLE XIV - SELF TEST CARD OUTPUT (Cont'd)

TEST CARD	TEST POINTS SELECTORS A	B	EXCEPTIONS TO SWITCHES SELECTED	APPLI-CABLE NOTES	TEST IDENT LAMPS	OUTPUT
ST 18	1-7	-	RESET/OFF PROFILE/OFF	1, 8	ST, 16, 2	4.5 ± 0.7 Vdc
	7	-	RESET/ON	1,8	ST, 16, 2	-0.4 to +0.4 Vdc
	8	-	RESET/OFF MATRIX Selector 2 positive 12	1, 8	ST, 16, 2	+8.7 ± 1 Vdc
ST 19	57	1		5	ST, 16, 2 1	45 to 55 ohms
	58	1		5	ST, 16, 2 1	45 to 55 ohms
	55	1	MATRIX Selector 1 position 1	5	ST, 16, 2 1	Infinite resistance
	55	1	position 2	5	ST, 16, 2 1	90 to 110 kohms
	55	1	position 3	5	ST, 16, 2 1	67.5 to 82.5 kohms
	55	1	position 4	5	ST, 16, 2 1	29.7 to 36.3 kohms
	55	1	position 5	5	ST, 16, 2 1	9 to 11 kohms
	55	1	position 6	5	ST, 16, 2 1	6.75 to 8.25 kohms

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TABLE XIV - SELF TEST CARD OUTPUT (Cont'd)

TEST CARD	TEST POINTS SELECTORS A	B	EXCEPTIONS TO SWITCHES SELECTED	APPLI-CABLE NOTES	TEST IDENT LAMPS	OUTPUT
ST 19	55	1	position 7	5	ST, 16, 2 1	2.97 to 3.63 kohms
	55	1	position 8	5	ST, 16, 2 1	900 to 1100 ohms
	55	1	MATRIX Selector 1 position 9	5	ST, 16, 2 1	675 to 825 ohms
	55	1	position 10	5	ST, 16, 2 1	297 to 363 ohms
	55	1	position 11	5	ST, 16, 2 1	90 to 110 ohms
	55	1	position 12	5	ST, 16, 2 1	less than 1 ohm
	56	1	MATRIX Selector 2 position 1	5	ST, 16, 2 1	Infinite resistance
	56	1	position 2	5	ST, 16, 2 1	90 to 110 kohms
	56	1	position 3	5	ST, 16, 2 1	67.5 to 82.5 kohms
	56	1	position 4	5	ST, 16, 2 1	29.7 to 36.3 kohms
	56	1	position 5	5	ST, 16, 2 1	10 ± 1K

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TABLE XIV - SELF TEST CARD OUTPUT (Cont'd)

TEST CARD	TEST POINTS SELECTORS A	B	EXCEPTIONS TO SWITCHES SELECTED	APPLI-CABLE NOTES	TEST IDENT LAMPS	OUTPUT
ST 19	56	1	position 6	5	ST, 16, 2 1	6.75 to 8.25 kohms
	56	1	position 7	5	ST, 16, 2 1	2.97 to 3.63 kohms
	56	1	position 8	5	ST, 16, 2 1	900 to 1100 ohms
	56	1	position 9	5	ST, 16, 2 1	675 to 825 ohms
	56	1	position 10	5	ST, 16, 2 1	297 to 363 ohms
	56	1	position 11	5	ST, 16, 2 1	90 to 110 ohms
	56	1	position 12	5	ST, 16, 2 1	less than 1 ohm
	2	1		5	ST, 16, 2, 1	less than 1 ohm
	28	1		5	ST, 16, 2, 1	less than 1 ohm
ST 20	57	1		5	ST, 16, 4,	450 ± 45 ohms
	58	1		5	ST, 16, 4	450 ± 45 ohms
	55	1	MATRIX Selector 1 position 12	5	ST, 16, 4	90 to 110 ohms

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TABLE XIV - SELF TEST CARD OUTPUT (Cont'd)

TEST CARD	TEST POINTS SELECTORS A	B	EXCEPTIONS TO SWITCHES SELECTED	APPLI-CABLE NOTES	TEST IDENT LAMPS	OUTPUT
ST 20	56	1	MATRIX Selector 2 position 12	5	ST, 16, 4	90 to 110 ohms
ST 21	57	1		5	ST, 16, 4 1	4.4 to 5.4 kohms
	58	1		5	ST, 16, 4, 1	4.4 to 5.4 kohms
	55	1	MATRIX Selector 1 position 12	5	ST, 16, 4, 1	990 to 1210 ohms
	56	1	MATRIX Selector 2 position 12	5	ST, 16, 4, 1	990 to 1210 ohms
ST 22	57	1		5	ST, 16, 4, 2	22.4 to 27.4 kohms
	58	1		5	ST, 16, 4, 2	22.4 to 27.4 kohms
	55	1	MATRIX Selector 1 position 12	5	ST, 16, 4, 2	11.1 ± 1.1 kohms
	50	1	MATRIX Selector 2 position 12	5	ST, 16, 4, 2	11.1 ± 1.1 kohms

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TABLE XIV - SELF TEST CARD OUTPUT (Cont'd)

TEST CARD	TEST POINTS SELECTORS A	B	EXCEPTIONS TO SWITCHES SELECTED	APPLI-CABLE NOTES	TEST IDENT LAMPS	OUTPUT
ST 23	1		0 to -10 Vdc applied between J2-A(-) and J2-B(+)	1, 11	ST, 16, 4, 2, 1	Equal to voltage applied up to -10 Vdc
	2		TEST/1	1, 11	ST, 16, 4, 2, 1	+24 ± 3 Vdc
	2		TEST/2	1, 11	ST, 16, 4, 2, 1	-0.4 to +0.4 Vdc
	4		TEST/1 RESET/ON	1, 11	ST, 16, 4, 2, 1	+24. ± 3 Vdc
	4		TEST/1 RESET/OFF	1, 8	ST, 16, 4, 2, 1	-0.4 to +0.4 Vdc
	43	43		8, 10	ST, 16, 4, 2, 1	NOTE ABSENCE OF 7.5 MC clock.
ST 24	1-58			7	ST, 16, 8	1 megohm or greater
ST 25	1-8		RESET/ON MATRIX Selector 2 position 12	7	ST, 16, 8, 1	1 megohm or greater
ST 26	1-8			7	ST, 16, 8, 2	1 megohm or greater
ST 21	58	1		5	ST, 16, 4, 1	4.4 to 5.4 kohms

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TABLE XIV - SELF TEST CARD OUTPUT (Cont'd)

TEST CARD	TEST POINTS SELECTORS A	B	EXCEPTIONS TO SWITCHES SELECTED	APPLI-CABLE NOTES	TEST IDENT LAMPS	OUTPUT
ST 21	55	1	MATRIX Selector 1 position 12	5	ST, 16, 4, 1	990 to 1210 ohms
	56	1	MATRIX Selector 2 position 12	5	ST, 16, 4, 1	990 to 1210 ohms
ST 22	57	1		5	ST, 16, 4, 2	22.4 to 27.4 kohms
	58	1		5	ST, 16, 4, 2	22.4 to 27.4 kohms
	55	1	MATRIX Selector 1 position 12	5	ST, 16, 4, 2	11.1 ± 1.1 kohms
	50	1	MATRIX Selector 2 position 12	5	ST, 16, 4, 2	11.1 ± 1.1 kohms
ST 23	1		0 to -10 Vdc applied between J2-A(-) and J2-B(+)	1, 11	ST, 16, 4, 2, 1	Equal to voltage applied up to -10 Vdc
	2		TEST/1	1, 11	ST, 16, 4, 2, 1	+24 ± 3 Vdc
	2		TEST/2	1, 11	ST, 16, 4, 2, 1	-0.4 to +0.4 Vdc
	4		TEST/1 RESET/ON	1, 11	ST, 16, 4, 2, 1	+24. ± 3 Vdc

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TABLE XIV - SELF TEST CARD OUTPUT (Cont'd)

TEST CARD	TEST POINTS SELECTORS A	B	EXCEPTIONS TO SWITCHES SELECTED	APPLI-CABLE NOTES	TEST IDENT LAMPS	OUTPUT
ST 23	4		TEST/1 RESET/OFF	1, 8	ST, 16, 4, 2, 1	-0.4 to +0.4 Vdc
	43	43		8, 10	ST, 16, 4, 2, 1	NOTE ABSENCE OF 7.5 MC clock.
ST 24	1-58			7	ST, 16, 8	1 megohm or greater
ST 25	1-8		RESET/ON MATRIX Selector 2 position 12	7	ST, 16, 8, 1	1 megohm or greater
ST 26	1-8			7	ST, 16, 8, 2	1 megohm or greater
ST 27	1-56			1, 8	ST, 16, 8, 2, 1	+4.5 to +5.5 Vdc
ST 28	1-56			1, 8	ST, 16, 8, 4	+4.5 to +5.5 Vdc
ST 29	1-56			1, 8	ST, 16, 8, 4, 1	+4.5 to +5.5 Vdc
ST 30	1-56			1, 8	ST, 16, 8, 4, 2	+4.5 to +5.5 Vdc
ST 31	1	1		8, 10 Fig. 4	ST, 16, 8, 4, 2, 1	T1
	2	1		8, 10 Fig. 4	ST, 16, 8, 4, 2, 1	$\overline{T1}$

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TABLE XIV - SET TEST CARD OUTPUT (Cont'd)

TEST CARD	TEST POINTS SELECTORS A	B	EXCEPTIONS TO SWITCHES SELECTED	APPLI-CABLE NOTES	TEST IDENT LAMPS	OUTPUT
ST 31	3	1		8, 10 Fig. 5	ST, 16, 8, 4, 2, 1	T2
	4	1		8, 10 Fig. 5	ST, 16, 8, 4, 2, 1	$\overline{T2}$
ST 32	1	1		8, 10 Fig. 5	ST, 32	T1
	2	1		8, 10 Fig. 5	ST, 32	T3
	3	1		8, 10 Fig. 5	ST, 32	$\overline{T3}$
	4	1		8, 10 Fig. 5	ST, 32	$\overline{T9}$
ST 33	19	19		8, 10 Fig. 6	ST, 32, 1	$\overline{T1/4KC}$
	1	19		8, 10 Fig. 5	ST, 32, 1	T1
	2	19		8, 10 Fig. 5	ST, 32, 1	$\overline{T10}$
	3	19		8, 10 Fig. 6	ST, 32, 1	$\overline{PRF}$
ST 34	24-26	05-07	OVERLOAD/ON 0 to 10 Vdc applied between J2-A(+) and J2-B(Ret)	2	ST, 32, 2	Equal to voltage applied up to 10 Vdc

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TABLE XIV - SELF TEST CARD OUTPUT (Cont'd)

TEST CARD	TEST POINTS SELECTORS A	B	EXCEPTIONS TO SWITCHES SELECTED	APPLI-CABLE NOTES	TEST IDENT LAMPS	OUTPUT
ST 34	05-07	24-26		2	ST, 32, 2	+6 ± 1 Vdc
	12-13	22-23, 34-35		2	ST, 32, 2	-23 ± 3 Vdc
	16-17	22-23, 34-35		2	ST, 32, 2	+23 ± 3 Vdc
	41-42	51-54		2	ST, 32, 2	-33 ± 5 Vdc
	45-46	51-54		2	ST, 32, 2	+33 ± 5 Vdc
	16-17	14-15		5, 6, 13	ST, 32, 2	Resistance ratio 10: 1 or greater
	36-37	14-15		5, 6, 13	ST, 32, 2	Resistance ratio 10:1 or greater
	12-13	10-11		5, 6, 13	ST, 32, 2	Resistance ratio 10:1 or greater
	20-21	10-11		5, 6, 13	ST, 32, 2	Resistance ratio 10:1 or greater
	45-46	47-48		5, 6, 13	ST, 32, 2	Resistance ratio 10:1 or greater
	55-56	47-48		5, 6, 13	ST, 32, 2	Resistance ratio 10:1 or greater
	41-42	43-44		5, 6, 13	ST, 32, 2	Resistance ratio 10:1 or greater
	49-50	43-44		5, 6, 13	ST, 32, 2	Resistance ratio 10:1 or greater

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TEST CARD	TEST POINTS SELECTORS A	B	EXCEPTIONS TO SWITCHES SELECTED	APPLI-CABLE NOTES	TEST IDENT LAMPS	OUTPUT
ST 34	16-17	36-37	OVERLOAD/ON	5, 6	ST, 32, 2	less than one ohm
	12-13	20-21	OVERLOAD/ON	5,6	ST, 32, 2	less than one ohm
	45-46	55-56	OVERLOAD/ON	5, 6	ST, 32, 2	less than one ohm
	41-42	49-50	OVERLOAD/ON	5, 6	ST, 32, 2	less than one ohm
ST 35	30-32	24-26		5, 6	ST, 32, 2 1	3.75 ± 1 ohm
	20-21	22-23, 34-35		5, 6	ST, 32, 2 1	30.2 ± 3 ohms
ST 35	21	22		5, 6	ST, 32, 2 1	30.2 ± 3 ohms
	36	22		5, 6	ST, 32, 2 1	30.2 ± 3 ohms
	37	22		5, 6	ST, 32, 2 1	30.2 ± 3 ohms
	49	51		5, 6	ST, 32, 2 1	50 ohms ± 10%
	50	51		5, 6	ST, 32, 2 1	50 ohms ± 10%
	55	53		5, 6	ST, 32, 2 1	50 ohms ± 10%
	56	53		5, 6	ST, 32, 2 1	50 ohms ± 10%

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NOTES:

1. Voltage available between TEST POINT A AND GND 1. GND 1 shall be reference point for voltage at TEST POINT A.

2. Voltage available between TPA and TPB. TPB shall be reference point for voltage or signal at TPA.

3. Voltage available between TEST POINT B AND GND 1. GND 1 shall be reference point for voltage at TEST POINT B.

4. Pulse repetition rate is 4 kHz.

5. Resistance stated exists between TEST POINT A and TEST POINT B.

6. Circuit breakers CB1 through CB5 open.

7. Resistance stated exists between TEST POINT A and GND 1.

8. 3 kohm resistive load required between TEST POINT A and GND 1.

9. 3 kohm resistive load required between TEST POINT A and TEST POINT B.

10. Signal at TPA referenced to GND 1, also sync. scope on TPB.

11. 3 kohm resistive load required between TEST POINT A and GND 2.

12. Signal of TPA referenced to GND 2, also sync. scope on TPB.

13. Measures front-to-back ratio of semiconductor. High resistance is with positive meter lead on TEST POINT A, negative meter lead on TEST POINT B. Low resistance is with leads reversed.

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3.5.8 Test Set, Subassembly MX-8051/TPM-22.- With TEST MODE switch S2 of power supply 1A1A4 set to PWR/SUP SERVO position, Test Set, Subassembly MX-8051/TPM-22 shall operate as follows.

3.5.8.1 POWER SUPPLY test section.

3.5.8.1.1 +24 Vdc switch S1.

3.5.8.1.1.1 OFF legend.- The OFF legend of S1 shall be illuminated when S1 is in normally closed position.

3.5.8.1.1.2 ON legend.- The ON legend of S1 shall be illuminated when S1 is in normally open position.

3.5.8.1.2 TEST MODE switch S4.

3.5.8.1.2.1 POWER SUPPLY CONV & SERVO legend.- The POWER SUPPLY CONV & SERVO legend of S4 shall be illuminated and 24 Vdc + 5% shall be present between J5-A(+) and J5-B (RET) under the following conditions:

a. ON legend of S1 illuminated.

b. S4 is in normally closed position.

3.5.8.1.2.2 REGULATOR legend.- The REGULATOR legend of S4 shall be illuminated and the output voltages listed in Table XIV shall be present at J2 under the following conditions:

a. ON legend of S1 is illuminated.

b. S4 is in normally open position.

3.5.8.1.3 DC circuit breaker CB1.- The DC circuit breaker (CB1) shall be closed when reset button is depressed so white band is not exposed. The circuit breaker shall be open when reset button is pulled out to expose white band.

3.5.8.1.4 INTERLOCK switch S2.

3.5.8.1.4.1 STANDBY legend.- The STANDBY legend of S2 shall be illuminated when 24 Vdc ± 5% is applied between J9-G and J9-H.

3.5.8.1.4.2 OPERATE legend.- The OPERATE legend of S2 shall be illuminated when 24 Vdc ± 5% is applied between J9-V and J9-a. Continuity shall exist between J9-F and J9-H when S2 is in normally open position. Resistance between J9-F and J9-H shall be infinite when S2 is in normally open position.

3.5.8.1.5 RADIATE COMMAND switch S3.

3.5.8.1.5.1 RADIATE COMMAND legend.- The RADIATE COMMAND legend of S3 shall be illuminated when 24 Vdc ± 5% is applied to J8-N and J8-P.

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3.5.8.1.5.2 RADIATE COMMAND switch.- With S3 in normally open position and conditions are as stated in 3.5.8.1.2.1, 24 Vdc ± 5% shall be present J6-U(+) and J6-V (Ret).

TABLE XV - OUTPUT VOLTAGES

OUTPUT TERMINALS J2		RESISTIVE LOAD OR OUTPUT CURRENT	OUTPUT VOLTAGE	WAVEFORM
FROM(+)	TO(-)			
BB,DD,r	HH,CC,s	10 ±0.3A	35 to 48 Vdc	N/A
N,AA	p,GG	2.5 ±0.3A	10 to 14 Vdc	N/A
N,AA	E	0.1A ±10%	20 to 24 Vdc	N/A
z	Y	1.0 ±0.1A	20 to 25 Vdc	N/A
D	C		20 to 25 Vdc	N/A
x	v		27 to 33 Vdc	N/A
H	m		27 to 33 Vdc	N/A
n	K		38 to 48 Vdc	N/A
i	q	1.0 ±0.1A	38 to 48 Vdc	N/A
t	Z	2k ohms	60V p-p	square wave, less than 1 usec rise/fall time
w	A			
a	b			
c	d	2k ohms	60V p-p	
e	f	2k ohms	60V p-p	square wave, less than 1 usec rise/fall time
g	h			
j	k			
L	M	2k ohms	60V p-p	

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3.5.8.1.6 Regulator loading (S6-S9, S12) and monitoring (M1 and M2).- Meters M1 and M2 shall indicate as shown in Table XVI under the following conditions:

a. S13 in POWER SUPPLY position.

b. Controls and conditions met in accordance with 3.5.8.1.2.1.

c. A resistive load of 2.9 ohms ± 10% between J5-A and J5-B.

d. Voltages applied to J8 and J9 as follows:

J8	Voltage
x	+15 V
FF	15 V Ret
Y	-15 V
q	+23 V
CC	23 V Ret
r	-23 V
v	+33 V
w	+33 V Ret
c	-33 V
u	-33 V Ret
B	+6 V
C	6 V Ret

J9	Voltage
X	+29 V
Y	29 V Ret

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TABLE XVI - METERS M1 AND M2 READINGS

SWITCH POSITION S12	S6	S7	S8	S9	AMPERES M1	VOLTAGE **M2	TERMINAL POINTS
OFF	-	-	-	-	0	0	
+24V, 0-50A	-	-	-	-	6.9 - 10.0	24 ±7%	N/A
+15V,	FL				1.0 ±8%	15 ±9%	E1(+)
	NL	-	-	-	0		
0-2A	CLL				1.2 ±7%		E2(Ret)
-15V,	FL				1.0 ±8%		E3(-)
	NL	-	-	-	0	15 ±9%	
0-2A	CLL				1.2 ±7%		E4(Ret)
+23V,		FL			1.2 ±7%		E5(+)
	-	NL	-	-	0	23 ±7%	
0-2A		CLL			1.36 ±7%		E6(Ret)
-23V,		FL			1.2 ±7%		E7(-)
	-	NL	-	-	0	23 ±7%	
0-2A		CLL			1.36 ±7%		E8(Ret)
+33V,			FL		0.75 ±9%		E9(-)
	-	-	NL	-	0	33 ±5%	
0-2A			CLL		1.0 ±7%		E10(Ret)
-33V,			FL		0.75 ±9%		E12(-)
	-	-	NL	-	0	33 ±5%	
0-2A			CLL		1.0 ±7%		E11(Ret)
+29V,				FL	10.0 ±8%		E13(+)
	-	-	-	NL	0	29 ±6%	
0-20A				CLL	11.6 ±8%		E14(Ret)
+6V,				FL	*2.47 ±15%		E15(+)
	-	-	-	NL	0	6 ±10%	
0-5A				CLL	*2.9 ±15%		E16(Ret)

* Readings made on 0-5 scale. All other M1 readings are on 0-2 scale.
** Voltages indicated on M2 are available on front panel at TP27(-) and TP28(+).

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3.5.8.1.7 OVERLOAD TEST circuitry, 0-2A LOAD.

3.5.8.1.7.1 MIN load.- Meters M1 and M2 shall indicate 0.018 amperes ±17% and 10V ±7%, respectively, under the following conditions:

a. TEST SELECTOR switch S13 in POWER SUPPLY position.

b. Controls and conditions met in accordance with 3.5.8.1.2.2.

c. 10V ±3% between J8-B(+) and J8-C(Ret).

d. POWER SUPPLY switch S12 in +6V, 0-2A position.

e. +29V, 6V REG LOAD switch S9 in NL position.

f. A shorting wire connected between E20 and E16.

g. A shorting wire connected between E19 and E15.

h. 0-2A LOAD MAX/MIN switch S11 in MIN position.

i. 0-2A LOAD INCREASE control R29 fully counterclockwise.

3.5.8.1.7.2 MAX load.- Meter M1 indication shall be variable from 0.19 to 2.5 amperes ±11% by operation of 0-2A INCREASE control R29 and meter M2 shall indicate 10V ±7% under the following conditions:

a. Controls and conditions met in accordance with 3.5.8.1.7.1 a through f.

b. A 4 ohms ±5% load connected between E19 and E15.

c. 0-2A LOAD MAX/MIN switch S11 in MAX position.

2.5 ampere meter indication shall occur when R29 is fully clockwise.

3.5.8.1.8 2A CB circuit breaker CB3.- The 2A CB circuit breaker (CB3) shall be closed when reset push-button is depressed so white band is not exposed. The circuit breaker shall be open when reset push-button is pulled out to expose white band.

3.5.8.1.9 OVERLOAD TEST circuitry, 0-15A LOAD.

3.5.8.1.9.1 MIN load.- Meters M1 and M2 shall indicate 0.73 amperes ±17% and 24V ±7%, respectively, under the following conditions:

a. TEST SELECTOR switch S13 in POWER SUPPLY position.

b. Controls and conditions met in accordance with 3.5.8.1.2.2.

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c. 24V ±3% between J9-X(+) and J9-Y(Ret).

d. POWER SUPPLY switch S12 in +29V, 0-20A position.

e. +29V, 6V REG LOAD switch S9 in NL position.

f. A shorting wire connected between E18 and E14.

g. A shorting wire connected between E17 and E13.

h. 0-15A LOAD MAX/MIN switch S10 in MIN position.

i. 0-15A LOAD INCREASE control R27 fully counterclockwise.

3.5.8.1.9.1 MAX load.- Meter M1 indication shall be variable from 4.8 to 12.0 amperes ±11% by operation of 0-15A INCREASE control R27 and meter M2 shall indicate 24V ±7% under the following conditions:

a. Controls and conditions met in accordance with 3.5.8.1.9.1a through g.

b. A 2 ohm ±5% load connected between E17 and E13.

c. 0-15A MAX/MIN switch in MAX position.

12.0 ampere meter indication shall occur when R27 is fully clockwise.

3.5.8.1.10 15A CB circuit breaker CB2.- The 15A CB circuit breaker (CB2) shall be closed when reset push-button is depressed so white band is not exposed. The circuit breaker shall be open when reset push-button is pulled out to expose white band.

3.5.8.1.11 SERVO 115V 400 Hz switch S21.

3.5.8.1.11.1 OFF legend.- The OFF legend of SERVO 115V, 400 Hz switch S21 shall be illuminated when S21 is in normally closed position.

3.5.8.1.11.2 ON legend.- The ON legend of S21 shall be illuminated and 115V ±2V, 400 Hz shall be available between J1-E/J1-F(Ret), J7-D/J1-F(Ret and J7-s/J7-i(Ret) when S21 is in normally open position.

3.5.8.1.12 AC circuit breaker CB4.- The AC circuit breaker CB4 shall be closed when reset push-button is depressed so white band is not exposed. The circuit breaker shall be open when reset push-button is pulled out to expose white band.

3.5.8.2 SERVO ELECTRONICS section.

3.5.8.2.1 SERVO ELECTRONICS switch circuitry.

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3.5.8.2.1.1 SERVO COMMAND switch S5.

3.5.8.2.1.1.1 SERVO COMMAND legend.- The SERVO COMMAND legend of S5 shall be illuminated when 24 Vdc ±5% is applied to J7-p(+) and J10-I(Ret).

3.5.8.2.1.1.2 SERVO COMMAND switch.- 24 Vdc ±5% shall be available between J6-W(+) and J6-X(Ret) under conditions stated in 3.5.8.1.2.1 and S5 in normally open position.

3.5.8.2.1.2 ELEVATION ON switch S22.- ELEVATION ON legend of S22 shall be illuminated and 24 Vdc ±5% shall be available between J7-V(+) and J1-p(Ret) when 24 Vdc ±5% is applied between J7-p(+) and J1-p(Ret) and S22 is in normally open position.

3.5.8.2.1.3 TRAVERSE ON switch S23.- TRAVERSE legend of S23 shall be illuminated and 24 Vdc ±5% shall be available between J7-H(+) and J1-p(Ret) when 24 Vdc ±5% is applied between J7-p(+) and J1-p(Ret) and S23 in normally open position.

3.5.8.2.1.4 MODE SELECTION switch S27.

3.5.8.2.1.4.1 NORMAL legend.- NORMAL legend of S27 shall be illuminated under conditions as stated in 3.5.8.1.2.1 and S27 in normally closed position.

3.5.8.2.1.4.2 STOW legend.- STOW legend of S27 and either legend of S28 shall be illuminated and continuity shall exist between points listed below under conditions as stated in 3.5.8.1.2.1 and S27 in normally open position.

a. J10-I and J7-G

b. J10-I and J2-U

3.5.8.2.1.5 MODE SELECTION SWITCH S28.

3.5.8.2.1.5.1 UP-RIGHT legend.- UP-RIGHT legend of S28 shall be illuminated and continuity shall exist between J7-d and J1-U when STOW legend of S27 is illuminated under conditions stated in 3.5.8.2.1.4.2 and S28 in normally closed position.

3.5.8.2.1.5.2 DOWN-LEFT legend.- DOWN-LEFT legend of S28 shall be illuminated when STOW legend of S27 is illuminated under conditions stated in 3.5.8.2.1.4.2 and S28 in normally open position.

3.5.8.2.1.6 MODE SELECTION switch S29.- The LIMIT ON legend of S29 shall be illuminated and continuity shall exist between points listed below under conditions stated in 3.5.8.1.2.1 and S29 in normally open position.

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3.5.8.2.1.7 LOAD switch S20.

3.5.8.2.1.7.1 LOW LOAD legend.- Under conditions stated in 3.5.8.1.2.1 and S20 in normally closed position, the following conditions exist:

a. LOW legend of S20 illuminated.

b. Resistance between J1-m and J1-r shall be 5 ±5% ohms.

3.5.8.2.1.7.2 HIGH LOAD legend.- Under conditions stated in 3.5.8.1.2.1, S20 in normally open position and a shorting wire between J1-Y and J1-A, the following conditions exist:

a. HIGH legend of S20 illuminated.

b. Resistance between J1-m and J1-r shall be 1 ohm ±2%.

3.5.8.2.1.8 AC SIG switch S24.

3.5.8.2.1.8.1 ELEVATION legend.- The ELEVATION legend of S24 shall be illuminated under the following conditions:

a. Controls and conditions met in accordance with 3.5.8.1.11.2.

b. A shorting wire between J7-E and J7-D.

c. S24 in normally closed position.

3.5.8.2.1.8.2 TRAVERSE legend.- The TRAVERSE legend of S24 shall be illuminated under the following conditions:

a. Controls and conditions met in accordance with 3.5.8.1.11.2.

b. A shorting wire between J7-E and J7-D.

c. S24 in normally open position.

3.5.8.2.1.9 DC SIG switch S25.

3.5.8.2.1.9.1 ELEVATION legend.- The ELEVATION legend of S25 shall be illuminated under the following conditions:

a. Controls and conditions met in accordance with 3.5.8.1.11.2.

b. A shorting wire between J7-E and J7-D.

c. S25 in normally closed position.

3.5.8.2.1.9.2 TRAVERSE legend.- The TRAVERSE legend of S25 shall be illuminated under the following conditions:

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a. Controls and conditions met in accordance with 3.5.8.1.11.2.

b. A shorting wire between J7-E and J7-D.

c. S25 in normally open position.

3.5.8.2.2 Signal sources (SERVO ELECTRONICS section).- The voltages shall be as indicated hereafter under the following conditions:

a. Controls and conditions met in accordance with 3.5.8.1.2.1.

b. S21 ON (3.5.8.1.11.2).

c. S24 ELEVATION legend illuminated.

d. S25 ELEVATION legend illuminated.

e. 18K ohm ±5%, resistive load between J7-X and J7-Y.

f. 4.7k ohm ±5%, resistive load between J7-A and J7-B.

3.5.8.2.2.1 AC signal source.

a. AC SIGNAL rotary switch S31 in 0° position.- With S31 in 0° position, the voltage across IN(TP23) and RET(TP24) terminals shall be variable by operation of AC SIGNAL potentiometer R58 from 0 to not less than 0.45 Vrms 400 Hz, in phase ±15° with respect to J10-C. 0 volts occurs at fully counterclockwise position of R58.

b. AC SIGNAL rotary switch S31 in 180° position.- With S31 in 180° position, the voltage across IN(TP23) and RET(TP24) terminals shall be variable by operation of AC SIGNAL potentiometer R58 from 0 to not less than 0.45 Vrms, 400 Hz, 180° ±15° out of phase with respect to J10-C.

c. AC SIGNAL switch S31 in SHORT position.- With S31 in SHORT position and R58 fully counterclockwise, the voltage across IN(TP23) and RET(TP24) terminals shall be less than 0.4 millivolt rms.

3.5.8.2.2.2 DC signal source.

a. DC SIGNAL rotary switch S26 in + position.- With S26 in + position, the voltage across IN(TP25) and RET(TP26) terminals shall be variable by operation of DC SIGNAL potentiometer R60 from 0 to not less than +4.25 Vdc. 0 volts is present when R60 is fully counterclockwise. The 400 Hz ripple on the dc signal shall be less than 0.2 volts p-p.

b. DC SIGNAL rotary switch S26 in - position.- With S26 in - position, the voltage across IN(TP25) and RET(TP26) terminals shall be variable by operation of DC SIGNAL potentiometer R60 from 0 to not less

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than -4.25 Vdc. 0 volts is present when R60 is fully counterclockwise. The 400 Hz ripple on the dc signal shall be less than 0.2 volts p-p.

c. DC SIGNAL switch S26 in SHORT position.- With S26 in SHORT position and R60 fully counterclockwise, the voltage across IN(TP25) and RET(TP26) terminals shall be less than 1.0 millivolt rms.

3.5.8.3 FIRE CONTROL SERVO ELECTRONICS section.

3.5.8.3.1 280V, 15V switch S32.- Continuity shall exist between J10-A and J11-F when S32 is in normally open position. Resistance between J10-A and J11-F shall be infinite when S32 is in normally closed position.

3.5.8.3.1.1 280V ØA, 280V ØB legend.- The 280V ØA and 280V ØB legends of S32 shall be illuminated when 115V ±2V, 400 Hz is applied between J11-B and J11-C.

3.5.8.3.1.2 +15V legend.- The +15V legend of S32 shall be illuminated and 15 Vdc shall be present between J4-T(+) and J4-e(Ret).

3.5.8.3.1.3 -15V legend.- The -15V legend of S32 shall be illuminated and 15 Vdc shall be present between J4-m(-) and J4-e(Ret).

3.5.8.3.2 280V p-p SQ WV 400 Hz ØA circuit breaker CB5.- The ØA circuit breaker CB5 shall be closed when reset push-button is depressed so white band is not exposed. The circuit breaker shall be open when reset push-button is pulled out to expose the white band.

3.5.8.3.3 280V p-p SQ WV 400 Hz ØB circuit breaker CB6.- The ØB circuit breaker CB6 shall be closed when reset push-button is depressed so white band is not exposed. The circuit breaker shall be open when reset push-button is pulled out to expose the white band.

3.5.8.3.4 +15V circuit breaker CB7.- The +15V circuit breaker CB7 shall be closed when reset push-button is depressed so white band is not exposed. The circuit breaker shall be open when reset push-button is pulled out to expose the white band.

3.5.8.3.5 -15V circuit breaker CB8.- The -15V circuit breaker CB8 shall be closed when reset push-button is depressed so white band is not exposed. The circuit breaker shall be open when reset push-button is pulled out to expose the white band.

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TABLE XVII - PULSE VOLTAGE CHARACTERISTICS

LINE	AMPLITUDE VDC P-P	LIMITS VDC	1/2 CYCLE PERIOD SECONDS NEG	1/2 CYCLE PERIOD SECONDS POS	RISE TIME MILLISEC	FALL TIME MILLISEC
1	+13	0 +1.0 -0 13 ±1	0.32 to 0.65	0.32 to 0.65	less than 0.1	less than 0.1
2	-13	0 ±1.5 13 ±1	0.32 to 0.65	0.32 to 0.65	less than 0.1	less than 0.1

3.5.8.3.6 Signal sources (FIRE CONTROL ELECTRONICS section).- The voltages across IN (TP29) and RET(TP30) terminals or J4-C and TP30 shall be as indicated hereafter with 56k ohm ±5%, 1/2 watt resistive load between J4-C and J4-s.

3.5.8.3.6.1 DC SIGNAL/PULSE rotary switch S30 in DC SIGNAL, +13 position.- With S30 in DC SIGNAL +13 position, the voltage across TP29(+) and TP30 (Ret) shall be variable by operation of AMPL DC INPUT control R67 from 0 to not less than 13 Vdc. 0 volts are present when R67 is fully counterclockwise.

3.5.8.3.6.2 DC SIGNAL/PULSE rotary switch S30 in DC SIGNAL, -13 position.- With S30 in DC SIGNAL -13 position, the voltage across TP29(+) and TP30(Ret) shall be variable by operation of AMPL DC INPUT control R67 from 0 to not less than -13 Vdc. 0 volts are present when R67 is fully counterclockwise.

3.5.8.3.6.3 DC SIGNAL/PULSE rotary switch S30 in SHORT position.- With S30 in SHORT position, 0 volts exist between TP29 and TP30 and continuity exists between J4-C and J11-T.

3.5.8.3.6.4 DC SIGNAL/PULSE rotary switch S30 in PULSE +13 position.- With S30 in PULSE +13 position, the voltage waveform between J4-C and TP30 shall be per line 1 of Table XVII.

3.5.8.3.6.5 DC SIGNAL/PULSE rotary switch S30 in PULSE -13 position.- With S30 in PULSE -13 position, the voltage waveform between J4-C and TP30 shall be per line 2 of Table XVII.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of one SMSS manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the Contracting Officer, inspection lot size, lot information, and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables XVII and XVIII herein. Examinations and test for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

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TABLE XVIII - CLASSIFICATION OF DEFECTS

100% INSPECTION

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101.	Electrical	3.4	4.6.2
102.	Performance	3.5	4.6.3
103.	Power requirements	3.5.1	4.6.3.1
104.	Power supply 1A1A4	3.5.2	4.6.3.2
105.	Test Set Subassembly MX-8046A/TPM-22	3.5.3	4.6.3.3
106.	Test Set Subassembly MX-8047/TPM-22	3.5.4	4.6.3.4
107.	Test Set Subassembly MX-8048/TPM-22	3.5.5	4.6.3.5
108.	Test Set Subassembly MX-8049/TPM-22	3.5.6	4.6.3.6
109.	Test Set Subassembly MX-8050/TPM-22	3.5.7	4.6.3.7
110.	Test Set Subassembly MX-8051/TPM-22	3.5.8	4.6.3.8

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.- One SMSS out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests in Table XVIII and shall then meet the requirements and tests in Table XIX.

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TABLE XIX - SPECIAL SAMPLING

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable dwg. visual
302.	General specifications	3.2	MIL-F-13926/ MIL-C-46357 visual
303.	Vibration	3.3.1	4.6.1.1
304.	Shock	3.3.2	4.6.1.2
305.	Altitude - operating	3.3.3.1	4.6.1.3
306.	Altitude - nonoperating	3.3.3.2	4.6.1.3
307.	Temperature	3.3.4	4.6.1.4
308.	Humidity - nonoperating	3.3.5	4.6.1.5

4.4.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government-furnished inspection equipment.- Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor-furnished inspection equipment.

4.5.2.1 Contractor design.- The contractor shall design and supply inspection equipment compatible with "Test Methods and Procedures" specified in 4.6 of this specification and with the "Component Inspection" procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is considered normally to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the

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prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to ensure that the test equipment will indicate reliable acceptability of a product that does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of the test equipment.

4.5.2.2 Test equipment.- In conjunction with 4.5.2.1, the following test equipment shall be used in the performance of the applicable test as specified in 4.6:

4.5.2.2.1 Test equipment for power requirements.-

Item No.	Item	Description
1.	Standard Measuring Equipment (SME) or Ohmmeter	To measure continuity
2.	Voltage Measuring Device	Range 0-30 Vdc 0-130 Vac Accuracy of ±3% on dc ±3% on ac Sensitivity of 20Kohms/Vdc 5Kohms/Vac

4.5.2.2.2 Test equipment for power supply 1A1A4.-

Item No.	Item	Description
1.	Voltmeter, digital	0-200 volts ac, 0-50 volts, dc, accuracies of ±2% on dc and ±3% on ac. Sensitivities of 20,000 ohms per volt dc and 5000 ohms per volt ac.

4.5.2.2.3 Test equipment for Test Set, Subassembly MX-8046A/TPM-22.

Item No.	Item	Description
1.	Voltage & Resistance measuring device	0-40Vdc ± .1% Continuity measuring capabilities
2.	Oscilloscope	Tektronix Type 454 or similar

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Item No.	Item	Description
3.	60 MHz source	Ord. No. 10548158
4.	Attenuator	80 db variable coaxial Jerrold AV-50-2 3 or similar
5.	BNC TEE	3, with 2 jacks and one plug each
6.	Pulse Generator	PRF-0 to 100 KHz Voltage - 0 to + 10V Pulse width .2 usec max. to 5 usec min. 50 ohm impedence
7.	Frequency counter	Systron Donner 1037 with plug in 1255A or similar
8.	Waveguide	X-Band from 6 to 12 inches
9.	Spectrum analyzer	To 10 GHz 60 db Range TEKTRONIX 491 or similar
10.	Signal Generator	600 Hz to 25 KHz square wave and Sine wave 5V p-p
11.	Attenuator	X-Band, 20 db fixed with N connectors
12.	Power Meter	Hewlett Packard Type 432A or similar
13.	Noise Tube	Standard, Calibrated for 25 db noise above KTBn
14.	Termination	BNC 95 ohms
15.	Termination	2 BNC 50 ohms
16.	Probe	Standard oscilloscope probe, Tektronix or similar
17.	X-Band Frequency source	Between 9.14 GHz and 9.245 GHz Ord. No. 10548182 with plug in crystal and Power Source
18.	Mixer Preamplifier X-Band	Ord. No. 10548162

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Item No.	Item	Description
19.	Stop watch	0-60 min ± .5 sec.
20.	Switch	SPST contact rate 20 amp
21.	Switch	SPST contact rate 20 amp
22.	Resistor	500 ohm 10%, 2 watts
23.	Test cable	Ord. No. 11739284
24.	Wide Band Amplifier	60 db gain
25.	Attenuator, Waveguide	Waveline 622, or similar
26.	Adapter	Waveguide to N jack

4.5.2.2.4 Test equipment for Test Set, Subassembly MX-8047/TPM-22.

Item No.	Item	Description
1.	Voltmeter	Range 0 to 40V, accuracy ±3%
2.	Resistive load	12 ohms ±1%, 150 watts
3.	Resistive load (2 required)	14 ohms ±1%, 100 watts
4.	Ohmmeter	Standard measuring equipment to indicate continuity or infinity
5.	Voltmeter	H.V. dc, 4 to 15 kVdc, electrostatic deflection, accuracy of 2%
6.	Pulse generator	Capable of providing a 0.5 usec, 20 kHz 18 volt pulse into 50 ohms
7.	Oscilloscope	Input sensitivity = 5 V per division Sweep period = 1.0 usec per division Input resistance = 10 megohms Vertical passband = dc to 10 MHz Input capacitance = less than 5 pF

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Item No.	Item	Description
8.	Oscilloscope probe	Voltage attenuation = 1000/1, compatible with oscilloscope item 7

4.5.2.2.5 Test equipment for Test Set, Subassembly MX-8048/TPM-22.

Item No.	Item	Description
1.	Standard Measuring Equipment (SME)	To indicate continuity from 0 to
2.	Oscilloscope	Dual channel. Vertical passband DC to 150 MHz. Rise time of scope plus measuring probe 3.5 nanoseconds or less. Sweep period 50 nano-seconds per cm or less. Input resistance greater than 1 megohm. Input capacitance less than 25 picofarads.
3.	Oscilloscope	Dual channel. Vertical passband DC to 15 MHz. Rise time of scope plus measuring probe 27 nanoseconds or less. Sweep period 100 nano-seconds per cm or less. Input resistance greater than 1 megohm. Input capacitance less than 25 picofarads.
4.	Electronic counter	Input sensitivity at least 200 mV rms. Registration of six places or more. Range 10.1 to 100 MHz. Input impedance 1 megohm or greater.
5.	DC digital voltmeter	Visual display 4 digits plus one overrange. Full scale range 100.00 volts or greater. Input resistance 10 megohms or greater. Response time 250 ms or less. Accuracy ±0.01% of reading and ±0.1% of full scale reading.

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Item No.	Item	Description
6.	TPS "T" connector	
7.	TPS to BNC adapter	
8.	Test harnesses	Per Figure 8.

PART LIST FOR TEST HARNESS, ITEM 8

DESIGNATION	PART	DESCRIPTION
P1	Connector	PT06SE-22-78SX, Bendix Sidney, N.Y., or equivalent
P2	Connector	PT06SE-22-78SW, Bendix Sidney, N.Y., or equivalent
P3	Connector	MS3116E-20-39P (SR)
P4	Connector	PT06SE-22-78S, Bendix, Sidney, N.Y., or equivalent

BNC connectors, cable, terminal strips, and wiring per Figure 8.

4.5.2.2.6 Test equipment for Test Set, Subassembly MX-8049/TPM-22.

Item No.	Item	Description
1.	Power supply	0-40 Vdc 1% regulation at 1 ampere
2.	Ohmmeter	To measure up to 10k ohms ±3%, continuity or infinity
3.	Voltmeter	DC, 0-35 volt range, accuracy of ±1% full scale
4.	Resistive load	100 ohms, 1/2 watt, ±5%
5.	Frequency counter	Sensitivity of 0.5V rms capable of counting 60 MHz ±1KHz
6.	Oscilloscope	Rise time of 0.1 nsec including probe. Sweep 5 nsec per division, sensitivity of 1V per division
7.	Signal generator 1	Capable of producing positive pulse, 0V to +6V, 0.3 ± 0.1 usec wide, less than 40 nsec rise time, at 20 KHz into 50 ohms

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Item No.	Item	Description
8.	Signal generator 2	Capable of producing positive pulse, 0V to +3V, 0.3 ± 0.1 usec wide externally triggered at 20 KHz
9.	Resistive load (2)	150 ohms ±10%, 1/2 watt

4.5.2.2.7 Test equipment for Test Set, Subassembly MX-8050/TPM-22.

Item No.	Item	Description
1.	Ohmmeter	To measure resistance, zero ohms to greater than 1 megohm
2.	Voltmeter	DC, 0 to 50 volts, accuracy of 1%
3.	Oscilloscope	Vertical sensitivity = 0.2 to 2.0V/div. Horizontal deflection = 0.1 usec/div Vertical passband = DC to 10 MHz
4.	Voltmeter	AC, 0 to 150 volts, accuracy 1%
5.	Resistor	17 ohms ±10%, not less than 68 watts
6.	Resistor	12 ohms ±10%, not less than 48 watts
7.	Resistor	1 ohm ±10%, not less than 36 watts
8.	Resistor	4 ohms ±10%, not less than 144 watts
9.	Resistor	63 ohms ±10%, not less than 252 watts

4.5.2.2.8 Test equipment for Test Set, Subassembly MX-8051/TPM-22.

Item No.	Item	Description
1.	Power supply	0-28 Vdc ±5%, 3A, unregulated, less than 5% ripple
2.	Power supply	0-40 Vdc ±3%, 17A, unregulated, less than 5% ripple
3.	Volt measuring device, ac-dc	Range - ac 0 to 120 - dc -0 to 40 Accuracies of ±2% on dc and ±3% on ac Sensitivities - 20,000 ohms per volt dc 5000 ohms per volt ac

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Item No.	Item	Description
4.	Resistive load (power)	2.9 ohms ±10%, 300 watt
5.	Oscilloscope	Vertical passband dc to 10 MHz. Input sensitivity - 1 mv per div. Horizontal deflection - 0.2 usec per div. to 0.5 sec per division.
6.	Ohmmeter	To measure up to 2K ohms ±1%, continuity or infinity
7.	Precision voltmeter dc ac/dc	CIMRON DIG.V.M. or equivalent
8.	Power supply dc	+15 Vdc ±3%, 0.350A, regulated, less than 1% ripple
9.	Power supply dc	-15 Vdc ±3%, 0.350A, regulated, less than 1% ripple
10.	Resistive load	4 ohms ±5%, 50 watt
11.	Resistive load (1/2W, ±5%)	(a) 18K ohm (b) 4.7K ohm (c) 56K ohm
12.	Resistive load	2 ohms ±5%, 500 watt
13.	Power supply	23 Vdc ±3%, 2A unregulated
14.	Power supply	23 Vdc ±3%, 2A unregulated
15.	Power supply	33 Vdc ±3%, 1.5A, unregulated
16.	Power supply	33 Vdc ±3%, 1.5A, unregulated
17.	Power supply	6 Vdc ±5%, unregulated
18.	Ammeter	DC, 0-15 amperes, accuracy of 1%
19.	Resistive load	Variable, 0 to 100 ohms 300 watt
20.	Resistive load	2K ohms, 2 watt, ±10%

4.6 Test methods and procedures.

4.6.1 Environmental.- Environmental test shall be conducted in accordance with the general requirements and test methods of

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MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

4.6.1.1 Vibration.- The test shall be performed as specified in Method 514, Procedure IX, Part I, of MIL-STD-810 except that Failure Criteria of Method 514 of MIL-STD-810 do not apply. After vibration, the SMSS shall be tested as given in the sub-paragraphs of 4.6.2 and 4.6.3 for compliance with 3.3.1.

4.6.1.2 Shock.- This test shall be performed as specified in Procedure I of Method 516 of MIL-STD-810 except that shock pulses shall be 11 ± 1 ms and 15 g's. The G value of each impulse shall be within ±10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5-1/2 milliseconds. After the shock impulses, the SMSS shall be tested as specified in 4.6.2 and 4.6.3 for compliance with 3.3.2.

4.6.1.3 Altitude - non-operating and operating.- This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 except that in Step 2 the altitude shall be limited to 40,000 ft. (5.5 in Hg). Operation of Step 3 shall be the performance tests of 4.6.3.

4.6.1.4 Temperature.

4.6.1.4.1 High (operating and non-operating).- This test shall be performed as specified in Procedure I, Method 501, of MIL-STD-810. The temperature of Step 2 shall be +185°F. The temperature of Step 4 shall be +130°F. In Step 5 the SMSS shall be tested in accordance with 4.6.2 and 4.6.3.

4.6.1.4.2 Low (operating and non-operating).- This test shall be performed as specified in Procedure I, Method 502 of MIL-STD-810. The temperature of Step 2 shall be -85°F maintained for 24 hours. The temperature of Step 4 shall be +32°F. In Steps 5 and 7, the SMSS shall be tested in accordance with 4.6.2 and 4.6.3.

4.6.1.5 Humidity-nonoperating.- This test shall be performed in accordance with Procedure I, Method 507 of MIL-STD-810 except that in Step 4, the relative humidity shall be maintained at 95%.

4.6.2 Electrical.- Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.6.3 Performance.

4.6.3.1 Power requirements.- References to test instruments are to those in 4.5.2.2.1. Connect power to the Shop Maintenance Support Set as specified in 3.5.1.1 and 3.5.1.2. This power will remain on

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for all subsequent tests.

4.6.3.1.1 Output power.- Using item 2 verify that the output voltages to the GE DS/GS equipment specified are available as specified in 3.5.1.3.

4.6.3.1.2 Interlock switch 1A1S1.- This test must be conducted prior to installing subassembly 1A1A3. Connect dc voltmeter, item 2, across 1A1P8-A(+) and 1A1P3(-). Push in shaft of switch 1A1S1. Dc voltmeter shall read 24 ± 0.5 volts. Release switch and observe that dc voltmeter reading drops to zero. Pull out switch shaft to its service position and observe that dc voltmeter reads 24 ± 0.5 volts and pilot light is illuminated.

4.6.3.1.3 Circuit breaker CB1.- This test must be conducted prior to the connection of blower subassembly 1A1A5.

a. First phase.- Close circuit breaker CB1 by depressing reset push-button so white band is not exposed. Connect voltmeter, item 2 from 1A1P21-A(+) to ground. Observe that power supply voltage registers on voltmeter. Pull out reset button and observe zero volts on voltmeter.

b. Second phase.- Repeat 4.6.3.1.3a except substitute 1A1P21-B for 1A1P21-A.

c. Third phase.- Repeat 4.6.3.1.3b except substitute 1A1P21-C for 1A1P21-A.

4.6.3.2 Power supply 1A1A4.- References to test instruments are to those in 4.5.2.2.2.

4.6.3.2.1 Circuit breakers.- Circuit breakers CB1 through CB5 shall be inspected in accordance with the following procedures for compliance with 3.5.2.1.

4.6.3.2.1.1 115V, 400 Hz circuit breaker (CB1).- Put TEST MODE switch S2 in any position except FC CONSOLE. Close 115V, 400 Hz circuit breaker CB1 and 24 Vdc CONTROL CKT circuit breaker CB4 by depressing their push-buttons so white bands are concealed. Either ON or OFF legend of 115V, 400 Hz push-button indicator switch (S1) shall light. If OFF legend lights, depress push-button and OFF legend shall be extinguished and ON legend shall light. Connect voltmeter, item 1, across ACVM test points (TP1 and TP2). Voltage shall be 115 volts rms, nominal. Open 115V, 400 Hz circuit breaker by pulling out reset button so white band is exposed. Voltmeter shall read 0 and ON legend of 115V, 400 Hz push-button indicator shall be extinguished.

4.6.3.2.1.2 PWR SUP 1 circuit breaker (CB2).- Put TEST MODE switch S2 in any position except FC CONSOLE. Turn PWR SUP switch

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S5 to 1. Close PWR SUP 1 circuit breaker CB2 and 24 Vdc CONTROL CKT circuit breaker CB4 by depressing their push-buttons so white bands are concealed. Either ON or OFF legend indicator DS3 associated with +24 Vdc switch S3 shall light. If OFF legend lights, depress switch S3 and OFF legend shall be extinguished and ON legend shall light. Turn VOLTAGE switch S4 to +24V and VOLTS meter M1 shall read +24 ±2 volts dc. Open PWR SUP 1 circuit breaker by pulling out reset button so white band is exposed. VOLTS meter M1 shall read 0 and ON legend of +24 Vdc push-button indicator shall be extinguished.

4.6.3.2.1.3 TESTER CKT circuit breaker (CB3).- Put TEST MODE switch S2 in any position except FC CONSOLE. Close PWR SUP 1 circuit breaker CB2 and 24 VDC CONTROL CKT circuit breaker CB4 by depressing their push-buttons so white bands are concealed. Close TESTER CKT circuit breaker CB3 by depressing push-button so white band is concealed. Push +24 Vdc switch S3 to ON (ON legend of DS3 illuminated). Connect voltmeter, item 1, between J3-p(+) and J3-R(RET). Voltage shall be +24 ± 2 volts dc. Open TESTER CKT circuit breaker by pulling out reset button so white band is exposed. Voltmeter shall read 0.

4.6.3.2.1.4 CONTROL CKT circuit breaker (CB4).- Turn TEST MODE switch S2 to the PWR SUP/SERVO position. Close CONTROL CKT circuit breaker CB4 by depressing push-button so white band is concealed. If indicator DS3 reads OFF, push switch S3 so that DS3 reads ON. Connect voltmeter, item 1, between J2-N(+) and J2-C(RET). Voltage shall be +24 ± 2 volts dc. Open CONTROL CKT circuit breaker by pulling out reset button so white band is exposed. Voltmeter shall read 0. ON legend of DS3 shall be extinguished.

4.6.3.2.1.5 PWR SUP 2 circuit breaker (CB5).- Put TEST MODE switch S2 in any position except FC CONSOLE. Turn PWR SUP switch S5 to 2. Close PWR SUP 1 circuit breaker CB2. 24 Vdc CONTROL CKT circuit breaker CB4, and PWR SUP 2 circuit breaker CB5 by pressing reset buttons so that their white bands are concealed. Either ON or OFF legend of indicator DS3 associated with switch S3 shall be illuminated. If OFF legend is illuminated, press switch S3 and OFF legend shall be extinguished and ON legend shall be illuminated. Turn VOLTAGE switch S4 to +24V and VOLTS meter M1 shall read +24 ±2 volts dc. Open PWR SUP 2 circuit breaker CB5 by pulling out reset button so white band is exposed. Voltmeter M1 shall read 0.

4.6.3.2.2 115V 400 Hz ON-OFF Switch S1.- Turn circuit breakers and switches as listed below prior to performing following test procedures.

Circuit Breakers	Reference Designation	Position
115V 400 Hz (10A)	CB1	Open (white band exposed)
PWR SUP 1 (35A)	CB2	Open (white band exposed)
TESTER CKT (10A)	CB3	Open (white band exposed)
CONTROL CKT (2A)	CB4	Open (white band exposed)
PWR SUP 2 (25A)	CB5	Open (white band exposed)

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Switches	Reference Designation	Position
TEST MODE rotary switch	S2	OFF (full CCW)
VOLTAGE rotary switch	S4	OFF (full CCW)
PWR SUP toggle switch	S5	1

Put TEST MODE switch S2 in any position except FC CONSOLE. Close 115V, 400 Hz circuit breaker CB1 and CONTROL CKT circuit breaker CB4 by depressing their push-buttons so white bands are concealed. Either ON or OFF legend of 115V, 400 Hz indicator DS1 shall light. If OFF legend lights, depress 115V 400 Hz switch S1 and OFF legend shall be extinguished and ON legend shall be illuminated. Connect voltmeter, item 1, across ACVM test points (TP1 and TP2). Voltage shall be 115 volts rms, nominal, when 115V, 400 Hz switch is ON, and zero when 115V, 400 Hz switch is OFF. Operate 115V, 400 Hz switch to ON-OFF three times.

4.6.3.2.3 +24 Vdc ON-OFF Switch S3.- Put TEST MODE switch S2 in any position except FC CONSOLE. Turn PWR SUP switch S5 to 1. Close PWR SUP 1 and CONTROL CKT circuit breakers (CB2 and CB4) by depressing push-buttons so white bands are concealed. Either ON or OFF legend of +24 Vdc indicator DS3 shall light. If OFF legend lights, depress switch S3 and OFF legend shall be extinguished and ON legend shall light. Turn VOLTAGE switch S4 to +24V and VOLTS meter M1 shall read +24 ± 2 volts dc. Depress +24 Vdc ON-OFF switch. ON legend shall be extinguished and OFF legend shall light and VOLTS meter M1 shall read 0. Operate switch to ON-OFF three times. Leave indicator DS3 reading ON.

4.6.3.2.4 Converter No. 1 voltages and metering.- After completing 4.6.3.2.3, turn VOLTAGE selector switch S4 to each of its 10 positions and at each position the voltage reading given in Table I shall register on VOLTS meter M1 in compliance with 3.5.2.4.

4.6.3.2.5 Converter No. 2 voltages and metering.- Reestablish conditions existing at the end of 4.6.3.2.3. Close PWR SUP 2 circuit breaker CB5 and turn PWR SUP switch S5 to 2. Turn VOLTAGE selector switch S4 to each of its 10 positions and at each position the voltage reading given in Table I shall register on VOLTS meter M1 in compliance with 3.5.2.5. Press S3 so that indicator DS3 reads OFF. Turn TEST MODE switch S2 to FC CONSOLE. The last reading observed on meter M1 shall reappear.

4.6.3.2.6 Dc output voltages.- With CB2, CB3, CB4, and CB5 closed, DS3 reading ON, and TEST MODE Switch S2 in any position except FC CONSOLE, use digital voltmeter, item 1, to measure polarity and amplitude of dc no-load output voltages at J3 and J4 terminals given in Table II for compliance with 3.5.2.6.

4.6.3.2.7 Power control voltage.- Close CONTROL CKT circuit breaker CB4. Use voltmeter, item 1, to measure 24 ± 2 volts dc from each terminal specified in Table III to J1-E, as TEST MODE switch S2 is set to the position specified in Table III.

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4.6.3.2.8 FC console voltages.- Turn TEST MODE switch S2 to FC CONSOLE. Close CONTROL CKT circuit breaker CB4 and PWR SUP 2 circuit breaker CB5. Use voltmeter, item 1, to measure voltage at test points and terminals given in Table IV for compliance with 3.5.2.8.

4.6.3.3 Test Set, Subassembly MX-8046A/TPM-22.- Reference to test instruments are those in 4.5.2.2.3, turn test mode switch of Power Supply 1A4 to RCVR/XMTR position.

4.6.3.3.1 Power Switch S1.- Using a voltmeter (Item 1) note that the following voltages exist between the following points. When power is supplied to tester. (See Figure 9). Item 20 and Item 21 shall be closed.

FROM	TO(-)	VOLTAGE
FL 3	FL 2	33 V ± .5%
FL 4	C 3	15 V ± .5%
C 3	FL 5	15 V ± .5%
FL 12	FL 8	24 V ± 4Vdc
FL 7	FL 8	24 V ± 4 Vdc
C 2	FL 8	24 V ± 4 Vdc

Depress POWER switch S1, note that the following takes place:

a. ON legend of POWER switch illuminates and OFF legend extinguishes.

b. INTERNAL legend of INTERNAL-EXTERNAL switch illuminates.

c. OFF legend of RADIATE switch illuminates.

d. All other indicators remain extinguished.

With the voltmeter, Item 1, measure the voltages listed in 3.5.3.1.2 (Table V) at the points indicated. When Item 20 is open the light will meet the requirement in paragraph 3.5.3.1.2f.

4.6.3.3.2 Radiate Switch S2.- Depress RADIATE switch. Note that the ON legend illuminates and OFF legend extinguishes. Note that after 2 min ± 12 sec the RADIATE legend illuminates and that a potential of 24 Vdc ± 4 Vdc appears between J4-R (+) J4-F (-). Verify using stop watch (item 19).

4.6.3.3.3 NOISE FIGURE Switch S3.- With RADIATE and ON legends of S2 illuminated. Depress NOISE FIGURE switch S3. Note that the following takes place:

a. NOISE FIGURE legend of S3 illuminates.

b. RADIATE legend of S2 extinguishes. Depress S3 to extinguish legend. After 2 min ± 12 sec the RADIATE legend shall reilluminate.

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4.6.3.3.4 Depress S4 (STALO switch) and note that legend illuminates. Depress again and note that legend extinguishes.

4.6.3.3.5 PULSE GENERATOR switch S5.- Note that the GEN DRIVER switch S10 is in the OFF position. Depress S5. Note that PULSE GEN and SIG legends illuminate. Set S10 to the ON position. Note that DRIVE legend of S5 illuminates.

4.6.3.3.6 INTERNAL - EXTERNAL switch S6.- Depress S6. Note that the INTERNAL legend extinguishes and EXTERNAL legend illuminates.

4.6.3.7 IF AMP switch S11.- Depress S11. Note that the green indicator illuminates. At completion of panel check out, set all switches in the positions such that the conditions stated in 4.6.3.3.1 exist.

4.6.3.3.8 AMPLIFIER A3.- Connect as shown in figure 10.

Oscilloscope settings:

Gain - CH 1 - 5 mv/div

CH 2 - 10 v/div

Channel select - Alt

Sync - int.

Sweep - 20 usec/div

Attenuate signal from 60 MHz oscillator with Item 5 until signal is less than 15 Mv peak to peak.

Depress IF AMP switch to illuminate green indicator. Channel 2 signal shall be not less than 15 V peak to peak.

4.6.3.3.9 AGC adjust R1.- Connect positive lead of voltmeter (Item 1) to center contact of AGC Jack of MX8046A panel, and connect negative lead to frame. Voltage shall vary continuously from NMT 0.1 Vdc at minimum settings of AGC to NLT + 6 Vdc at maximum setting.

4.6.3.3.10 Pulse generator A5.- Using a standard coaxial cable connect PULSE GEN jack to channel 1 of oscilloscope. (Item 2).

Oscilloscope Settings:

Gain - 2V/div

Sweep - 50 usec/div

Sync - int.

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Depress PULSE GEN switch S5 to illuminate PULSE GEN and signal legends set GENERATOR DRIVER Toggle switch to OFF position and note that legend of PULSE GEN switch is not illuminated. Set Sync toggle switch to INT. Pulses visible shall be 3.25 ± .75 divisions high at maximum setting of LEVEL ADJ control and NMT 1/4 division high at minimum setting.

Adjust Freq ADJ until the number of pulses visible on oscilloscope reaches a minimum. At this setting, the time between pulses shall be NLT 250 usec. Adjust FREQ ADJ until the number of pulses visible reaches a maximum. Set the scope sweep to 20 usec/div. At this setting the time between pulses shall be NLT 27.8 usec NMT 29.4 usec. Set oscilloscope sweep to .2 usec/div and locate a pulse. The pulse visible shall not be less than 1 division wide or more than 2 divisions wide. Connect a BNC TEE (Item 5) to output jack of pulse generator (Item 6) and connect TEE jacks to SYNC jack of MX8046A and channel 2 jack of oscilloscope. Set oscilloscope sweep to 50 usec/div and channel selector to Alt. Set PULSE WIDTH on pulse generator (Item 6) to approximately 0.3 usec. Set SYNC toggle switch on MX8046A to EXT. Note that with a PRF of less than 5 pulses per division, the (100 KHz) pulses on channel 1 and channel 2 are syncronized.

4.6.3.3.11 MICROWAVE POWER SOURCE.

a. Power Source frequency measurement. Connect J3 (TRANSMIT PULSE OUT) to Frequency counter (Item 7) using W80 (Item 23). Depress STALO switch S4, set MOD MODE switch to ON and verify that the frequencies generated by the power source are the same as those listed in 3.5.3.13.1b for all twelve channels.

b. Power Source Spectrum Analysis. Disconnect W80 (Item 23) from frequency counter (Item 7) and connect to Spectrum Analyzer (Item 9). With STALO switch depressed and legend illuminated and with MOD MODE switch set to ON, Cw signal shall appear as a narrow spike, with no sideband noise greater than 50 db below the carrier.

c. Connect frequency generator (Item 10) to Ø MOD jack on tester panel. With conditions as described in b above and a square wave with a frequency of between 600 Hz and 20 KHz 4 V peak to peak generated by frequency generator, there shall be side-band noise visible on spectrum analyzer screen. (See Figure 3b).

d. Using frequency generator (Item 10) insert a 5 KHz sine wave 10 V peak to peak into Phase Ø MOD jack on tester panel. A frequency modulated (FM) signal shall be visible on analyzer screen. (See Fig. 3a).

e. Disconnect frequency generator and set MOD MODE switch to PULSE. Depress PULSE-GEN/SIG/DRIVE switch to illuminate PULSE GEN and SIG legends. Set GEN DRIVER switch to ON. Spectrum visible on analyzer shall be that characteristic of a pulse modulated signal. (See Fig. 3a).

4.6.3.3.12 Power Output and microwave chassis loss.

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a. Connect power meter (Item 12) with 20 db attenuator (Item 11) to J3 (TRANSMIT PULSE OUT). With conditions in 4.6.3.3.11b and attenuator AT-1 Set at 0 db, power meter shall read 0.25 mW minimum indicating a minimum power output of 25 mW.

b. Connect power meter to MONITOR Jack J9. With conditions in 4.6.3.3.11b the power output of J9 shall be .5 mW minimum.

c. Connect power meter to SPECTRUM ANALYZER Jack (J24). Power output of J24 shall be .05 mW minimum.

d. Set MOD MODE switch to OFF position. With STALO legend of STALO switch illuminated. Connect Power meter to J3. Power output shall be .005 mW minimum.

e. Using a frequency source (Item 17) of between 9.140 GHz and 9.245 GHz, insert a 50 mW ± 1% signal into J1 ATTENUATOR jack. Connect Power meter and 20 db Attenuator (Item 11) to J3 of tester panel. MOD MODE switch back to ON. Depress S6 to illuminate EXTERNAL legend. Minimum Power reading shall be .19 mW with AT 1 set of 0 db.

f. Rotate AT-1 control slowly between Od Band maximum settings. Power reading shall decrease continuously. The maximum variation between AT 1 Reading and measured attenuation shall be ± 1 db.

4.6.3.3.13 Calibration of MX8046A-TPM-22 noise source.- Connect apparatus as shown in Figure 12. Before operation, set waveguide attenuator (Item 25) so that the output of the X-band source is 10 mW (Measure with Power Meter (Item 12) and attenuator (Item 4). Supply power to all units and set noise source current such that the noise figure is 25 db. Record power meter reading.

a. Remove test cable (Item 23) from noise source and connect to J3 of MX8046A. Depress S3 of MX8046A to illustrate NOISE FIGURE legend and note Power Meter reading.

b. If power meter differs from reading recorded in 4.6.3.3.13 by more than ± 0.5 db, locate miniature attenuator A2AT-1 on MX8046A microwave chassis and adjust until Power Meter reading agrees with that recorded in 4.6.3.3.13.

4.6.3.3.14 MOD TRIGGER, 60 MHz, and To connections.- MOD TRIGGER, S11, S15, J5-A1. Attach BNC TEE's (Item 5) to J11, and J15, to the TEE at J11 connect signal generator (Item 10) and Channel 1 of oscilloscope (Item 2). To TEE at J15 connect 95 ohm load (Item 14). Set frequency generator to produce a 20 KHz square wave signal. Using a probe (Item 16) connected to CH2 of oscilloscope, compare the signal at J15 and J5-A1 to that ON Channel 1 to ascertain that there is no amplitude or phase distortion.

4.6.3.3.15 60 MHz CW J12 and J16.- Connect BNC TEE's (Item 5) to J12 and J16. To TEE and J12 connect 60 MHz source (Item 3) and

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Channel 1 of oscilloscope TEE at J16 connect 50 Ohm load (Item 15) and Channel 2 of oscilloscope. Set sweep on oscilloscope so that actual 60 MHz wavegorm is discernable to ascertain that there is no amplitude or phase distortion between the signal at J12 and that at J16.

4.6.3.3.16 TO TRIGGER J17, J18.- Perform same test as described in 4.6.3.3.14 with input at J14 and 50 ohm loads (Item 15) at both J17 and J18.

4.6.3.3.17 +24 Vdc Control Out.- With conditions as described in 4.6.3.3.1 measured with voltmeter (Item 1) there shall be + 24 ± 4 Vdc across Item 22. (See Figure 5).

4.6.3.4 Test Set, Subassembly, MX-8047/TPM-22.- References to test instruments are to those in 4.5.2.2.4. Turn TEST MODE switch S2 of power supply 1A1A4 to MICROWAVE/KLYSTRON PWR SUP position.

4.6.3.4.1 MICROWAVE CHASSIS TEST section, 2A1.

4.6.3.4.1.1 Interlock indicator lamp (DS7).- Under the conditions stated in 3.5.4.1.1a through d, verify that continuity exists between J8-S and TB2-4. If any one of the conditions are not met, verify that continuity does not exist. Verify that DS7 is illuminated when S9 is pulled out in manual position and extinguished when S9 is pushed in or returned to its center (OFF) position.

4.6.3.4.1.2 Blower motor (B1).- Under conditions stated in 3.5.4.1.2, verify that B1 operates.

4.6.3.4.1.3 POWER switch (S4).

4.6.3.4.1.3.1 OFF legend.- Under conditions as stated in 3.5.4.1.3.1, verify that OFF legend of S4 is illuminated.

4.6.3.4.1.3.2 STANDBY legend.- Under conditions stated in 3.5.4.1.3.2, operate S4 to the normally open position and verify that STANDBY legend of S4 is illuminated.

4.6.3.4.1.4 Output voltages (J1).- With STANDBY legend of S4 illuminated in compliance with conditions stated in 3.5.4.1.4, verify using item 1 that the following voltages are available at J1:

a. 28 Vdc ±6% between J1-1(+) and J1-2(Ret).

b. 28 Vdc ±6% between J1-3(+) and J1-2(Ret).

c. 28 Vdc ±6% between J1-5(+) and J1-6(Ret).

d. 33 Vdc ±6% between J1-7(-) and J1-8(Ret).

e. 0 to 6 Vdc ±5%, variable by operation of AGC ADJUST R1, between J1-A2(+) and J1-4(Ret).

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4.6.3.4.1.5 RADIATE/STANDBY switch (S5).

4.6.3.4.1.5.1 STANDBY legend.- Under conditions stated in 3.5.4.1.5.1, verify that STANDBY legend of S5 is illuminated.

4.6.3.4.1.5.2 STANDBY mode.- Connect P2 and J5. Refer to schematic 10554570 and connect voltmeter item 5 across C8, positive lead to junction of R3 and C8. Set up conditions as stated in 3.5.4.1.5.1. Set S10 to TEST "A" position. Close circuit breaker CB1. After 20 ±2 seconds verify using item 5, that voltage across C8 is 4.3 $^{+0.5}_{-0.2}$ kVdc

4.6.3.4.1.5.3 RADIATE legend and radiate mode.- Remove all input voltages and use shorting bar to remove residual voltages from high voltage points. Connect P2 into J5. Close circuit breaker CB1. Connect a shorting wire between K5 - normally open and normally closed contacts. Set S10 to TEST "B" position. Connect oscilloscope, item 7 with High Voltage probe, item 8 to junction of C8 and P2-E. Set up conditions in 3.5.4.1.4. Verify that STANDBY legend of S5 is illuminated. After 2 minutes ±12 seconds, operate S5 to normally open position and verify that the following occur:

a. RADIATE legend of S5 is illuminated.

b. STANDBY legend of S5 is extinguished.

c. Pulse displayed on oscilloscope, item 7 conforms to characteristics in 3.5.4.1.5.3e and 3.5.4.1.5.3f.

After completion of test, disconnect P2 from J5.

4.6.3.4.1.6 KLYSTRON FILAMENT CURRENT meter (M1).- Set up conditions as specified in 3.5.4.1.5.2. Verify that initial indication of M1 is 1.5 amperes ±5%. Verify that after 20 ±2 seconds, meter M1 indication is 2.0 amperes ±5%.

4.6.3.4.1.7 2A1 OVERLOAD indicator (S6).- Disconnect P1 from subassembly A1.

4.6.3.4.1.7.1 High voltage power supply overload.- Apply 28 Vdc ±3% between P1-1(+) and chassis ground (Ret). Verify that HIGH VOLTAGE POWER SUPPLY legend of S6 is illuminated.

4.6.3.4.1.7.2 Modulator overload.- Apply 28 Vdc ±3% between P1-e(+) and chassis ground (Ret). Verify that MODULATOR legend of S6 is illuminated.

4.6.3.4.1.8 2A1 OVERLOAD reset switch (S6).- With STANDBY legend of S5 illuminated in compliance with conditions stated in 3.5.4.1.3.2, actuate and hold S6 in the normally open position and verify that 28 Vdc ±3% is available between P1-2(+) and chassis ground (Ret). Use item 1 to measure voltage.

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4.6.3.4.2 KLYSTRON POWER SUPPLY TEST section 2A2.

4.6.3.4.2.1 Interlock indicator lamp (DS7).- Under conditions stated in 3.5.4.2.1a through f, verify that continuity exists between J8-S and TB2-4. If any one of the conditions is not met, verify that continuity does not exists. Verify that DS7 is illuminated when S9 is pulled out in manual position and extinguished when S9 is pushed in or returned to its center (OFF) position.

4.6.3.4.2.2 POWER switch (S1).

4.6.3.4.2.2.1 OFF legend.- Under conditions stated in 3.5.4.2.2.1, verify that OFF legend of S1 is illuminated.

4.6.3.4.2.2.2 STANDBY legend.- Under conditions stated in 3.5.4.2.2.2, operate S1 to the normally open position and verify that STANDBY legend of S1 is illuminated. Verify that 28 Vdc ±3% is available between J3-A3(+) and J3-A2(Ret). Use item 1 to measure voltage.

4.6.3.4.2.3 OPERATE/STANDBY switch (S2).

4.6.3.4.2.3.1 STANDBY legend and standby mode.- Under conditions stated in 3.5.4.2.2.2, operate S2 to the normally closed position and verify that the STANDBY legend of S2 is illuminated.

4.6.3.4.2.3.2 OPERATE legend and operate mode.- Under conditions listed in a through e below, verify that conditions listed in f through g exist.

a. Controls and conditions met in accordance with 3.5.4.2.2.2.

b. Switch S2 set to normally open position.

c. 185 volt peak-to-peak symmetrical square wave applied between J3-13 and J3-14 using Item 6.

d. Switch S10 set to "B" position.

e. OPERATE legend of S2 illuminated.

f. 24 Vdc ±5% available between J3-4(+), J3-A4(+) and J3-5(Ret).

g. 9.0 $^{+0.5}_{-0}$ kVdc available between J2(-) and chassis ground using Item 5.

4.6.3.4.2.4 KLYSTRON POWER SUPPLY CURRENT meter (M2).

4.6.3.4.2.4.1 Standby current.- Apply a resistive load of 14 ohms ±1%, 100 watts (Item 3) between J3-A3 and J3-A2. With conditions

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as stated in 3.5.4.2.2.2, verify that meter M2 indicates 2.0 amperes ±6%.

4.6.3.4.2.5 2A2 OVERLOAD indicator (S3).

4.6.3.4.2.5.1 High voltage power supply overload.- Apply 28 Vdc ±3% to J3-1(+) and J3-16(Ret). Verify that HIGH VOLTAGE POWER SUPPLY legend of S3 is illuminated.

4.6.3.4.2.5.2 Modulator overload.- Apply 28 Vdc ±3% to J3-3(+) and J13-6(Ret). Verify that MODULATOR legend of S3 is illuminated.

4.6.3.4.2.6 2A2 OVERLOAD reset switch (S3).- With conditions as stated in 3.5.4.2.2.2, 3.5.4.2.3.1 or 3.5.4.2.3.2, actuate and hold S6 in the normally open position. Verify that 28 Vdc ±3% is available between (J3-2(+) and chassis ground (Ret). Use item 1 to measure voltage.

4.6.3.4.2.7 TEST A/B switch S10.- Apply voltages to J8 as specified in 4.6.3.4. Operate S10 to TEST A position and set up conditions as stated in 3.5.4.2.3.2a through c. Verify using item 4 that continuity exists between P2-C and chassis ground.

4.6.3.5 Test Set, Subassembly, MX-8048/TPM-22.- References to test instruments are to those in 4.5.2.2.5. Turn TEST MODE switch S2 of power supply 1A4 to RCVR position. Attach test harness (Item 8) to receiver test panel as shown in Figure 8. Terminate TB3 test harness terminals J1-D, J4-A, and J4-G with 50 ohms, and terminate the following receiver test panel terminals with 50 ohms: J11, J12, J13, J14 and J20. Terminate the following receiver test panel terminals with 150 ohms: J8, J9, J10, and J21. Turn IF DELAY ADJ switch (COARSE) to MIN position. Turn IF DELAY ADJ control (FINE) fully counterclockwise.

4.6.3.5.1 POWER switch S1.- Either the OFF or ON legend of POWER indicator DS1 shall be illuminated. Depress POWER switch S1 three times and verify that it causes OFF legend to be extinguished and ON legend to be illuminated, and the reverse. Leave POWER switch S1 ON.

a. MODULATOR TRIGGER.- Connect test equipment as in Figure 10. Verify that the amplitude and width of the pulses conform to the requirements of MODULATOR TRIGGER, Table V. Verify that signal disappears when POWER switch S1 is OFF. Connect test equipment as in Figure 11. Adjust electronic counter, item 4, to verify that frequency of MODULATOR TRIGGER is 19,979 ±1 Hz. Return to the setup of Figure 10.

b. PRE RANGE TRIGGER.- Remove 150-ohm termination from test harness terminal TB3-J2-B. Connect this terminal to coaxial lead z of Figure 10. Adjust oscilloscope to display approximately ten cycles on each channel. Verify that pulses on channel B occur in a one-for-one relationship to those on channel A. Verify that the amplitude, width, and delay of the pulses on channel B conform to

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the requirements of PRE RANGE TRIGGER, Table V. Remove coaxial lead z from TB3-J2-B and restore 150-ohm termination at that point.

c. E/L TRIGGER.- Remove 150-ohm termination from test harness terminal TB3-J2-D. Connect this terminal to coaxial lead z of Figure 10. Adjust oscilloscope to display approximately ten cycles on each channel. Verify that pulses on channel B occur in a one-for-one relationship to those on channel A. Verify that amplitude, width, and delay of the pulses on channel B conform to the requirements of E/L TRIGGER, Table V. Remove coaxial lead z from TB3-J2-D and restore 150-ohm termination at that point.

d. MODULATOR TRIGGER Unit 3.- Remove 150-ohm termination from test harness terminal TB3-J2-E. Connect this terminal to coaxial lead z of Figure 10. Adjust oscilloscope to display approximately ten cycles on each channel. Verify that pulses on channel B occur in a one-for-one relationship to those on channel A. Verify that the amplitude width, and delay of the pulses on channel B conform to the requirements of MODULATOR TRIGGER, Unit 3, Table V. Remove coaxial lead z from TB3-J2-E and restore 150-ohm termination at that point.

e. VIDEO PULSE 3A2.- Remove 150-ohm termination from receiver test panel terminal J8. Connect this terminal to coaxial lead z of Figure 10. Adjust oscilloscope to display approximately ten cycles on each channel. Verify that pulses on channel B occur in a one-for-one relationship to those on channel A. Verify that the amplitude width, and delay of the pulses on channel B conform to the requirements of VIDEO PULSE 3A2, Table V. Verify that VIDEO ADJUST control R1 affects pulse amplitude as specified in VIDEO PULSE 3A2, Table VI. Remove coaxial lead z from J8 and restore 150-ohm termination at that point.

f. 3A2/3A5 TRIGGER.- Remove 150-ohm termination from receiver test panel terminal J10. Connect this terminal to coaxial lead z of Figure 10. Adjust oscilloscope to display approximately ten cycles on each channel. Verify that pulses on channel B occur in a one-for-one relationship to those on channel A. Verify that the amplitude, width, and delay of the pulses on channel B conform to the requirements of 3A2/3A5 TRIGGER, Table V. Verify that TRIGGER ADJUST control R2 affects pulse amplitude as specified in 3A2/3A5 TRIGGER, Table V. Remove coaxial lead z from J10 andrestore 150-ohm termination at that point.

g. 3A9 TRIGGER.- Remove 150-ohm termination from receiver test panel terminal J21. Connect this terminal to coaxial lead z of Figure 10. Adjust oscilloscope to display approximately ten cycles on each channel. Verify that pulses on channel B occur in a one-for-one relationship to those on channel A. Verify that the amplitude, width, and delay of the pulses on channel B conform to the requirements of 3A9 TRIGGER, Table V. Remove coaxial lead z from J21 and restore 150-ohm termination at that point.

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h. STC/DC UNIT 3.- Remove 150-ohms termination from test harness terminal TB3-J2-C. Remove coaxial lead Y (Figure10) from receiver test panel terminal J22. Attach coaxial lead Y to TB3-J2-C. With AGC SELECT switch S6 in DC position, verify that the signal on the oscilloscope conforms to the requirements of STC/DC UNIT 3, part 1), Table V. With AGC SELECT switch S6 in STC position, verify that the shape and level of the signal conform to the requirements of STC/DC UNIT 3, part 2), Table V. Re-establish the setup of Figure10 except with coaxial lead on test harness terminal TB3-J2-C. Verify that frequency of signal conforms to the requirements of STC/DC UNIT 3, part 2), Table V. Remove coaxial lead from TB3-J2-C and restore 150-ohm termination at that point.

i. +15V 5% REG.- Use digital voltmeter, item 5 to verify that voltage between test harness terminals TB2-J3-g(+) and TB2-J3-r conforms to the requirement of +15V 5% REG, Table V.

j. -15V 5% REG.- Use digital voltmeter, item 5 to verify that voltage between test harness terminals TB2-J3-h and TB2-J3-r conforms to the requirements of -15V 5% REG, Table V.

k. STALO OSC.- Connect test equipment as in Figure 12. Verify that amplitude of signal is constant and that it conforms to the requirements of STALO OSC, Table V. Reconnect test equipment as in Figure 11, except with coaxial lead connected to test harness terminal TB3-J4-A. Verify that frequency of signal conforms to requirement of STALO OSC, Table V. Remove coaxial lead from TB3-J4-A and restore 50-ohm termination at that point.

l. STC/DC 3A2.- Return to the setup of Figure10. Remove 150-ohm termination from receiver test panel terminal J9. Remove coaxial lead Y from receiver test panel terminal J22. Attach coaxial lead Y to J9. With AGC SELECT switch S6 in DC position, verify that the signal on the oscilloscope conforms to the requirements of STC/DC UNIT 3, part 1), Table V. With AGC SELECT switch S6 in STC position, verify that the shape and level of the signal conform to the requirements of STC/DC UNIT 3, part 2), Table V. Reestablish the setup of Figure 11 except with coaxial lead on receiver test panel terminal J9. Verify that frequency of signal conforms to the requirement of STC/DC UNIT 3, part 2), Table V. Remove coaxial lead from J9 and restore 150-ohm termination at that point.

m. LO OUT (1).- Remove 50-ohm termination from receiver test panel terminal J13. Connect test equipment as in Figure 13. Verify that amplitude of signal is constant and that it conforms to the requirement of LO OUT (1), Table V. Reconnect test equipment as in Figure10, except with coaxial lead connected to receiver test panel terminal J13. Verify that frequency of signal conforms to requirement of LO OUT (1), Table V. Remove coaxial lead from J13 and restore 50-ohm termination at that point.

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n. LO OUT (2).- Repeat step (m) above, except substitute J14 for J13 throughout, and LO OUT (2) for LO OUT (1).

o. LO 3A5.- Remove 50-ohm termination from receiver test panel J23. Repeat step (m) above, except substitute J20 for J13 throughout, and LO 3A5 for LO OUT (1). Restore 50-ohm termination at J23.

p. +25V REG.- Use digital voltmeter, item 5, to verify that voltage between test harness terminals TB2-J3-M(+) and TB2-J3-N (RET) conforms to the requirements of +25V REG, Table VI. Verify that when OFF legend of POWER indicator DS1 is illuminated, +25V REG does not appear at TB2-J3-M.

q. -15V REG.- Use digital voltmeter, item 5, to verify that voltage between test harness terminals TB2-J3-a(-) and TB2-J3-m (RET) conforms to the requirement of -15V REG, Table VI. Verify that when OFF legend of POWER indicator DS1 is illuminated, -15V REG does not appear at TB2-J3-a.

r. +15V REG.- Use digital voltmeter, item 5 to verify that voltage between test harness terminals TB2-J3-B(+) and TB2-J3-m(RET) conforms to the requirement of +15V REG, Table VI. Verify that when OFF legend of POWER indicator DS1 is illuminated, +15V REG does not appear at TB2-J3-b. Leave ON legend of POWER indicator DS1 illuminated.

4.6.3.5.2 CLUTTER LOCK ON switch S2.- Depress CLUTTER LOCK ON switch S2 three times and verify that CLUTTER CLOCK ON indicator DS2 alternately is illuminated and extinguished. Use digital voltmeter, item 5, to verify that +24 Vdc ±5% signal appears between test harness terminals TB2-J3-W(+) and TB2-J3-T(RET) when CLUTTER LOCK ON indicator DS2 is illuminated.

4.6.3.5.3 MID RANGE CALIBRATION switch S3.- Depress MID RANGE CALIBRATION switch S3 three times and verify that MID RANGE CALIBRATION indicator DS3 alternately is illuminated and extinguished.

a. MID RANGE CALIBRATION CONTROL.- Use digital voltmeter, item 5, to verify that voltage between test harness terminal TB2-J3-A(+) and TB2-J3-T(RET) conforms to the requirement of MID RANGE CALIBRATION CONTROL, Table VII.

b. RANGE CAL RF INPUT.- Connect test equipment as in Figure 14. Remove 50-ohm termination from test harness terminal TB3-J4-G and attach coaxial lead Z to this terminal. With indicator DS3 illuminated, verify that the width, amplitude, and delay of the bursts on channel B conform to the requirements of RANGE CAL RF INPUT, part 1), Table VII. Verify that signal on B channel disappears when indicator DS3 is extinguished. Remove coaxial lead Z from TB3-J4-G and restore 50-ohm termination.

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4.6.3.5.4 RECEIVER NOISE switch S4.- Depress RECEIVER NOISE switch S4 three times and verify that RECEIVER NOISE indicator DS4 is alternately illuminated and extinguished.

4.6.3.5.5 3A6 TEST switch S5.- Depress 3A6 TEST switch S5 three times and verify that 3A6 TEST indicator DS5 is alternately illuminated and extinguished.

4.6.3.5.6 Combinations of switches S2, S3, S4, and S5.- Turn IF AMPLITUDE ADJ control AT4 to the zero (00) position.

a. Connect test equipment as in Figure 14. Remove 50-ohm termination from test harness terminal TB3-J1-D and attached coaxial lead Z to this terminal. For each of the switch combinations G,L,N,P of Table VII verify that the width, amplitude and delay of the bursts on channel B conform to the requirements of IF INPUT UNIT 3, Table VIII. Verify for switch combinations G,L,N that 59.936 MHz excursions on B channel are synchronous (successive recurrences superimposed). Verify for switch combination P that 59.941 MHz excursions on channel B are synchronous. Remove coaxial lead Z from TB3-J1-D and restore 50-ohm termination.

b. Remove 50-ohm termination from receiver test panel terminal J11 and attach coaxial lead Z to this terminal. For each of the switch combinations D,H,K,0 of Table VIII, verify that the width, amplitude and delay of the bursts on channel B conform to the requirements of IF OUT, Table IX. Verify for switch combinations D,H,K that 59.936 MHz excursions on B channel are synchronous (successive recurrences superimposed). Verify for switch combination 0 that 59.941 MHz excursions on channel B are synchronous. Remove coaxial lead Z from J11 and restore 50-ohm termination.

c. Remove 50-ohm termination from receiver test panel terminal J12 and attach coaxial lead Z to this terminal. For each of the switch combinations D,H,K,0 of Table VIII, verify that the width, amplitude and delay of the bursts on channel B conform to the requirement of IF OUT 3A6, Table IX. Verify for switch combinations D,H,K that 59.936 MHz excursions on B channel are synchronous (successive recurrences superimposed). Verify for switch combination 0 that 59.941 MHz excursions on channel B are synchronous. Remove coaxial lead Z from J12 and restore 50-ohm termination.

d. Inside the receiver test panel, open the connection between P50 and J33, at the output of test oscillator 10554934 and insert a TPS T connector. Attach TPS-to-BNC adapter to the third port of the connector, and complete connections to set up test equipment according to Figure 15. Remove 50-ohm termination from test harness terminal TB3-J1-D and attach coaxial lead Z to this terminal. Set up switch combination P of Table VIII. Verify that 59.941 MHz excursions on both channels are synchronous, but that envelope of bursts on B channel is stable. Disconnect P50 from the TPS T connector in the receiver test panel and verify that signal on B channel only disappears. Reconnect P50 to the TPS T connector.

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Remove coaxial lead Z from TB3-J1-D. Using standard measuring equipment (item 1) verify for each of the switch combinations A,C,D,E,H,J,K,0 of Table VIII that the resistance from TB3-J1-D to ground is not more than 2 ohms. Verify for each of the switch combinations B,F,I,M that the resistance from TB3-J1-D to ground is not less than 40 ohms. Restore 50-ohm termination at TB3-J1-D.

e. Remove 50-ohm termination from receiver test panel terminal J11 and attach coaxial lead Z to this terminal. Set up switch combination 0 of Table VIII. Verify that 59.941 MHz excursions on both channels are asynchronous, but that envelope of bursts on B channel is stable. Disconnect P50 from the TPS T connector in the receiver test panel and verify that signal on B channel only disappears. Reconnect P50 to the TPS T connector. Remove coaxial lead Z from J11. Using standard measuring equipment (item 1) verify for each of the switch combinations B,F,G,I,L,M,N,P of Table VIII that the resistance from TB3-J1-D to ground is not more than 2 ohms. Verify for each of the switch combinations A,C,E,J that the resistance from TB3-J1-D to ground is not less than 40 ohms. Restore 50-ohm termination at J11.

f. Remove 50-ohm termination from receiver test panel terminal J12 and attach coaxial lead Z to this terminal. Set up switch combination 0 of Table VIII. Verify that 59.941 MHz excursions on both channels are asynchronous, but that envelope of bursts on B channel is stable. Disconnect P50 from the TPS T connector in the receiver test panel and verify that signal on B channel only disappears. Reconnect P50 to the TPS T connector. Remove coaxial lead Z from J12. Using standard measuring equipment (item 1) verify for each of the switch combinations B,F,G,I,L,M,N,P of Table VIII that the resistance from TB3-J1-D to ground is more than 2 ohms. Verify for each of the switch combinations A,C,E,J that the resistance from TB3-J1-D to ground is not less than 40 ohms. Restore 50-ohm termination at J12.

g. Remove coaxial lead Y from TPS to BNC adapter (see Figure 18.) Attached electronic counter, item 4, to adapter. Verify that frequency of signal is 59.941 ± 0.001 MHz.

4.6.3.5.7 IF DELAY ADJ controls.

a. Reestablish conditions of Figure 8. Verify that IF DELAY ADJ controls affect the delay of signal on B channel of oscilloscope in accordance with the requirements of 3.5.5.7.

b. Reestablish conditions existing at the end of the third sentence of 4.6.3.5.3b. Verify that IF DELAY ADJ controls affect the delay of signal on B channel of oscilloscope in accordance with the requirements of 3.5.5.7.

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oscilloscope in accordance with the requirements of 3.5.5.7.

b. Reestablish conditions existing at the end of the third sentence of 4.6.3.5.3b. Verify that IF DELAY ADJ controls affect the delay of signal on B channel of oscilloscope in accordance with the requirements of 3.5.5.7.

c. Reestablish conditions existing at the end of the third sentence of 4.6.3.5.6a. Verify that IF DELAY ADJ controls affect the delay of signal on B channel of oscilloscope in accordance with the requirements of 3.5.5.7.

d. Reestablish conditions existing at the end of the second sentence of 4.6.3.5.6b. Verify that IF DELAY ADJ controls affect the delay of signal on B channel of oscilloscope in accordance with the requirements of 3.5.5.7.

e. Reestablish conditions existing at the end of the second sentence of 4.6.3.5.6c. Verify that IF DELAY ADJ controls affect the delay of signal on B channel of oscilloscope in accordance with the requirements of 3.5.5.7.

f. Reestablish conditions existing at the end of the fifth sentence of 4.6.3.5.1e. Verify that IF DELAY ADJ controls affect the delay of signal on B channel of oscilloscope in accordance with the requirements of 3.5.5.7.

4.6.3.5.8 IF AMPLITUDE ADJ control.

a. Reestablish conditions existing at the end of the third sentence of 4.6.3.5.6a. Verify that IF AMPLITUDE ADJ control affects the amplitude of signal on B channel of oscilloscope in accordance with the requirements of 3.5.5.8

b. Reestablish conditions existing at the end of the second sentence of 4.6.3.5.6b. Verify that IF AMPLITUDE ADJ control affects the amplitude of signal on B channel of oscilloscope in accordance with the requirements of 3.5.5.8.

c. Reestablish conditions existing at the end of the second sentence of 4.6.3.5.6c. Verify that IF AMPLITUDE ADJ control affects the amplitude of signal on B channel of oscilloscope in accordance with the requirements of 3.5.5.8.

4.6.3.6 Test Set, Subassembly MX-8049/TPM-22.- References to test equipment are to those in 4.5.2.2.6 set TEST MODE switch S2 of power supply 1A4 to RANGE COMPUTER position.

4.6.3.6.1 POWER switch S1.

4.6.3.6.1.1 OFF.- Depress and release S1 three times. Verify that OFF legend of POWER switch is illuminated on alternate operations of S1.

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4.6.3.6.1.2 ON.- Verify that OFF or ON legend of POWER switch is illuminated. Depress and release S1 three times. Verify that OFF and ON legends of POWER switch are illuminated alternately with operation of S1. Using item 3, verify that 24 Vdc ±5% is present at J5-D(+), J5-B(+), J5-T(+) and J8-P(Ret) when ON legend is illuminated.

4.6.3.6.2 24V circuit breaker CB1.- Leave POWER switch in ON position. Open CB1 by pulling out reset button so that white band is exposed. Observe that ON legend of POWER switch is extinguished. Reset circuit breaker. Verify that ON legend is illuminated.

4.6.3.6.3 POWER indicator DS3 legends.

4.6.3.6.3.1 +33V legend.- Set up conditions as stated in 3.5.6.1.2. Verify that +33V legend of DS3 is illuminated.

4.6.3.6.3.2 -33V legend.- Set up conditions as stated in 3.5.6.1.2. Verify that -33V legend of DS3 is illuminated.

4.6.3.6.3.3 +23V legend.- Set up conditions stated in 3.5.6.1.2. Verify that +23V legend of DS3 is illuminated.

4.6.3.6.3.4 -23V legend.- Set up conditions as stated in 3.5.6.1.2. Verify that -23V legend of DS3 is illuminated.

4.6.3.6.4 RADIATE switch S2.- Set up conditions as stated in 3.5.6.1.2. Verify that OFF or ON legend of RADIATE indicator DS5 is illuminated. Depress and release RADIATE switch S2 three times. Verify that OFF and ON legend are illuminated alternately with operation of S2. Verify, using item 3, that 24 Vdc ±5% is present between J5-J(+) and J5-K(Ret) when RADIATE ON legend is illuminated.

4.6.3.6.5 TARGET/NO TARGET switch S3.- Set up conditions as stated in 3.5.6.1.2. Depress and release TARGET/NO TARGET switch three times. Verify that TARGET and NO TARGET legend are illuminated alternately with operation of S3.

4.6.3.6.6 SEARCH/TRACK indicator DS2.

4.6.3.6.6.1 DS2 TRACK legend.- Apply 24 Vdc ±5% item 1 to J5-W(+) and J8-P(Ret). Verify that TRACK legend of DS2 is illuminated.

4.6.3.6.6.2 DS2 SEARCH legend.- Apply 24 Vdc ±5%, item 1, to J5-C(+) and J8-P(Ret). Verify that SEARCH legend of DS2 is illuminated.

4.6.3.6.7 DOPPLER indicator DS1.

4.6.3.6.7.1 DS1 NO ECM legend.- Apply 24 Vdc ±5%, item 1, to J5-A(+) and J8-P(Ret). Verify that NO ECM legend of DS1 is illuminated.

4.6.3.6.7.2 DS1 ECM legend.- Apply 24 Vdc ±5%, item 1 to J5-V(+) and J8-P(Ret). Verify that ECM legend of DS1 is illuminated.

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4.6.3.6.8 TARGET REJECT switch S4.- Set up conditions as stated in 3.5.6.1.2. Verify that TARGET REJECT legend is illuminated white. Depress and release S4 three times. Verify that TARGET REJECT legend is illuminated red while S4 is depressed and returns to white when S4 is released. Verify using item 3 that 24 Vdc ±5% is present between J5-G(+) and J5-H(Ret) while S4 is depressed.

4.6.3.6.9 60 MHz oscillator.- Set up conditions stated in 3.5.6.1.2. Connect frequency counter item 5 and oscilloscope item 6 to J4-A with 50 ohm coax. Terminate coax at oscilloscope with 100 ohm load item 4. Verify that frequency counter registers 59.939 MHz ± 1000Hz at 2.0 V peak-to-peak ±1 volt. Place S1 in the OFF position. The output at J4-A shall go to zero.

4.6.3.6.10 Output signals.

4.6.3.6.10.1 Simulated target.- Connect test equipment to the range computer as shown in Figure 16. Apply signal 1 to J4-E using item 7 as specified in Table XII. Close circuit breaker CB1 and operate S1 to illuminate POWER ON legend.

4.6.3.6.10.1.1 COARSE RANGE switch S9.- Set FINE RANGE control R1 to its approximate center position. Set COARSE RANGE switch S9 to MIN position. Set OSCILLOSCOPE CHAN A switch S5 to position 9 and verify using item 6 that simulated target appears on B channel 1 to 13 usec after signal on A channel. Simulated target shall be 0.3 ± 0.1 usec wide, from 0 to +3 Vdc (minimum). Set COARSE RANGE switch to MID position and verify that simulated target appears 6 to 22 usec after signal A channel. Set COARSE RANGE switch to MAX position and verify that simulated target appears 12 to 28 usec after signal A channel.

4.6.3.6.10.1.2 FINE RANGE control R1.- Set COARSE RANGE switch S9 to MIN position. Verify using item 6 that operation of FINE RANGE control R1 from fully counterclockwise (ccw) to fully clockwise (cw) causes delay time between leading edge of simulated target on B channel of item 6 and leading edge of signal on A channel of item 6 to increase continuously from 1 ± 0.5 to 12 ± 1 usec. Set COARSE RANGE switch to MID position and repeat operation of R1 from ccw to cw. Delay time shall increase continuously from 7 ± 1 to 21 ± 1 usec. Set COARSE RANGE switch to MAX position and repeat operation of R1 from ccw to cw. Delay time shall increase continuously from 13 ± 1 to 27 ± 1 usec.

4.6.3.6.10.2 Sum and difference doppler.- Connect test equipment to the range computer as shown in Figure 17. Adjust test equipment as follows:

a. As listed in 4.6.3.6.10.1.

b. Apply signal 2 to J4-D using item 8 as specified in Table XII delay from signal 1 by 15 ± 3 usec.

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Close circuit breaker CB1 and operate S1 to illuminate POWER ON legend. Set SUM LEVEL control R2 and DIFFERENCE LEVEL control R3 fully clockwise. Operate TARGET/NO TARGET switch S3 to illuminate NO TARGET legend. Verify, using item 6 that no signal is present at J4-F and J4-G. Operate TARGET/NO TARGET switch S3 to illuminate TARGET legend. Set COARSE RANGE switch S9 to MID position. Operate FINE RANGE control R1 from fully counterclockwise to fully clockwise position. Verify that signals on channel A and B of item 6 are 2.5 KHz ± 25 Hz, amplitude equal to or greater than 1.5V (p-p) in phase when R1 is fully ccw and 180 degrees out of phase when R1 is fully cw.

4.6.3.6.10.2.1 SUM level control R2.- Under conditions stated in 4.6.3.6.10.2, operate SUM level control R2 from fully cw to fully ccw. Verify that amplitude of sum doppler signal on channel A varies continuously from greater than 1.5 V (p-p) Vdc to less than .05V (p-p).

4.6.3.6.10.2.2 DIFFERENCE level control R3.- Under conditions stated in 4.6.3.6.10.2, operate DIFFERENCE level control R3 from fully cw to fully ccw. Verify that amplitude of difference doppler signal on channel B varies continuously from greater than 1.5V (p-p) to less than .05V (p-p).

4.6.3.7 Test Set, Subassembly MX-8050/TPM-22.- References to test instruments are to those in 4.5.2.2.7. Turn TEST MODE switch S2 of power supply 1A4 to PC CARDS position.

4.6.3.7.1 POWER/ON/OFF switch S1.

4.6.3.7.1.1 POWER/OFF legend.- Set S1 to normally closed position. Verify that POWER/OFF legend is illuminated.

4.6.3.7.2 POWER/ON legend.- Under conditions as stated in 4.6.3.7.1.1, operate S1 to the normally open position and verify that POWER/ON legend is illuminated.

4.6.3.7.2 Circuit breakers.- Use voltmeter, item 2, to verify that circuit breakers CB1 through CB7 are closed when reset buttons are depressed so white bands are concealed, and open when reset buttons are pulled out to expose white bands. Use voltmeter, item 4, to perform same test on circuit breaker CB8. With circuit breaker CB1 through CB8 closed, use resistor, item 5 through 9, to check overload protection of circuit breakers as follows: Connect resistor indicated below from load side of circuit breaker to ground, and verify that breaker trips within 20 seconds.

CB1	Item 5
CB2	Item 5
CB3	Item 6
CB4	Item 6
CB5	Item 7
CB6	Item 6
CB7	Item 8
CB8	Item 9

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4.6.3.7.3 LEVEL/AC/DC switch S2.

4.6.3.7.3.1 LEVEL/DC legend.- Set up conditions as stated in 3.5.7.1.2. Close +24V EXT circuit CB7 and operate S2 to normally closed position. Verify that LEVEL/DC legend is illuminated.

4.6.3.7.3.2 LEVEL/AC legend.- Under conditions as stated in 4.6.3.7.2.1 operate S2 to the normally open position and verify that LEVEL/AC legend is illuminated.

4.6.3.7.4 OVERLOAD/OFF/ON switch S7.

4.6.3.7.4.1 OVERLOAD/OFF legend.- Set up conditions as stated 3.5.7.3.1a. Set S7 to normally closed position and verify that OVERLOAD/OFF legend is illuminated.

4.6.3.7.4.2 OVERLOAD/ON legend.- Set up conditions as stated 4.6.3.7.4.1, operate S7 to normally open position and verify that OVERLOAD legend is illuminated.

4.6.3.7.5 PROFILE/OFF/ON switch S3.

4.6.3.7.5.1 PROFILE/OFF legend.- Set up conditions as stated in 3.5.7.3.1a. Set S3 to normally closed position and verify that PROFILE/OFF legend is illuminated.

4.6.3.7.5.2 PROFILE/ON legend.- Set up conditions as stated in 4.6.3.7.5.1, operate S3 to normally open position and verify that PROFILE/ON legend is illuminated.

4.6.3.7.6 NORMAL/HI/LOW switch S4.

4.6.3.7.6.1 NORMAL legend.- Set up conditions as stated in 3.5.7.3.1a. Set S3 to illuminate PROFILE/OFF legend. Verify that NORMAL legend of S4 is illuminated.

4.6.3.7.6.2 HI legend.- Set up conditions as stated in 4.6.3.7.6.1. Operate S3 to illuminate PROFILE/ON legend. Set S4 to normally closed position and verify that HI legend of S4 is illuminated.

4.6.3.7.6.3 LOW legend.- Set up conditions as stated in 4.6.3.7.6.2. Operate S4 to normally open position and verify that LOW legend of S4 is illuminated.

4.6.3.7.7 TEST/1/2 switch S5.

4.6.3.7.7.1 TEST/1 legend.- Set up conditions as stated in 3.5.7.3.1a. Set S5 to the normally closed position and verify that TEST/1 legend of S5 is illuminated.

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4.6.3.7.7.2 TEST/2 legend.- Set up conditions as stated in 4.6.3.7.7.1. Operate S5 to the normally open position and verify that TEST/2 legend of S5 is illuminated.

4.6.3.7.8 RESET/OFF/ON switch S6.

4.6.3.7.8.1 RESET/OFF legend.- Set up conditions as stated in 3.5.7.3.1a. Set S6 to normally closed position. Insert Adapter CP6 into test sockets A and B. Insert Self Test Card #1 into CARD READER ASSEMBLY and operate handle. Verify that RESET/OFF legend of S6 is illuminated.

4.6.3.7.8.2 RESET/ON legend.- Set up conditions as stated in 4.6.3.7.8.1. Operate S6 to normally open position and verify that RESET/ON legend of S6 is illuminated.

4.6.3.7.9 POWER indicators.- Set up conditions as stated in 3.5.7.3.1a. Insert Adapter CB6 into test sockets A and B and insert any self test card, 1 through 33, into the CARD READER ASSEMBLY. Verify that power indicators +6.6V, -15V, +15V, +24V EXT, +24V INT, -25V, +25V and LINE are illuminated.

4.6.3.7.10 Digiswitch lamps.- Set up conditions stated in 4.6.3.7.9. Verify that two thumbwheel switch lamps are illuminated in each of TEST POINT A and TEST POINT B groups in conformance with 3.5.7.10.

4.6.3.7.11 TEST POINT A & B, and TEST IDENT lamp output.- Set up conditions as stated in 3.5.7.11 and 4.6.3.7.2. Insert Self Test Card #1 into CARD READER ASSEMBLY. Using items 1, 2, and 3, verify that outputs exist as stated in Table XIII and 3.5.7.11. Remove Self Test Card #1 and verify all outputs in Table XIV and 3.5.7.11 by inserting Self Test Cards 2 through 33 in turn. Remove Adapter CP6 and insert Adapter CP4 into test sockets B and C. Verify outputs for Self Test Cards 34 and 35.

4.6.3.8 Test Set, Subassembly MX-8051/TPM-22.- References to test instruments are to those in 4.5.2.2.8. Turn TEST MODE switch S2 of power supply 1A1A4 to PWR/SUP SERVO position.

4.6.3.8.1 POWER SUPPLY test section.

4.6.3.8.1.1 +24 Vdc switch S1.- Verify that OFF or ON legend of S1 is illuminated. Depress and release S1 three times. Verify that OFF and ON legends are illuminated alternately with operation of S1.

4.6.3.8.1.2 TEST MODE switch S4.- Under conditions stated in 3.5.8.1.2.1 verify that POWER SUPPLY CONV & SERVO legend of S4 is illuminated. Depress and release S4 three times. Verify that POWER SUPPLY CONV & SERVO legend and REGULATOR legend are illuminated

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alternately with operation of S4. Using item 3, verify that 24 Vdc ±5% is present between J5-A(+) and J5-B(Ret) when POWER SUPPLY CONV & SERVO legend of S4 is illuminated. When REGULATOR legend of S4 is illuminated, use test equipment as shown in Figure 18 to verify that dc voltages are present at J2 as listed in Table XV. Use test equipment as shown in Figure 19 to verify that square wave outputs are present at J2 as listed in Table XV.

4.6.3.8.1.3 DC circuit breaker CB1.- Set up conditions as stated in 3.5.8.1.2.1. Observe that either legend of S4 is illuminated. Open circuit breaker CB1 by pulling out reset push-button so that white band is exposed. Observe that both legends of S4 are extinguished. Reset circuit breaker, and turn on dc power. Either legend of S4 shall again be illuminated.

4.6.3.8.1.4 INTERLOCK switch S2.

4.6.3.8.1.4.1 STANDBY legend.- Apply 24 Vdc ±5%, using item 1 between J9-G and J9-H. Verify that STANDBY legend of S2 is illuminated.

4.6.3.8.1.4.2 OPERATE legend.- Apply 24 Vdc ±5%, item 1 between J9-V and J9-a. Verify that OPERATE legend of S2 is illuminated. Using item 6, measure resistance between J9-H and J9-F. Depress and release S2 three times. Verify that resistance changes from 0 to infinity alternately with operation of S2.

4.6.3.8.1.5 RADIATE COMMAND switch S3.

4.6.3.8.1.5.1 RADIATE COMMAND legend.- Apply 24 Vdc using item 1 between J8-N and J9-P. Verify that RADIATE COMMAND legend of S3 is illuminated.

4.6.3.8.1.5.2 RADIATE COMMAND switch.- Apply voltages using item 1 and set up conditions as stated in 3.5.8.1.2.1. Depress and release S3 three times. Using item 3, verify that 24 Vdc ±5% and 0 volts appears between J6-U(+) and J6-V(Ret) on alternate operations of S3.

4.6.3.8.1.6 Regulator loading (S6-S9, S12 and monitoring (M1 and M2).- Apply voltages to J8 and J9 using items 2, 8, 9, 13, 14, 15, 16 and 17 as indicated in Figure 20. Connect a resistive load, item 4, between J5-A and J5-B. Set TEST SELECTOR switch S13 to POWER SUPPLY position. Set up conditions as stated in 3.5.8.1.2.1. Verify voltage readings at terminal points using item 3 and meter readings as given in Table XVI.

4.6.3.8.1.7 OVERLOAD TEST circuitry, 0-2A LOAD.

4.6.3.8.1.7.1 MIN load.- Apply power to J8 and J10 as stated in 3.5.8.1.7.1c and 3.5.8.1.2.2, using item 1 to supply 24 Vdc and item 2 to supply 10 Vdc. Set up conditions as stated in 3.5.8.1.7.1 and verify that M1 and M2 indicate 0.018 amperes ±17% and 10V ±7%, respectively.

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4.6.3.8.1.7.2 MAX load.- Apply voltages to J8 as stated in 3.5.8.1.7.2 and using item 2 to supply 10 Vdc. Connect a 4 ohm resistive load, item 10, between E19 and E15. Set up conditions as stated in 3.5.8.1.7.2 and verify that M1 indication varies continuously from 0.19 to 2.5A ±11% when R29 is varied from fully counterclockwise to fully clockwise. Verify that M2 indication is 10V ±7%.

4.6.3.8.1.8 2A CB circuit breaker CB3.- Set up conditions stated in 4.6.3.8.1.7.2. Observe that meters M1 and M2 indicate 2.5 amperes ±11% and 10V ±7%, respectively. Open circuit breaker CB3 by pulling out reset push-button so that white band is exposed and observe that M1 and M2 indicate 0. Reset circuit breaker. Observe that M1 and M2 indicate 2.5 amperes ±11% and 10V ±7%, respectively.

4.6.3.8.1.9 OVERLOAD TEST circuitry, 0-15A LOAD.

4.6.3.8.1.9.1 MIN load.- Apply power to J9 and J10 as stated in 3.5.8.1.9.1c using item 2 to supply 24 Vdc ±3%. Set up conditions stated in 3.5.8.1.9.1 and verify that M1 and M2 indicated 0.73 amperes ±17% and 24V ±7%, respectively.

4.6.3.8.1.9.2 MAX load.- Apply voltages to J9 as stated in 3.5.8.1.9.2 using item 2 to supply 24 Vdc ±3%. Connect a 2 ohm resistive load, item 12 between E17 and E13. Set up conditions stated in 3.5.8.1.9.2 and verify that M1 indication varies continuously from 4.8 to 12 amperes ±11% when R27 is varied from fully counterclockwise to fully clockwise. Verify that M2 indication is 24V ±7%.

4.6.3.8.1.10 15A CB circuit breaker CB2.- Set up conditions stated in 4.6.3.8.1.9.2. Observe that meters M1 and M2 indicate 12 amperes ±11% and 24V ±7%, respectively. Open circuit breaker CB3 by pulling out reset push-button so white band is exposed and observe that M1 and M2 indicate 0. Reset circuit breaker. Observe that M1 and M2 indicate 12 amperes ±11% and 24V ±7%, respectively.

4.6.3.8.1.11 SERVO 115V 400 Hz switch S21.

4.6.3.8.1.11.1 OFF legend.- Set up conditions stated in 3.5.8.1.11.1. Verify that OFF legend of S21 is illuminated.

4.6.3.8.1.11.2 ON legend.- Set up conditions stated in 3.5.8.1.11.2. Verify that ON legend of S21 is illuminated. Depress and release S21 three times. Verify that OFF and ON legend of S21 are illuminated alternately with operation of S21. When the ON legend is illuminated, verify, using item 3, that 115 Vac ±5% is present between the following points:

a. J1-E and J1-F(Ret).

b. J7-D and J1-F(Ret).

c. J7-s and J7-i(Ret).

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4.6.3.8.1.12 AC circuit breaker CB4.- Set up conditions stated in 4.6.3.8.1.11.2. Observe that ON legend of S21 is illuminated and using item 3, verify that 115V ac is present between J1-E and J1-F(Ret). Open circuit breaker CB4 by pulling out reset push-button so white band is exposed and observe that ON legend of S21 is extinguished and no voltage is present at J1-E and J1-F(Ret). Reset circuit breaker. Observe that ON legend of S21 is illuminated and 115 Vac is again present between J1-E and J1-F(Ret).

4.6.3.8.2 SERVO ELECTRONICS test section.

4.6.3.8.2.1 SERVO ELECTRONICS switch circuity.

4.6.3.8.2.1.1 SERVO COMMAND switch S5.

4.6.3.8.2.1.1.1 SERVO COMMAND legend.- Apply 24 Vdc ±5%, item 1 between J7-p(+) and J10-I(Ret). Verify that SERVO COMMAND legend of S5 is illuminated.

4.6.3.8.2.1.1.2 SERVO COMMAND switch.- Set up conditions stated in 3.5.8.1.2.1 and 3.5.8.2.1.1.2. Verify that 24 Vdc ±5% is present between J6-W(+) and J6-X(Ret) using item 3. Depress and release S5 three times. Verify that 24 Vdc and 0 Vdc is present alternately with operation of S5.

4.6.3.8.2.1.2 ELEVATION ON switch S22.- Apply 24 Vdc ±5%, item 1, between J7-p(+) and J1-P(Ret). Depress and release S22 three times. Verify that ELEVATION ON legend of S22 is illuminated and extinguished alternately with operation of S22. Also verify, using item 3, that 24 Vdc ±5% is present between J7-V(+) and J1-p(Ret) when ELEVATION ON legend is illuminated.

4.6.3.8.2.1.3 TRAVERSE ON switch S23.- Apply 24 Vdc ±5%, item 1, between J7-p(+) and J1-p(Ret). Depress and release S23 three times. Verify that TRAVERSE ON legend of S23 is illuminated and extinguished alternately with operation of S23. Also verify, using item 3, that 24 Vdc ±5% is present between J7-H(+) and J1-p(Ret) when TRAVERSE ON legend is illuminated.

4.6.3.8.2.1.4 MODE SELECTION switch S27.- Set up conditions stated in 3.5.8.1.2.1. Verify that either NORMAL or STOW legend of S27 is illuminated. Depress and release S27 three times. Verify that NORMAL and STOW legends of S27 are illuminated alternately with operation of S27. Verify that the following exists when STOW legend of S27 is illuminated:

a. UP-RIGHT or DOWN-LEFT legend of S28 illuminated.

b. Continuity between J10-I and J7-G.

c. Continuity between J10-I and J1-U.

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Use item 6 for continuity test. Verify that a through c do not occur when NORMAL legend of S27 is illuminated.

4.6.3.8.2.1.5 MODE SELECTION switch S28.- Set up conditions stated in 3.5.8.1.2.1. Set S27 to STOW position (STOW legend illuminated). Verify that either UP-RIGHT or DOWN-LEFT legend of S28 is illuminated. Depress and release S28 three times. Verify that UP-RIGHT and DOWN-LEFT legends of S28 are illuminated alternately with operation of S28. Verify that continuity exists between J7-d and J1-U when UP-RIGHT legend of S28 is illuminated and resistance is infinite when DOWN-LEFT is illuminated. Use item 6 for continuity test.

4.6.3.8.2.1.6 MODE SELECTION switch S29.- Set up conditions stated in 3.5.8.1.2.1. Depress and release LIMIT ON switch S29 three times. Verify that LIMIT ON legend is illuminated and extinguished alternately with operation of S29. Verify, using item 6, that continuity exists between J10-I and each of the following when LIMIT ON legend is illuminated:

a. J1-S

b. J7-W

c. J7-J

Also verify that resistance is infinite between those points when LIMIT ON legend is extinguished.

4.6.3.8.2.1.7 LOAD switch S20.- Set up conditions stated in 3.5.8.1.2.1. Connect a shorting wire between J1-Y and J1-A. Depress and release LOAD switch S20 three times. Verify that HIGH legend of S20 is illuminated while switch is depressed and LOW legend is illuminated when S20 is released. Verify using item 6 that resistance between J1-m and J1-r is 5 ohms ±5% when LOW legend of S20 is illuminated and 1 ohm ±2% when HIGH legend of S20 is illuminated.

4.6.3.8.2.1.8 AC SIG switch S24.- Set up conditions stated in 3.5.8.1.11.2. Connect a shorting wire between J7-D and J7-E. Either ELEVATION or TRAVERSE legend of S24 shall be illuminated. Depress and release AC SIG switch S24 three times. Verify that ELEVATION and TRAVERSE legends of S24 are illuminated alternately with operation of S24.

4.6.3.8.2.1.9 DC SIG switch S25.- Set up conditions stated in 3.5.8.1.11.2. Connect a shorting wire between J7-D and J7-E. Either ELEVATION or TRAVERSE legend of S25 shall be illuminated. Depress and release DC SIG switch, S25, three times. Verify that ELEVATION and TRAVERSE legends of S25 are illuminated alternately with operation of S25.

4.6.3.8.2.2 Signal sources (SERVO ELECTRONICS section).- Connect a shorting wire between J7-D and J7-E. Connect a resistive load, item 11(a) between J7-X and J7-Y. Connect a resistive load, item 11(b)

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between J7-A and J7-B. Set SERVO 115V 400 Hz switch S21 to illuminate ON legend. Set AC SIG switch S24 and DC SIG switch S25 to illuminate ELEVATION legends.

4.6.3.8.2.2.1 AC signal source.

a. AC SIGNAL rotary switch S32 in 0° position.- Set S31 to 0° position. Rotate AC SIGNAL potentiometer R58 from fully counterclockwise to fully clockwise. Verify, using item 5, that signal between IN(TP23) and RET(TP24) terminals varies from 0 to not less than 1.26 volts peak-to-peak 400 Hz, in phase ± 15° with respect to J10-C.

b. AC SIGNAL rotary switch S31 in 180° position.- Set S31 to 180° position. Rotate AC SIGNAL potentiometer R58 from fully counterclockwise to fully clockwise. Verify, using item 5, that signal between IN(TP23) and RET(TP24) terminals varies from 0 volts to not less than 1.26 volts peak-to-peak, 400 Hz, 180° out of phase ±15° with respect to J10-C.

c. AC SIGNAL rotary switch S31 in SHORT position.- Set S31 to SHORT position. Set R58 to fully counterclockwise position. Verify, using item 5, that voltage between IN(TP23) and RET(TP24) terminals is less than 0.33 mV peak-to-peak.

4.6.3.8.2.2.2 DC signal source.

a. DC SIGNAL rotary switch S26 in + position.- Set S26 to + position. Rotate DC SIGNAL potentiometer R60 from fully counterclockwise to fully clockwise. Verify, using item 7 that voltage between IN(TP25) and RET(TP26) terminals varies from 0 volts to not less than +4.25 Vdc. Verify, using item 5, that 400 Hz ripple content on the dc signal is less than 0.2 volts peak-to-peak.

b. DC SIGNAL rotary switch S26 in - position.- Set S26 to - position. Rotate DC SIGNAL potentiometer R60 from fully counterclockwise to fully clockwise. Verify, using item 7, that voltage between IN(TP25) and RET(TP26) terminals varies from 0 volts to not less than -4.25 Vdc. Verify, using item 5, that 400 Hz ripple content on the dc signal is less than 0.2 volts peak-to-peak.

c. DC SIGNAL rotary switch S26 in SHORT position.- Set S26 to SHORT position and DC SIGNAL potentiometer R60 to fully counterclockwise. Verify, using item 7, that voltage across IN(TP25) and RET(TP26) terminals is less than 1.0 millivolt rms.

4.6.3.8.3 FIRE CONTROL SERVO ELECTRONICS section.

4.6.3.8.3.1 280V, 15V switch S32.- Connect an ohmmeter, item 6, between J10-A and J11-F. Depress and release S32 three times. Verify that resistance changes from 0 to infinity alternately with operation of S32.

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4.6.3.8.3.1.1 280V ØA, 280V ØB legend.- Verify that 280V ØA and 280V ØB legends of S32 are illuminated.

4.6.3.8.3.1.2 +15V legend.- Verify that +15V legend of S32 is illuminated. Verify, using item 3, that +15 Vdc ±3% is present between J4-t(+) and J4-e(Ret).

4.6.3.8.3.1.3 -15V legend.- Verify that -15V legend of S32 is illuminated. Verify, using item 3, that +15 Vdc ±3% is present between J4-m(-) and J4-e(Ret).

4.6.3.8.3.2 280V P-P SQ WV 400 Hz ØA circuit breaker CB5.- Set up conditions stated in 3.5.8.3.1.1. Observe that 280V ØA legend of S32 is illuminated. Open circuit breaker CB5 by pulling out reset push-button so white band is exposed and observe that 280V ØA legend of S32 is extinguished. Reset circuit breaker. 280V ØA legend of S32 shall again be illuminated.

4.6.3.8.3.3 280V P-P SQ WV 400 Hz ØB circuit breaker CB6.- Set up conditions stated in 3.5.8.3.1.1. Observe that 280V ØB legend of S32 is illuminated. Open circuit breaker CB6 by pulling out reset push-button so white band is exposed and observe that 280V ØB legend of S32 is extinguished. Reset circuit breaker. 280V ØB legend of S32 shall again be illuminated.

4.6.3.8.3.4 +15V circuit breaker CB7.- Apply 15 Vdc using item 8 as stated in 4.6.3.8.3.1.2. Observe that +15V legend of S32 is illuminated. Open circuit breaker CB7 by pulling out reset push-button so white band is exposed and observe that +15V legend is extinguished. Reset circuit breaker. +15V legend of S32 shall again be illuminated.

4.6.3.8.3.5 -15V circuit breaker CB8.- Apply 15Vdc using item 9 as stated in 4.6.3.8.3.1.3. Observe that -15V legend of S32 is illuminated. Open circuit breaker CB8 by pulling out reset push-button so white band is exposed and observe that -15V legend is extinguished. Reset circuit breaker. -15V legend of S32 shall again be illuminated.

4.6.3.8.3.6 Signal sources (FIRE CONTROL ELECTRONICS section).- Connect a resistive load, item 11(c), between J4-C and J4-s. Close circuit breakers CB7 and CB8.

4.6.3.8.3.6.1 DC SIGNAL/PULSE rotary switch S30 in DC SIGNAL, +13 position.- Rotate AMPL DC INPUT control R67 from fully counterclockwise to clockwise position. Verify, using item 7, that voltage between TP29(+) and TP30(Ret) varies from 0 to not less than 13 Vdc.

4.6.3.8.3.6.2 DC SIGNAL/PULSE rotary switch S30 in DC SIGNAL, -13 position.- Rotate AMPL DC INPUT control from fully counterclockwise to clockwise position. Verify, using item 7, that voltage between TP29(-) and TP30(Ret) varies from 0 to not less than 13 Vdc.

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4.6.3.8.3.6.3 DC SIGNAL/PULSE rotary switch S30 in SHORT position.- Verify, using item 7, that 0 Vdc exist between TP29 and TB30 and also verify using item 6, that continuity exists between J11-T and J4-C.

4.6.3.8.3.6.4 DC SIGNAL/PULSE rotary switch S30 in PULSE, +13 position.- Verify, using item 5, that signal between J4-C(+) and TP30(Ret) conforms to data on line 1, Table XVI.

4.6.3.8.3.6.5 DC SIGNAL/PULSE rotary switch S30 in PULSE, -13 position.- Verify, using item 5, that signal between J4-C(+) and TP30(Ret) conforms to data on line 2, of Table XVI.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10554524. The level of protection shall be as specified in the contract.

6. NOTES

6.1 Intended use.- The SMSS is used by direct support (3rd echelon) and general support (4th echelon) maintenance to locate, verify, and troubleshoot defective units, assemblies, and subassemblies of Radar Set AN/VPS-2, except for Antenna AS-2049/VPS-2. SMSS consists of vertical console assembly 1A1, overhead console assembly 1A2, and bench console assembly 1A3. In these console assemblies are installed the major subassemblies, which are used to test radar items as described below:

a. Test Set, Subassembly MX-8046A/TPM-22 (1A1A1).- This subassembly provides facilities for testing and troubleshooting Receiver-Transmitter, Unit 2 of Radar Set AN/VPS-2 and its repairable subassemblies.

b. Test Set, Subassembly MX-8047/TPM-22 (1A1A2).- This subassembly provides facilities for testing and troubleshooting the microwave chassis and klystron power supply assemblies (2A1 and 2A2) of Radar Set AN/VPS-2.

c. Test Set, Subassembly MX-8051/TPM-22 1A1A3).- This subassembly provides facilities for testing and troubleshooting of the Power Supply PP-4812/VPS-2 of Radar Set AN/VPS-2 and its repairable assemblies. In addition, servo amplifiers of Weapon Systems XM-163 and XM-167 are tested by this subassembly.

d. Power supply subassembly (1A1A4).- This subassembly supplies power for the subassemblies of the SMSS that require power except the Blower subassembly (1A1A5).

e. Blower subassembly (1A1A5).- This subassembly supplies cooling for the vertical console assembly.

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f. Test Set, Subassembly MX-8048/TPM-22 (1A2A1).- This subassembly provides facilities for testing, aligning, and troubleshooting the Receiver, Radar R-1475/VPS-2 of Radar Set AN/VPS-2.

g. RF Millivoltmeter Subassembly (1A2A2).- This meter is for general use in measuring RF voltages in the SMSS and Radar Set AN/VPS-2.

h. Test Set, Subassembly MX-8049/TPM-22 (1A2A3).- This subassembly provides voltages and signals needed to test and troubleshoot the Computer, Range CP-888/VPS-2 of the Radar Set AN/VPS-2.

i. Power Supply, Subassembly (1A2A4).- This power supply furnishes an adjustable voltage for general use in troubleshooting Radar Set AN/VPS-2.

j. Storage Cabinet Subassembly (1A3A1).- This subassembly contains drawers for storing test fixtures and hand tools.

k. Test Set, Subassembly MX-8050/TPM-22 (1A3A2).- This subassembly provides facilities for testing plug-in printed circuit cards that are used in Radar Set AN/VPS-2 and Weapon System XM-163 and XM-167.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet (see 5.1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Block of serial numbers to identify each SMSS.

g. Provisions for First Article testing.

Custodian

Army - MU

Preparing Activity

Army - MU

Project No. 4931-A118

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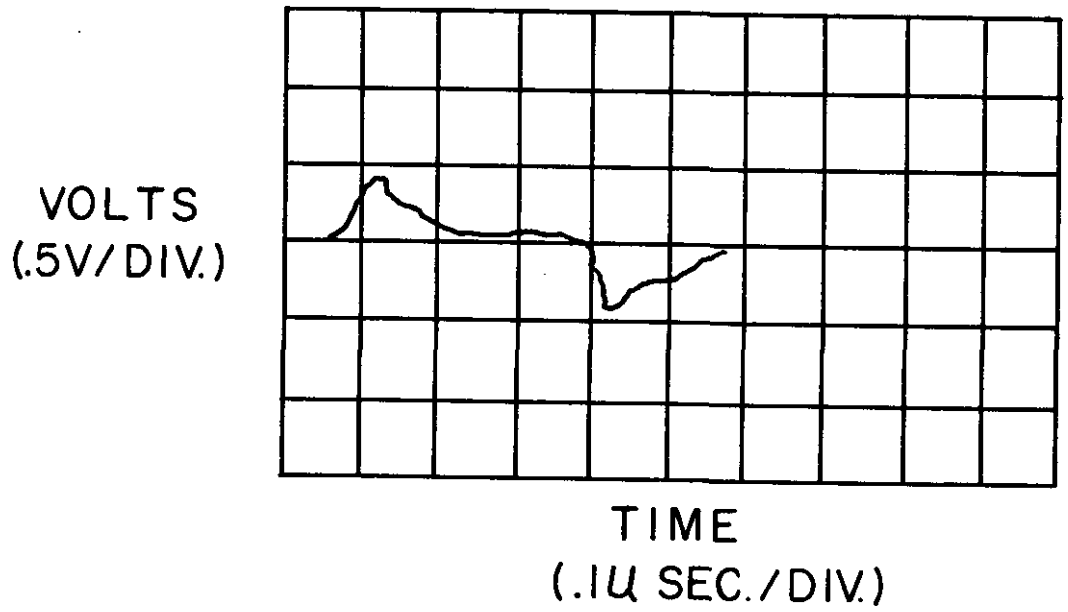


FIG. 1

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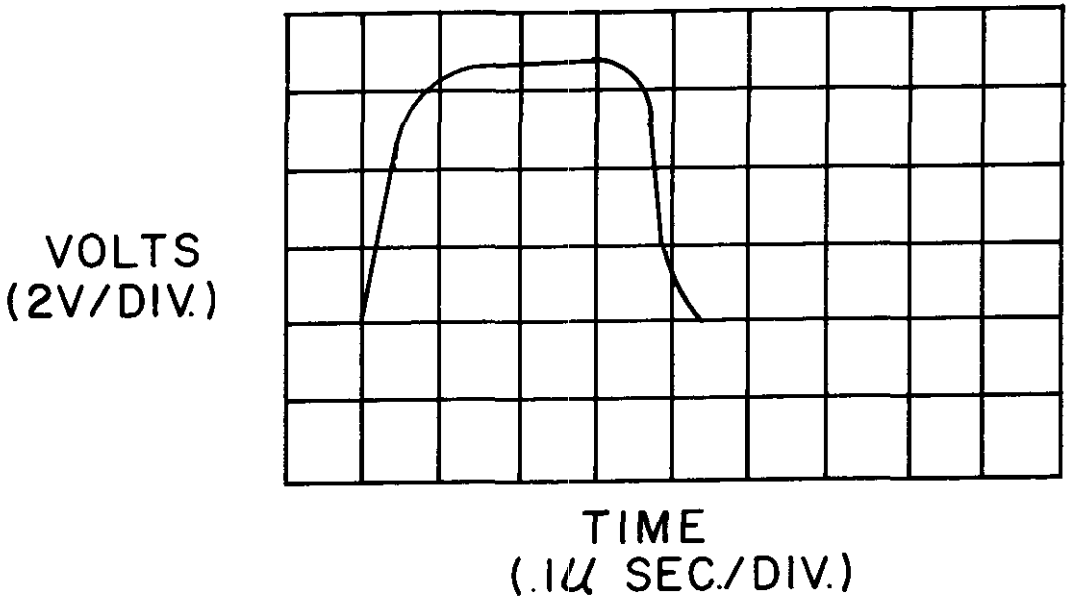


FIG. 2

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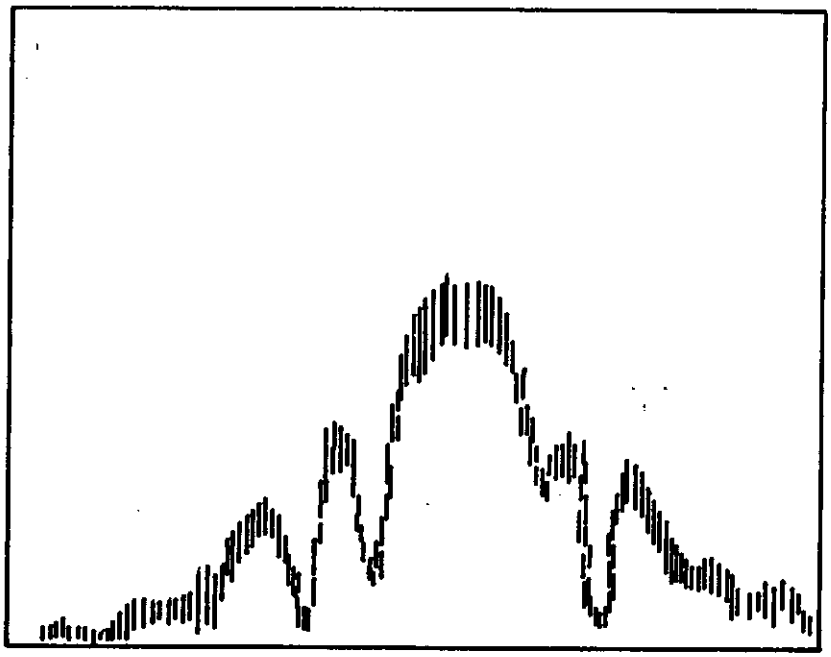


FIG. 3-a

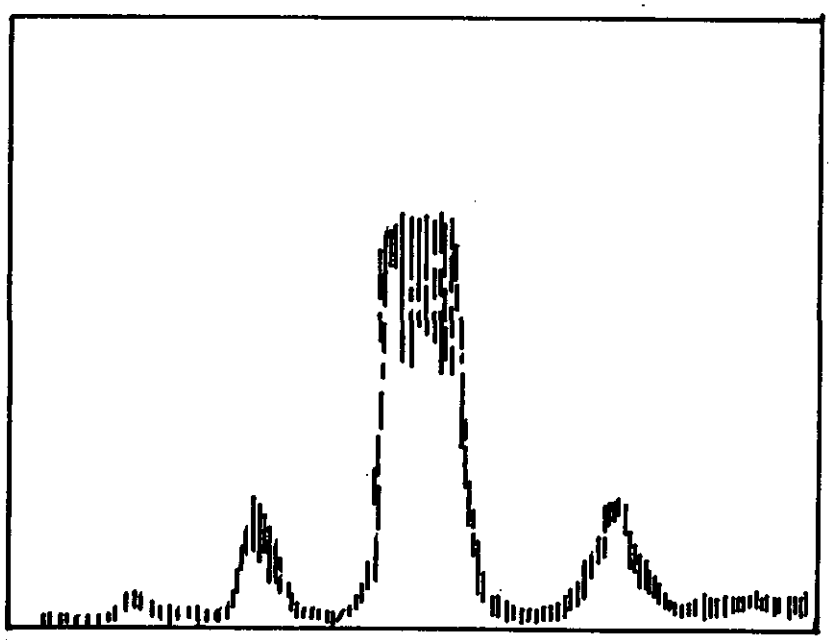


FIG. 3-b

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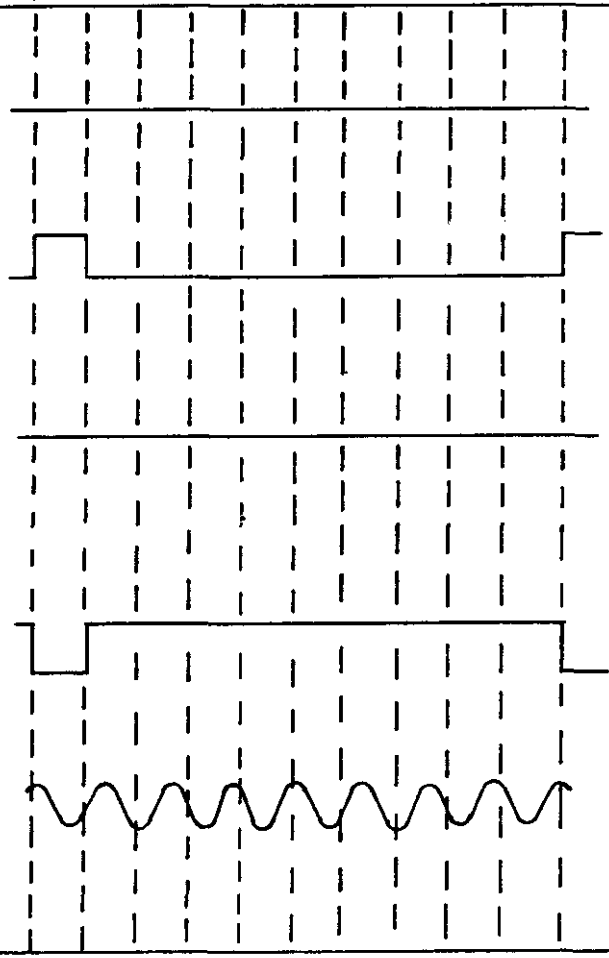
SIGNAL NAME	OSCILLOSCOPE SYNC	OSCILLOSCOPE SWEEP PER/DIV	OSCILLOSCOPE HORIZONTAL SCALE (DIV) 0 1 2 3 4 5 6 7 8 9 10	REMARKS
T PULSE	POS	5.0 μsec		1 PULSE EVERY 50 ± 5.0 μsec 0.8 ± 0.4 μsec wide
T PULSE	POS	1.0 μsec		
$\overline{T}$ PULSE	NEG	5.0 μsec		1 PULSE EVERY 50 ± 5.0 μsec
$\overline{T}$ PULSE	NEG	1.0 μsec		0.8 ± 0.4 μsec wide
7.5 MHz CLOCK	POS	0.1 μsec		1 CYCLE EVERY 0.133 μsec

FIGURE 4. WAVEFORMS

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:22Z
Check the source to verify that this is the current version before use.

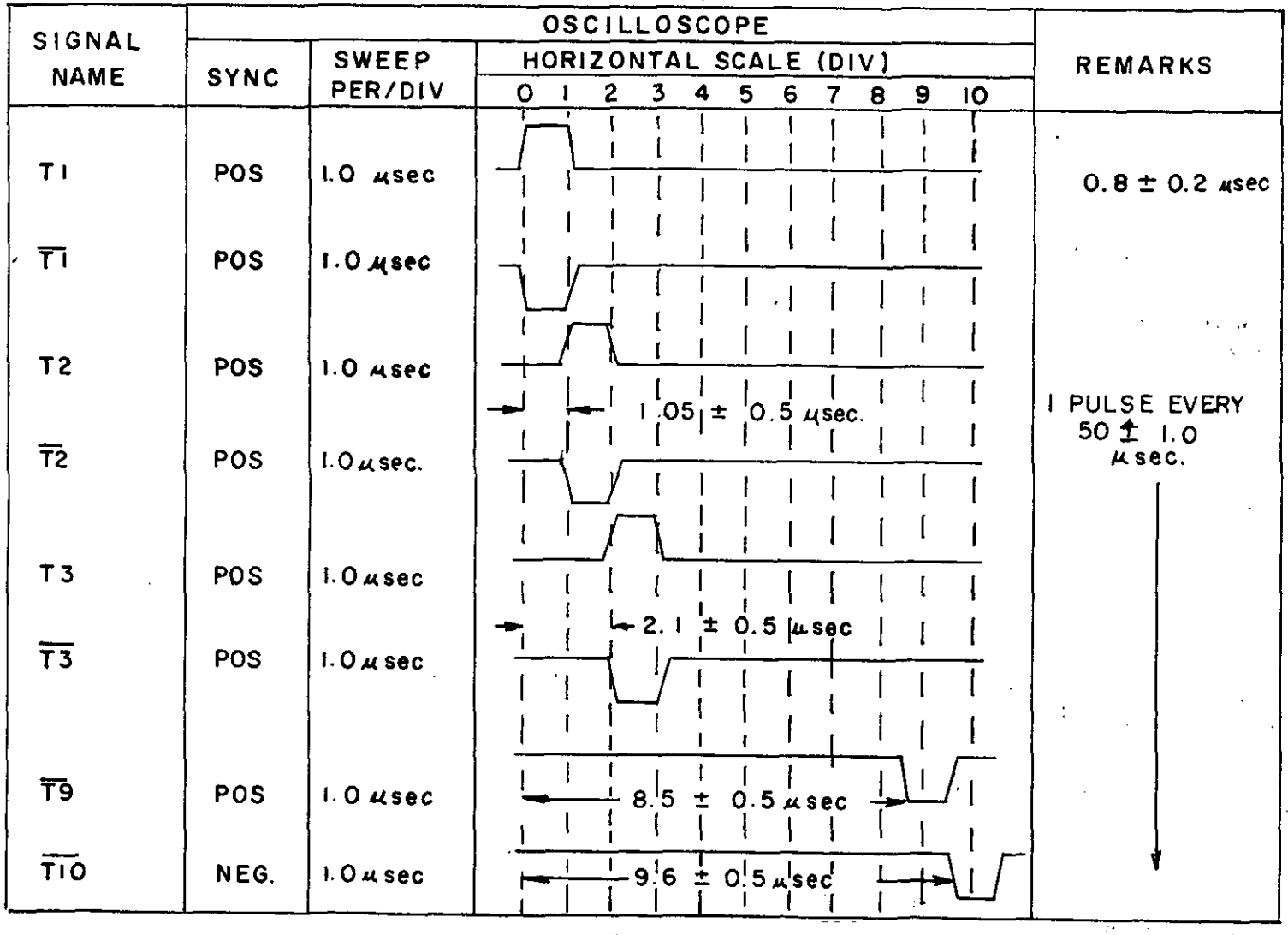


FIGURE 5. WAVEFORMS

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

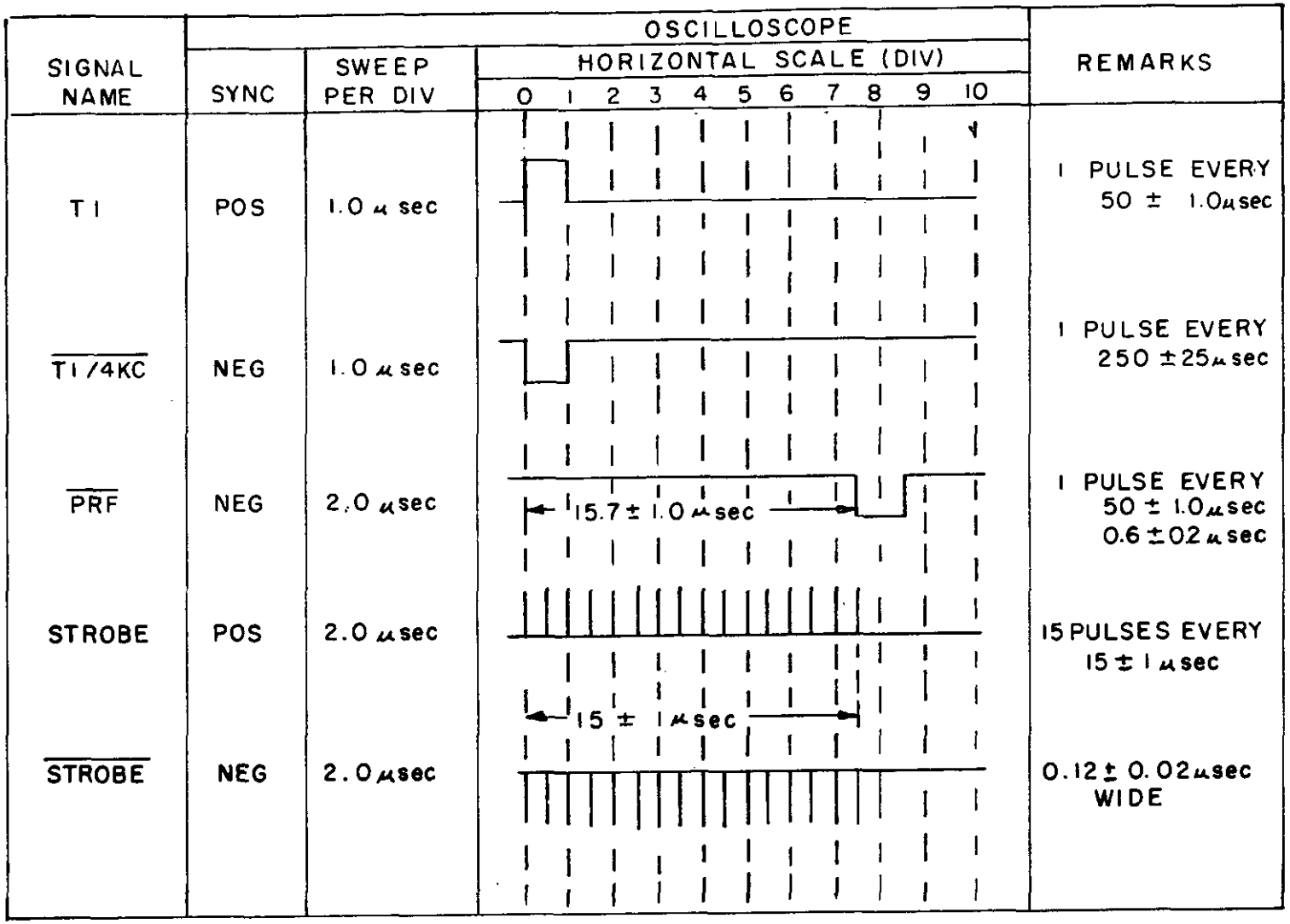


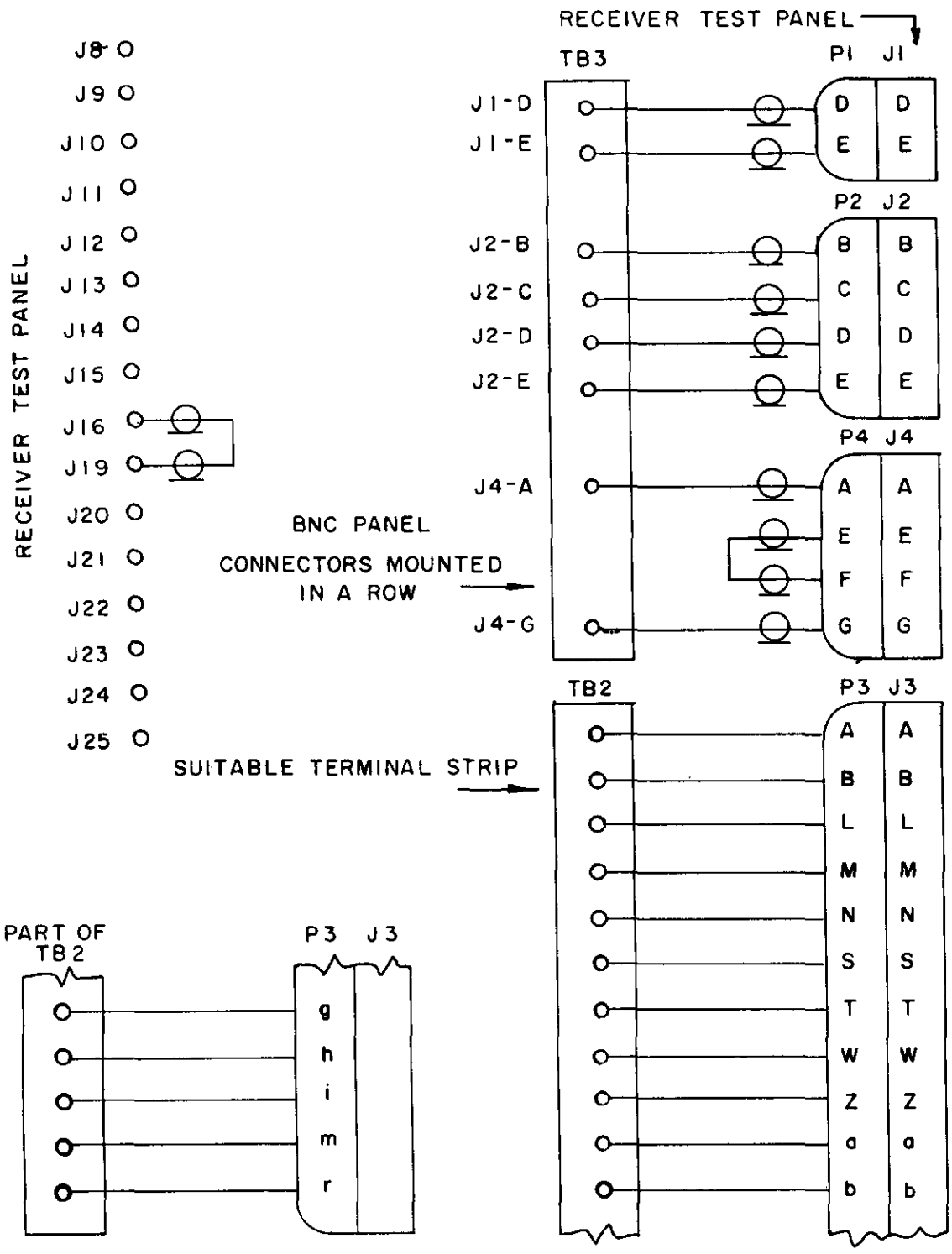
FIGURE 6. WAVEFORMS

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:22Z
Check the source to verify that this is the current version before use.

SIGNAL NAME	OSCILLOSCOPE SYNC	SWEEP PER/DIV	HORIZONTAL SCALE (DIV) 0 1 2 3 4 5 6 7 8 9 10	REMARKS
F1	NEG	50 μsec		
$\overline{F1}$	NEG	50 μsec		
F2	NEG	50 μsec		
$\overline{F2}$	NEG	50 μSec		
$\overline{F4}$	NEG	50 μsec		

FIGURE 7. WAVEFORMS

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PART OF TEST HARNESS ITEM

FIGURE 8. TEST HARNESS

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:22Z
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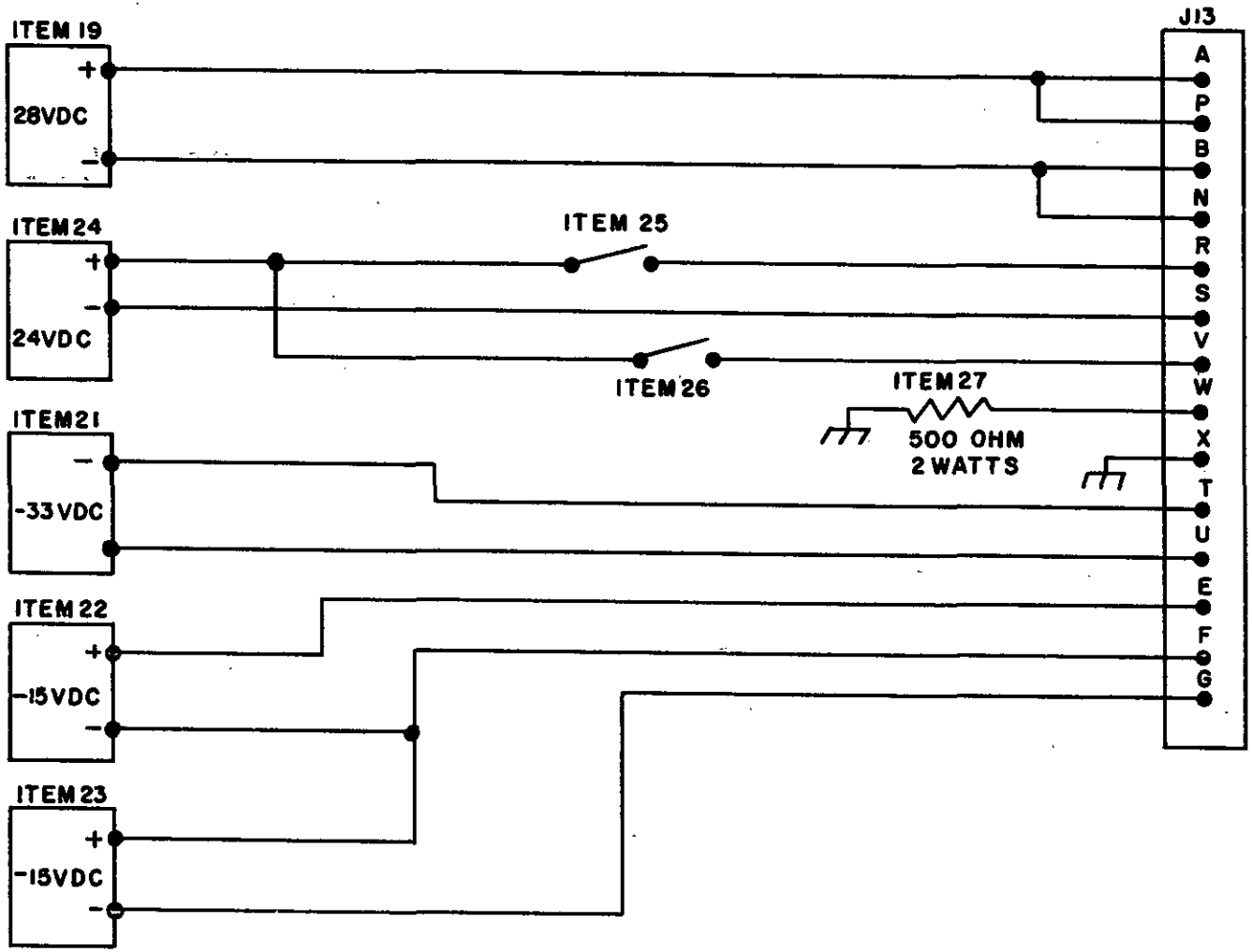


FIGURE 9.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:22Z
Check the source to verify that this is the current version before use.

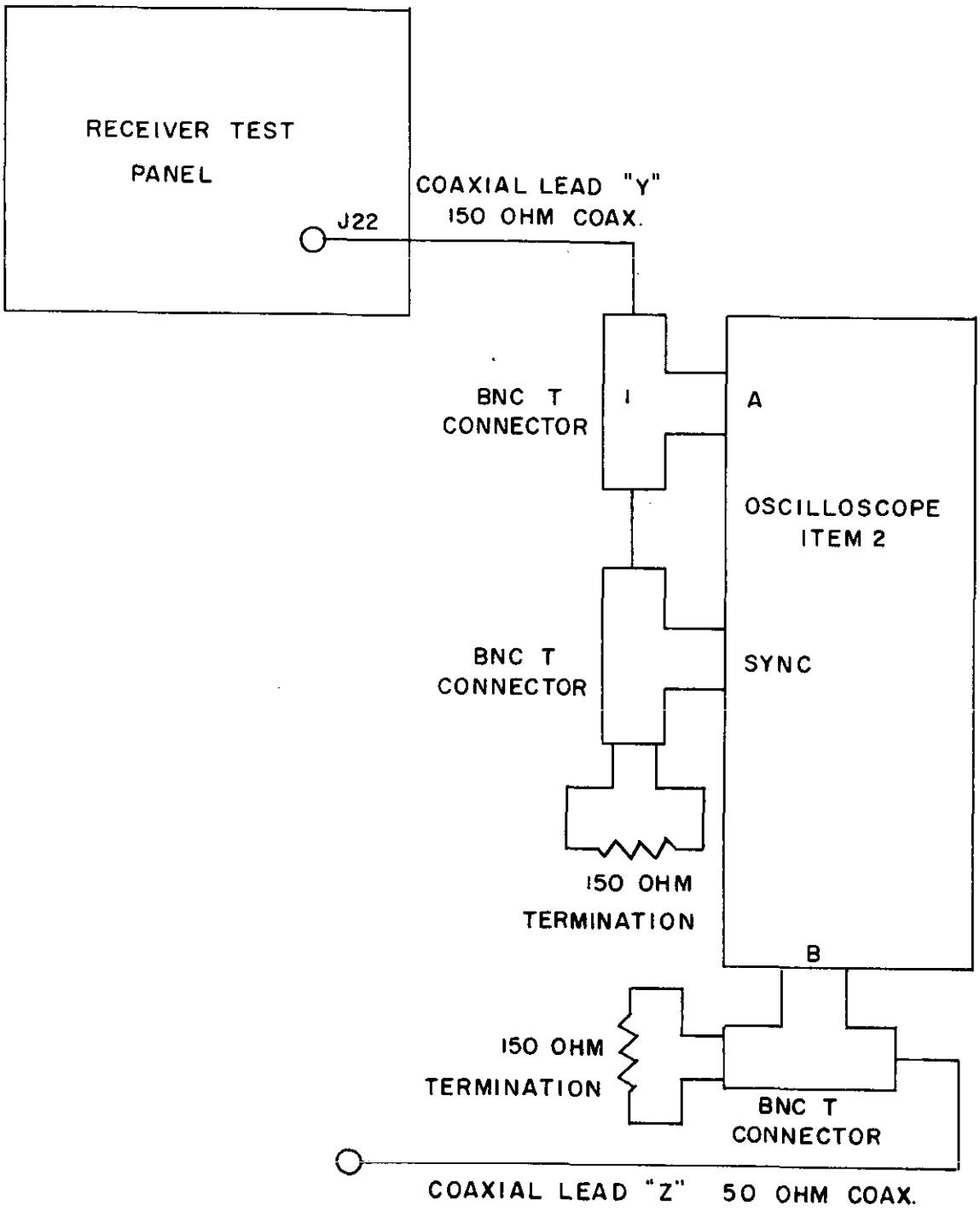


FIGURE 10. MODULATOR TRIGGER CHECK

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:22Z
Check the source to verify that this is the current version before use.

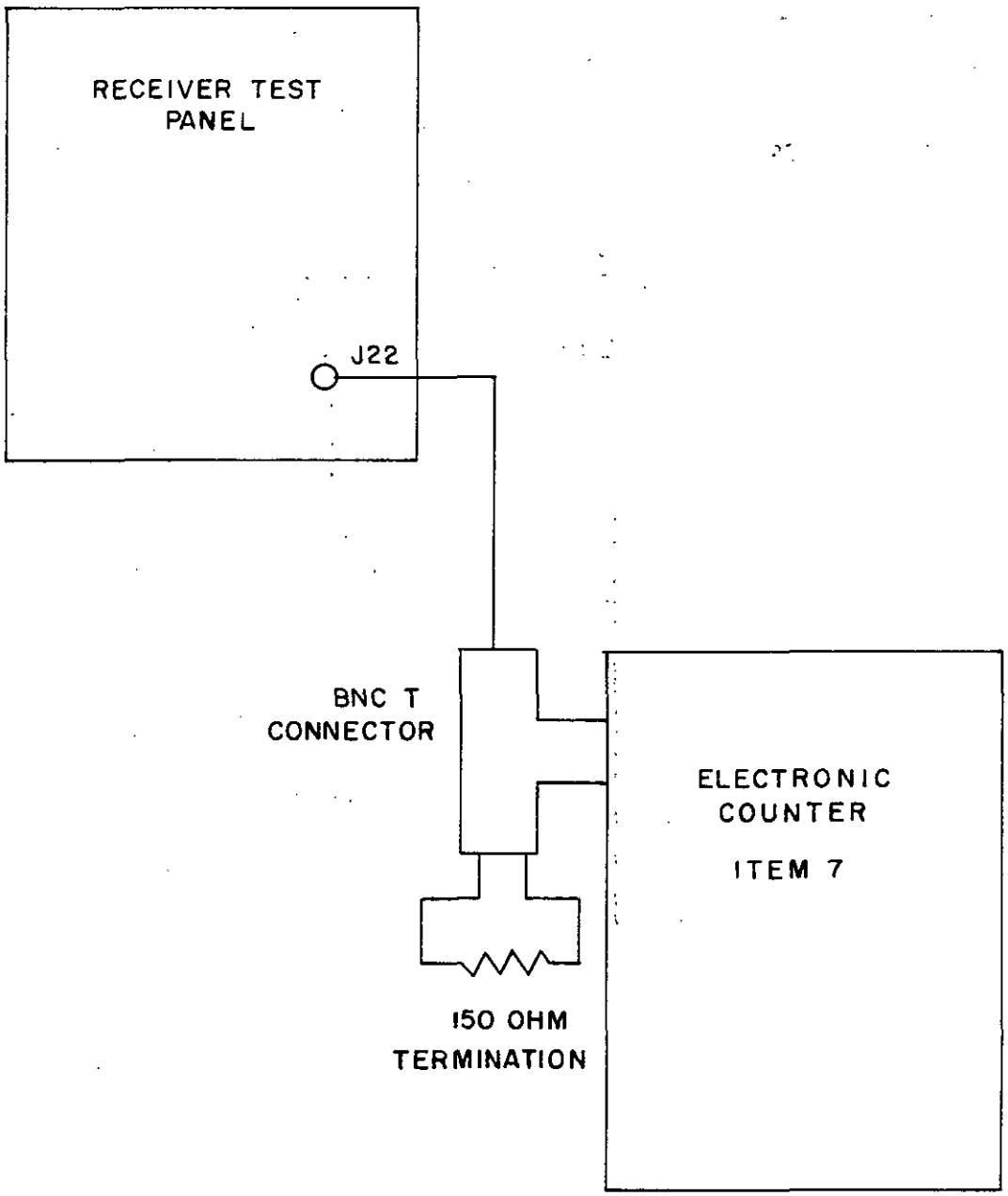


FIGURE 11. MODULATOR TRIGGER FREQUENCY CHECK

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:22Z
Check the source to verify that this is the current version before use.

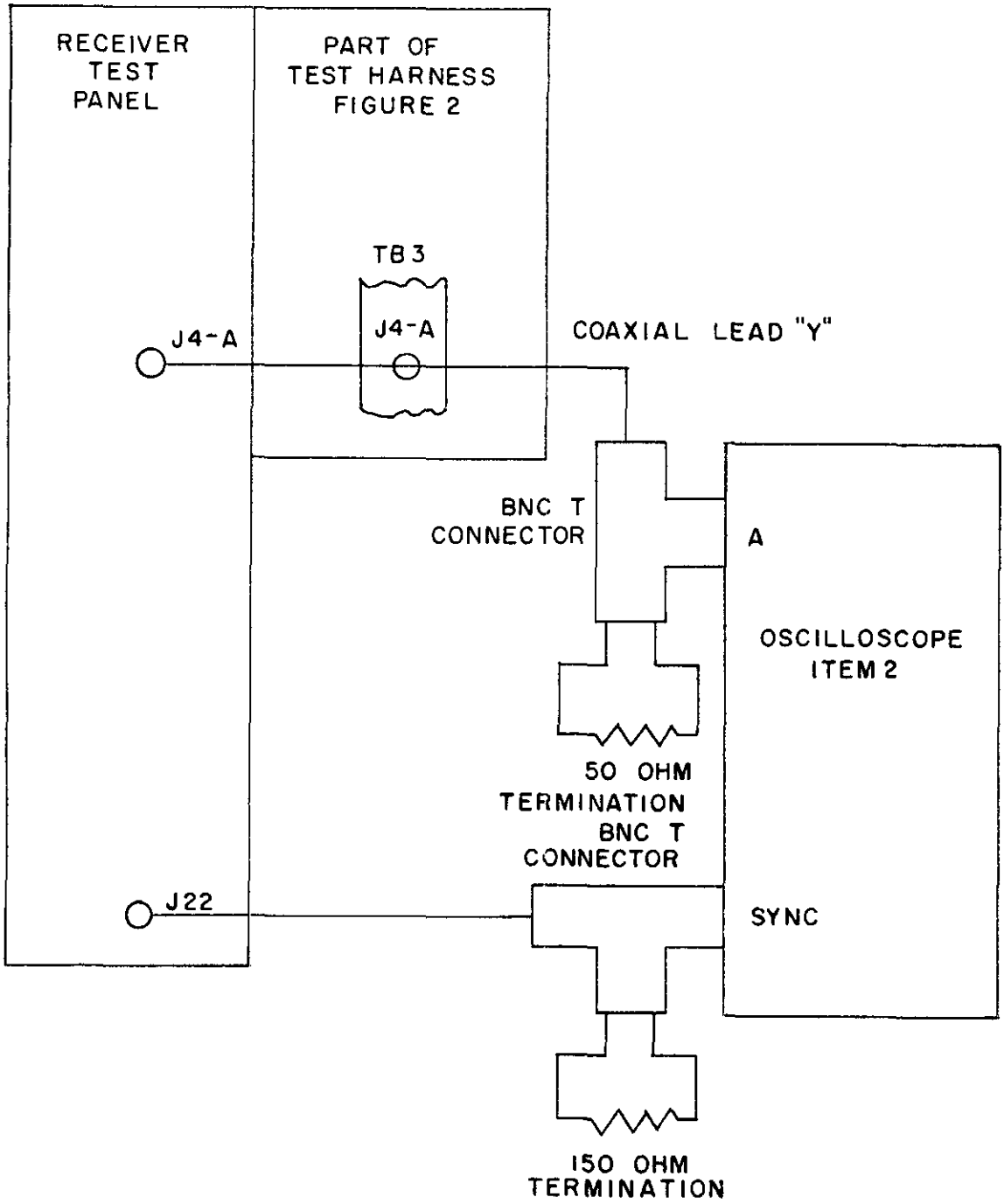


FIGURE 12. STALO OSC. CHECK

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:22Z
Check the source to verify that this is the current version before use.

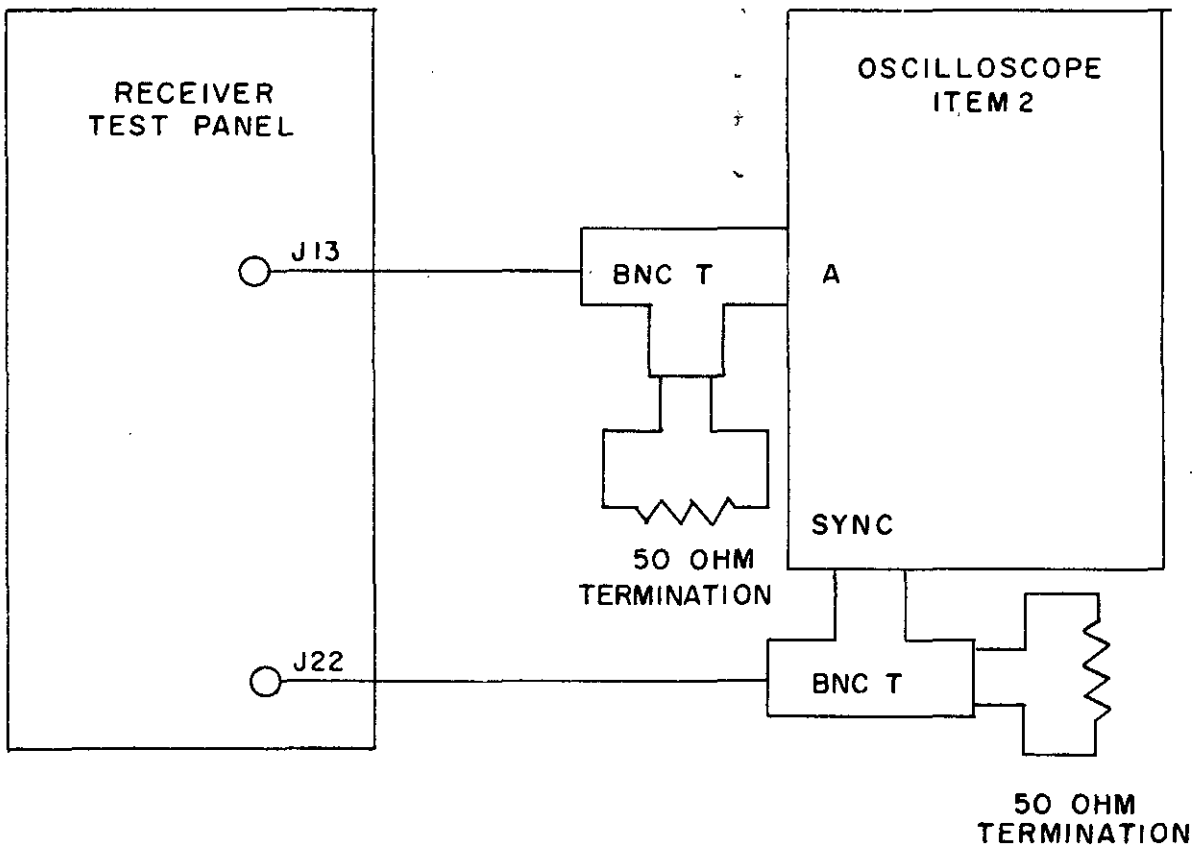


FIGURE 13. LO OUT CHECK

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:22Z
Check the source to verify that this is the current version before use.

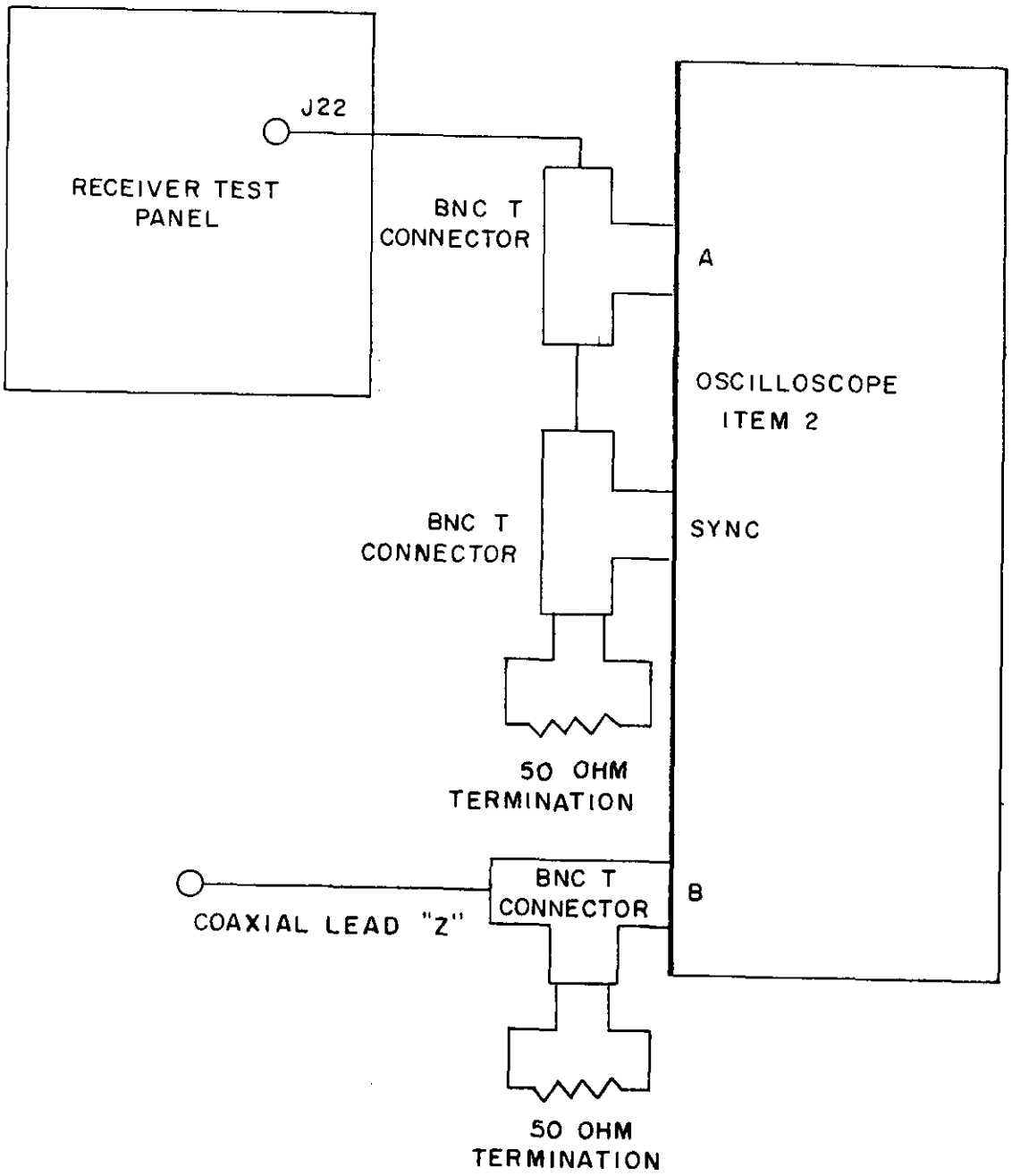


FIGURE 14. GATED STALO OSC. CHECK #1

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:22Z
Check the source to verify that this is the current version before use.

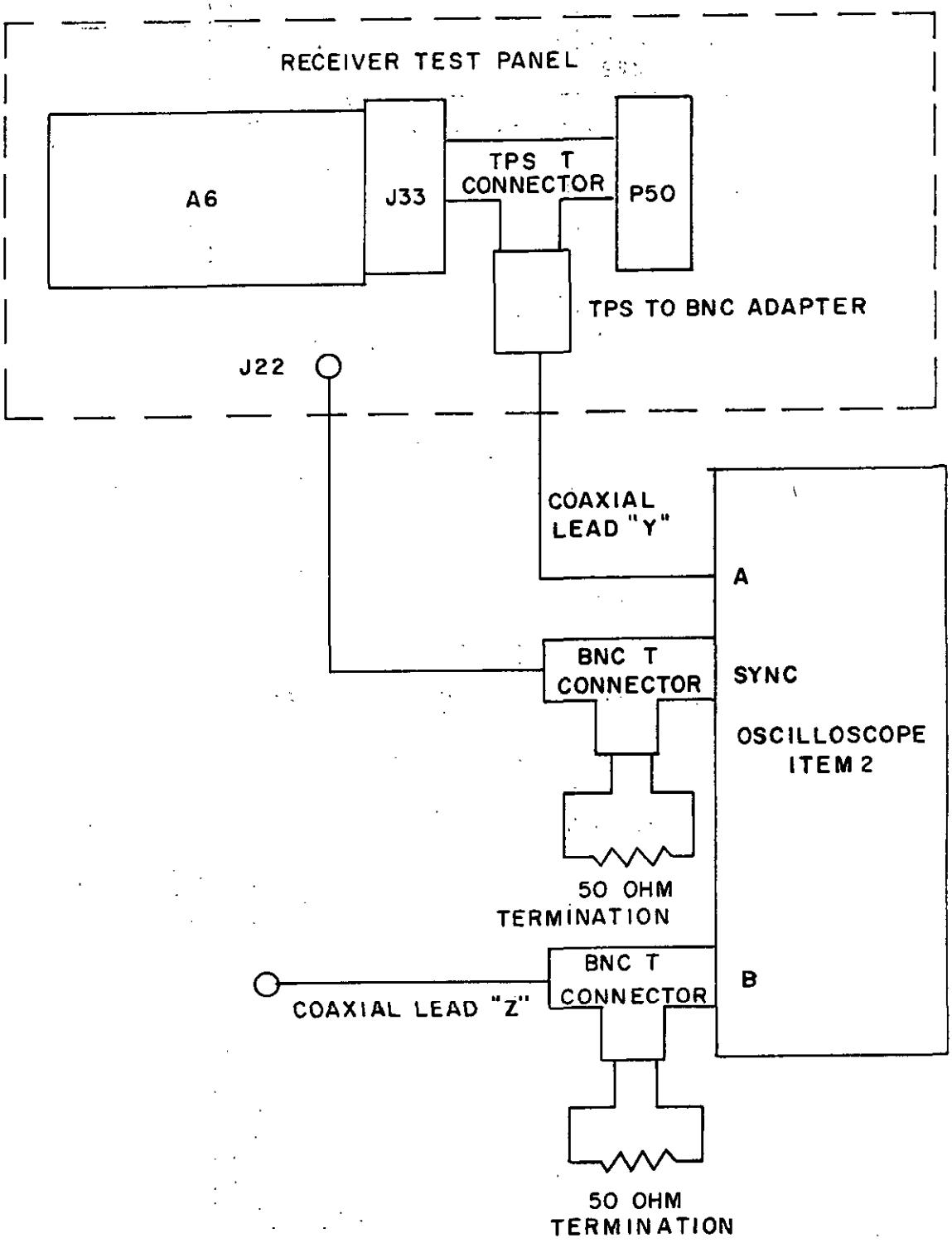


FIGURE 15. 59.941 MHz BURST CHECK

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Check the source to verify that this is the current version before use.

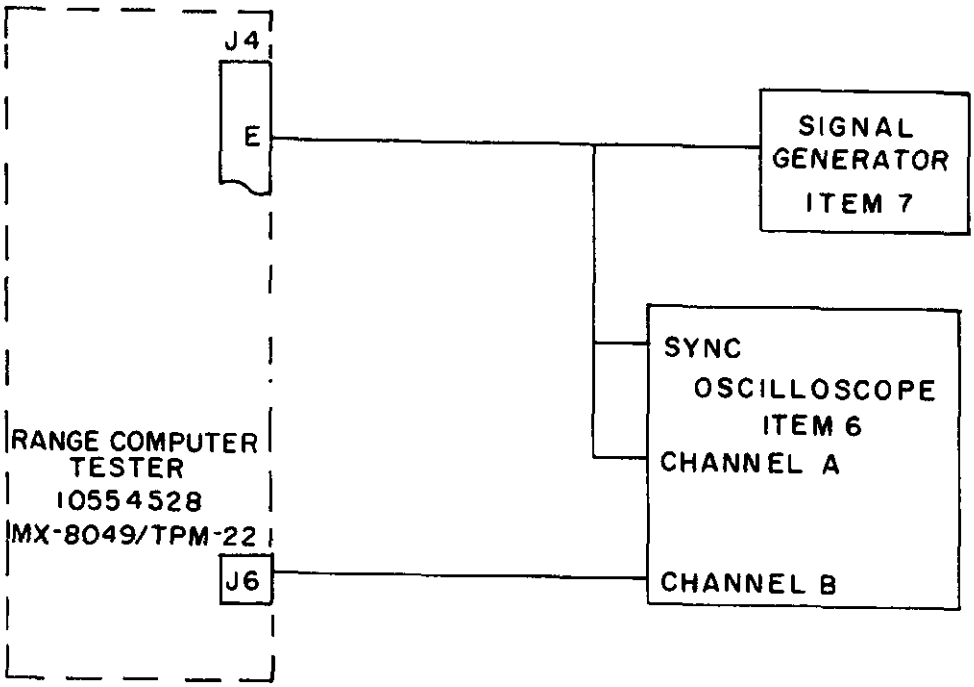


FIGURE 16. SIMULATED TARGET TEST SET UP

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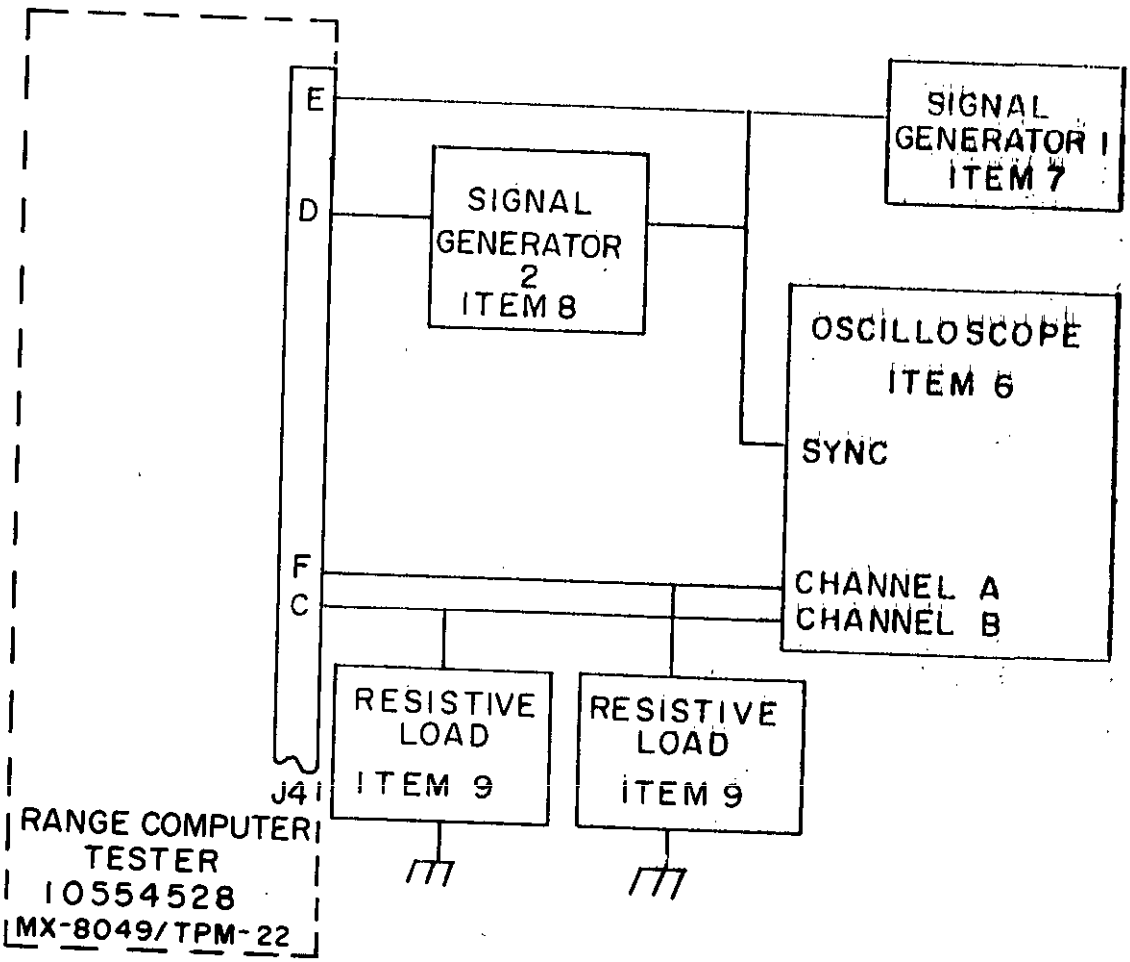


FIGURE 17. DOPPLER SETUP

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Check the source to verify that this is the current version before use.

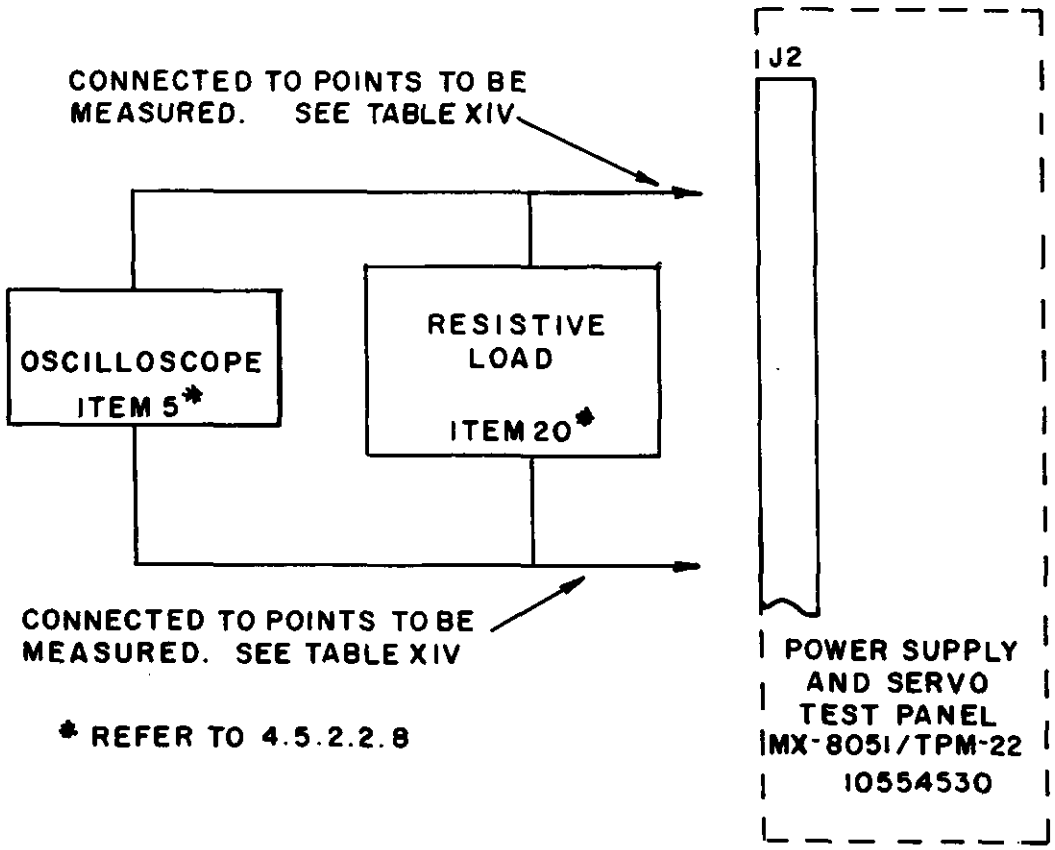


FIGURE 18. POWER SUPPLY AND SERVO TEST PANEL TEST SETUP

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Check the source to verify that this is the current version before use.

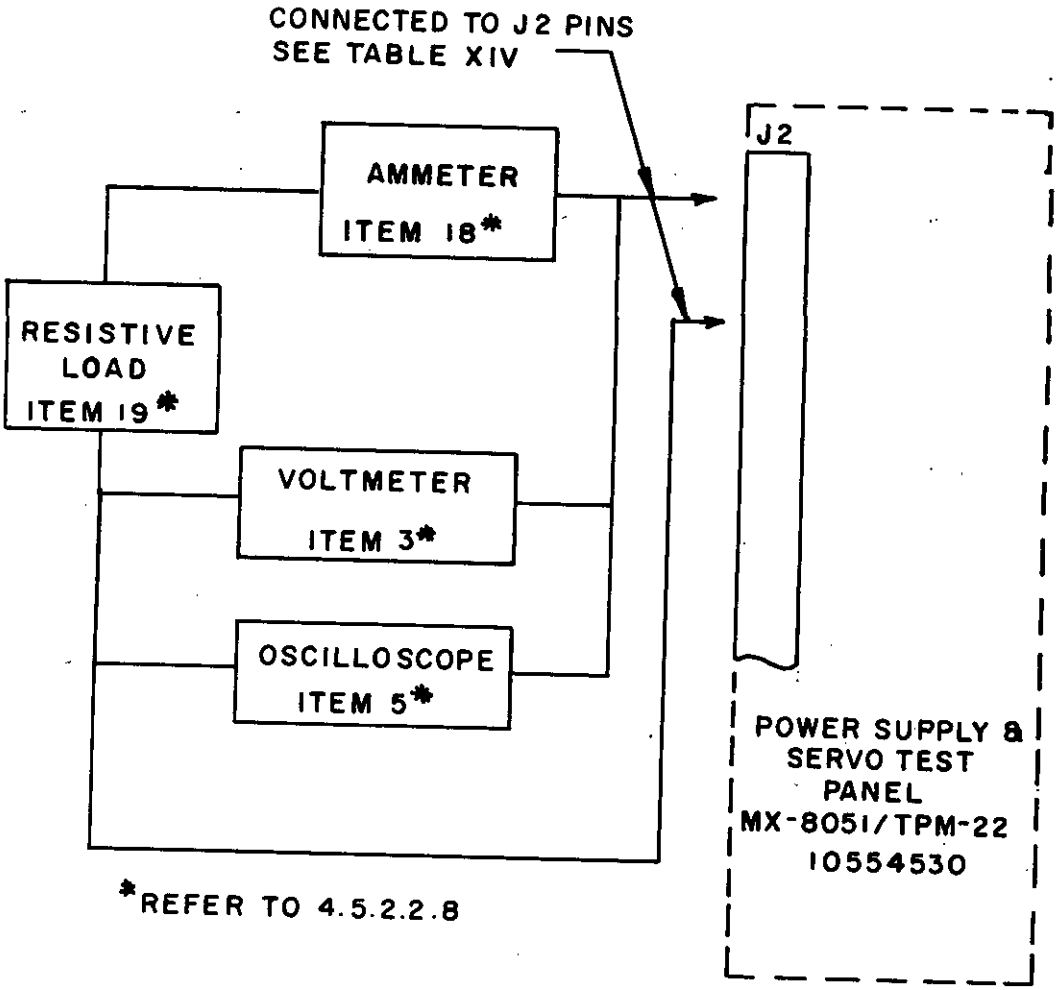


FIGURE 19. POWER SUPPLY AND SERVO TEST PANEL TEST SETUP

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

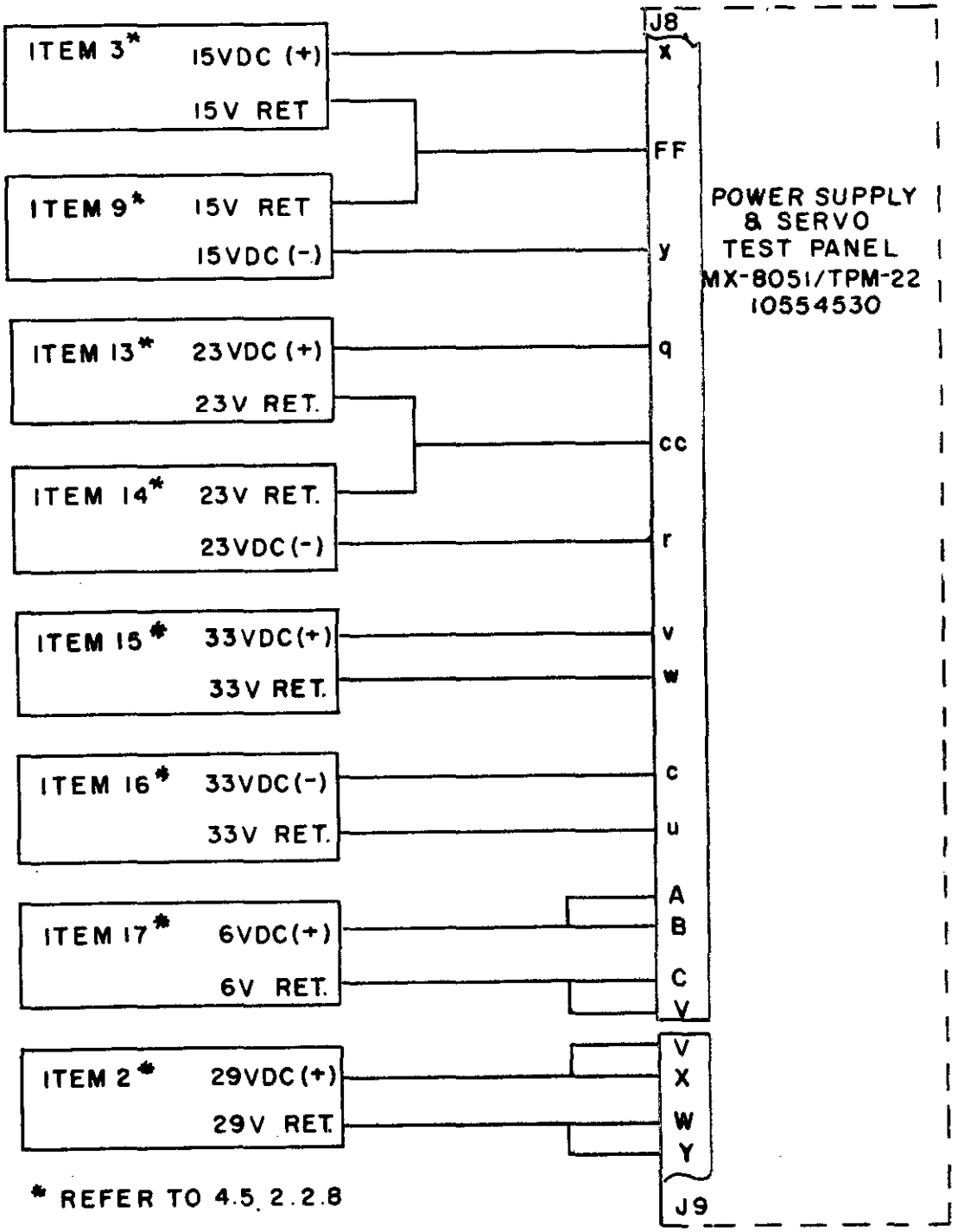


FIGURE 20. REGULATOR LOADING AND MONITORING TEST SETUP.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

129

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